

Finally, we note that MATLAB's random numbers are only seemingly unpredictable. In fact, MATLAB maintains a seed value that determines the subsequent "random" numbers that will be returned. This seed is controlled by the `rng` function; `s=rng` saves the current seed and `rng(s)` restores a previously saved seed. Initializing the random number generator with the same seed always generates the same sequence:

Example 1.26

```
>> s=rng;
>> 50+randi(50,1,12)
ans =
    89    76    80    80    72    92    58    56    77    78    59    58
>> rng(s);
>> 50+randi(50,1,12)
ans =
    89    76    80    80    72    92    58    56    77    78    59    58
```

When you run a simulation that uses `rand`, it normally doesn't matter how the `rng` seed is initialized. However, it can be instructive to use the same repeatable sequence of `rand` values when you are debugging your simulation.

Quiz 1.7

The number of characters in a tweet is equally likely to be any integer between 1 and 140. Simulate an experiment that generates 1000 tweets and counts the number of "long" tweets that have over 120 characters. Repeat this experiment 5 times.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

1.1.1 ● Continuing Quiz 1.1, write Gerlanda's entire menu in words (supply prices if you wish).

1.1.2 ● For Gerlanda's pizza in Quiz 1.1, answer these questions:

- Are N and M mutually exclusive?
- Are N , T , and M collectively exhaustive?
- Are T and O mutually exclusive? State this condition in words.
- Does Gerlanda's make Tuscan pizzas with mushrooms and onions?
- Does Gerlanda's make Neapolitan pizzas that have neither mushrooms nor onions?

1.1.3 ● Ricardo's offers customers two kinds of pizza crust, Roman (R) and Neapolitan (N). All pizzas have cheese but not all pizzas have tomato sauce. Roman pizzas can have tomato sauce or they can be white (W); Neapolitan pizzas always have tomato sauce. It is possible to order a Roman pizza with mushrooms (M) added. A Neapolitan pizza can contain mushrooms or onions (O) or both, in addition to the tomato sauce and cheese. Draw a Venn diagram that shows the relationship among the ingredients N , M , O , T , and W in the menu of Ricardo's pizzeria.

1.2.1 ● A hypothetical wi-fi transmission can take place at any of three speeds

depending on the condition of the radio channel between a laptop and an access point. The speeds are high (h) at 54 Mb/s, medium (m) at 11 Mb/s, and low (l) at 1 Mb/s. A user of the wi-fi connection can transmit a short signal corresponding to a mouse click (c), or a long signal corresponding to a tweet (t). Consider the experiment of monitoring wi-fi signals and observing the transmission speed and the length. An observation is a two-letter word, for example, a high-speed, mouse-click transmission is hm .

- What is the sample space of the experiment?
- Let A_1 be the event “medium speed connection.” What are the outcomes in A_1 ?
- Let A_2 be the event “mouse click.” What are the outcomes in A_2 ?
- Let A_3 be the event “high speed connection or low speed connection.” What are the outcomes in A_3 ?
- Are A_1 , A_2 , and A_3 mutually exclusive?
- Are A_1 , A_2 , and A_3 collectively exhaustive?

1.2.2 ● An integrated circuit factory has three machines X , Y , and Z . Test one integrated circuit produced by each machine. Either a circuit is acceptable (a) or it fails (f). An observation is a sequence of three test results corresponding to the circuits from machines X , Y , and Z , respectively. For example, aaf is the observation that the circuits from X and Y pass the test and the circuit from Z fails the test.

- What are the elements of the sample space of this experiment?
- What are the elements of the sets

$$Z_F = \{\text{circuit from } Z \text{ fails}\},$$

$$X_A = \{\text{circuit from } X \text{ is acceptable}\}.$$

- Are Z_F and X_A mutually exclusive?
- Are Z_F and X_A collectively exhaustive?

- What are the elements of the sets

$$C = \{\text{more than one circuit acceptable}\},$$

$$D = \{\text{at least two circuits fail}\}.$$

- Are C and D mutually exclusive?
- Are C and D collectively exhaustive?

1.2.3 ● Shuffle a deck of cards and turn over the first card. What is the sample space of this experiment? How many outcomes are in the event that the first card is a heart?

1.2.4 ● Find out the birthday (month and day but not year) of a randomly chosen person. What is the sample space of the experiment? How many outcomes are in the event that the person is born in July?

1.2.5 ● The sample space of an experiment consists of all undergraduates at a university. Give four examples of partitions.

1.2.6 ● The sample space of an experiment consists of the measured resistances of two resistors. Give four examples of partitions.

1.3.1 ● Find $P[B]$ in each case:

- Events A and B are a partition and $P[A] = 3P[B]$.
- For events A and B , $P[A \cup B] = P[A]$ and $P[A \cap B] = 0$.
- For events A and B , $P[A \cup B] = P[A] - P[B]$.

1.3.2 ● You roll two fair six-sided dice; one die is red, the other is white. Let R_i be the event that the red die rolls i . Let W_j be the event that the white die rolls j .

- What is $P[R_3W_2]$?
- What is the $P[S_5]$ that the sum of the two rolls is 5?

1.3.3 ● You roll two fair six-sided dice. Find the probability $P[D_3]$ that the absolute value of the difference of the dice is 3.

1.3.4 ● Indicate whether each statement is true or false.

- If $P[A] = 2P[A^c]$, then $P[A] = 1/2$.
- For all A and B , $P[AB] \leq P[A]P[B]$.

- (c) If $P[A] < P[B]$, then $P[AB] < P[B]$.
 (d) If $P[A \cap B] = P[A]$, then $P[A] \geq P[B]$.

1.3.5 ● Computer programs are classified by the length of the source code and by the execution time. Programs with more than 150 lines in the source code are big (B). Programs with ≤ 150 lines are little (L). Fast programs (F) run in less than 0.1 seconds. Slow programs (W) require at least 0.1 seconds. Monitor a program executed by a computer. Observe the length of the source code and the run time. The probability model for this experiment contains the following information: $P[LF] = 0.5$, $P[BF] = 0.2$, and $P[BW] = 0.2$. What is the sample space of the experiment? Calculate the following probabilities: $P[W]$, $P[B]$, and $P[W \cup B]$.

1.3.6 ● There are two types of cellular phones, handheld phones (H) that you carry and mobile phones (M) that are mounted in vehicles. Phone calls can be classified by the traveling speed of the user as fast (F) or slow (W). Monitor a cellular phone call and observe the type of telephone and the speed of the user. The probability model for this experiment has the following information: $P[F] = 0.5$, $P[HF] = 0.2$, $P[MW] = 0.1$. What is the sample space of the experiment? Find the following probabilities $P[W]$, $P[MF]$, and $P[H]$.

1.3.7 ● Shuffle a deck of cards and turn over the first card. What is the probability that the first card is a heart?

1.3.8 ● You have a six-sided die that you roll once and observe the number of dots facing upwards. What is the sample space? What is the probability of each sample outcome? What is the probability of E , the event that the roll is even?

1.3.9 ● A student's score on a 10-point quiz is equally likely to be any integer between 0 and 10. What is the probability of an A , which requires the student to get a score of 9 or more? What is the probability the student gets an F by getting less than 4?

1.3.10 ■ Use Theorem 1.4 to prove the following facts:

- (a) $P[A \cup B] \geq P[A]$
 (b) $P[A \cup B] \geq P[B]$
 (c) $P[A \cap B] \leq P[A]$
 (d) $P[A \cap B] \leq P[B]$

1.3.11 ■ Use Theorem 1.4 to prove by induction the *union bound*: For any collection of events $A_1, \dots, A_n$,

$$P[A_1 \cup A_2 \cup \dots \cup A_n] \leq \sum_{i=1}^n P[A_i].$$

1.3.12 ♦ Using *only* the three axioms of probability, prove $P[\emptyset] = 0$.

1.3.13 ♦ Using the three axioms of probability and the fact that $P[\emptyset] = 0$, prove Theorem 1.3. Hint: Define $A_i = B_i$ for $i = 1, \dots, m$ and $A_i = \emptyset$ for $i > m$.

1.3.14 ♦♦ For each fact stated in Theorem 1.4, determine which of the three axioms of probability are needed to prove the fact.

1.4.1 ● Mobile telephones perform *handoffs* as they move from cell to cell. During a call, a telephone either performs zero handoffs (H_0), one handoff (H_1), or more than one handoff (H_2). In addition, each call is either long (L), if it lasts more than three minutes, or brief (B). The following table describes the probabilities of the possible types of calls.

	H_0	H_1	H_2
L	0.1	0.1	0.2
B	0.4	0.1	0.1

- (a) What is the probability that a brief call will have no handoffs?
 (b) What is the probability that a call with one handoff will be long?
 (c) What is the probability that a long call will have one or more handoffs?

1.4.2 ● You have a six-sided die that you roll once. Let R_i denote the event that the roll is i . Let G_j denote the event that

the roll is greater than j . Let E denote the event that the roll of the die is even-numbered.

- What is $P[R_3|G_1]$, the conditional probability that 3 is rolled given that the roll is greater than 1?
- What is the conditional probability that 6 is rolled given that the roll is greater than 3?
- What is $P[G_3|E]$, the conditional probability that the roll is greater than 3 given that the roll is even?
- Given that the roll is greater than 3, what is the conditional probability that the roll is even?

1.4.3 ● You have a shuffled deck of three cards: 2, 3, and 4. You draw one card. Let C_i denote the event that card i is picked. Let E denote the event that the card chosen is an even-numbered card.

- What is $P[C_2|E]$, the probability that the 2 is picked given that an even-numbered card is chosen?
- What is the conditional probability that an even-numbered card is picked given that the 2 is picked?

1.4.4 ■ Phonesmart is having a sale on Bananas. If you buy one Banana at full price, you get a second at half price. When couples come in to buy a pair of phones, sales of Apricots and Bananas are equally likely. Moreover, given that the first phone sold is a Banana, the second phone is twice as likely to be a Banana rather than an Apricot. What is the probability that a couple buys a pair of Bananas?

1.4.5 ■ The basic rules of genetics were discovered in mid-1800s by Mendel, who found that each characteristic of a pea plant, such as whether the seeds were green or yellow, is determined by two genes, one from each parent. In his pea plants, Mendel found that yellow seeds were a dominant trait over green seeds. A yy pea with two yellow genes has yellow seeds; a gg pea with two recessive genes has green seeds; a hybrid gy or yg

pea has yellow seeds. In one of Mendel's experiments, he started with a parental generation in which half the pea plants were yy and half the plants were gg . The two groups were crossbred so that each pea plant in the first generation was gy . In the second generation, each pea plant was equally likely to inherit a y or a g gene from each first-generation parent. What is the probability $P[Y]$ that a randomly chosen pea plant in the second generation has yellow seeds?

1.4.6 ■ From Problem 1.4.5, what is the conditional probability of yy , that a pea plant has two dominant genes given the event Y that it has yellow seeds?

1.4.7 ■ You have a shuffled deck of three cards: 2, 3, and 4, and you deal out the three cards. Let E_i denote the event that i th card dealt is even numbered.

- What is $P[E_2|E_1]$, the probability the second card is even given that the first card is even?
- What is the conditional probability that the first two cards are even given that the third card is even?
- Let O_i represent the event that the i th card dealt is odd numbered. What is $P[E_2|O_1]$, the conditional probability that the second card is even given that the first card is odd?
- What is the conditional probability that the second card is odd given that the first card is odd?

1.4.8 ♦ Deer ticks can carry both Lyme disease and human granulocytic ehrlichiosis (HGE). In a study of ticks in the Midwest, it was found that 16% carried Lyme disease, 10% had HGE, and that 10% of the ticks that had either Lyme disease or HGE carried both diseases.

- What is the probability $P[LH]$ that a tick carries both Lyme disease (L) and HGE (H)?
- What is the conditional probability that a tick has HGE given that it has Lyme disease?

1.5.1 ● Given the model of handoffs and call lengths in Problem 1.4.1,

- (a) What is the probability $P[H_0]$ that a phone makes no handoffs?
- (b) What is the probability a call is brief?
- (c) What is the probability a call is long or there are at least two handoffs?

1.5.2 ● For the telephone usage model of Example 1.18, let B_m denote the event that a call is billed for m minutes. To generate a phone bill, observe the duration of the call in integer minutes (rounding up). Charge for M minutes $M = 1, 2, 3, \dots$ if the exact duration T is $M - 1 < t \leq M$. A more complete probability model shows that for $m = 1, 2, \dots$ the probability of each event B_m is

$$P[B_m] = \alpha(1 - \alpha)^{m-1}$$

where $\alpha = 1 - (0.57)^{1/3} = 0.171$.

- (a) Classify a call as long, L , if the call lasts more than three minutes. What is $P[L]$?
- (b) What is the probability that a call will be billed for nine minutes or less?

1.5.3 ♦ Suppose a cellular telephone is equally likely to make zero handoffs (H_0), one handoff (H_1), or more than one handoff (H_2). Also, a caller is either on foot (F) with probability $5/12$ or in a vehicle (V).

- (a) Given the preceding information, find three ways to fill in the following probability table:

	H_0	H_1	H_2
F			
V			

- (b) Suppose we also learn that $1/4$ of all callers are on foot making calls with no handoffs and that $1/6$ of all callers are vehicle users making calls with a single handoff. Given these additional facts, find all possible ways to fill in the table of probabilities.

1.6.1 ● Is it possible for A and B to be independent events yet satisfy $A = B$?

1.6.2 ● Events A and B are equiprobable, mutually exclusive, and independent. What is $P[A]$?

1.6.3 ● At a Phonesmart store, each phone sold is twice as likely to be an Apricot as a Banana. Also each phone sale is independent of any other phone sale. If you monitor the sale of two phones, what is the probability that the two phones sold are the same?

1.6.4 ■ Use a Venn diagram in which the event areas are proportional to their probabilities to illustrate two events A and B that are independent.

1.6.5 ■ In an experiment, A and B are mutually exclusive events with probabilities $P[A] = 1/4$ and $P[B] = 1/8$.

- (a) Find $P[A \cap B]$, $P[A \cup B]$, $P[A \cap B^c]$, and $P[A \cup B^c]$.
- (b) Are A and B independent?

1.6.6 ■ In an experiment, C and D are independent events with probabilities $P[C] = 5/8$ and $P[D] = 3/8$.

- (a) Determine the probabilities $P[C \cap D]$, $P[C \cap D^c]$, and $P[C^c \cap D^c]$.
- (b) Are C^c and D^c independent?

1.6.7 ■ In an experiment, A and B are mutually exclusive events with probabilities $P[A \cup B] = 5/8$ and $P[A] = 3/8$.

- (a) Find $P[B]$, $P[A \cap B^c]$, and $P[A \cup B^c]$.
- (b) Are A and B independent?

1.6.8 ■ In an experiment, C , and D are independent events with probabilities $P[C \cap D] = 1/3$, and $P[C] = 1/2$.

- (a) Find $P[D]$, $P[C \cap D^c]$, and $P[C^c \cap D^c]$.
- (b) Find $P[C \cup D]$ and $P[C \cup D^c]$.
- (c) Are C and D^c independent?

1.6.9 ■ In an experiment with equiprobable outcomes, the sample space is $S = \{1, 2, 3, 4\}$ and $P[s] = 1/4$ for all $s \in S$. Find three events in S that are pairwise independent but are not independent. (Note:

Pairwise independent events meet the first three conditions of Definition 1.7).

1.6.10 (Continuation of Problem 1.4.5) One of Mendel's most significant results was the conclusion that genes determining different characteristics are transmitted independently. In pea plants, Mendel found that round peas (r) are a dominant trait over wrinkled peas (w). Mendel crossbred a group of (rr, yy) peas with a group of (ww, gg) peas. In this notation, rr denotes a pea with two "round" genes and ww denotes a pea with two "wrinkled" genes. The first generation were either (rw, yg), (rw, gy), (wr, yg), or (wr, gy) plants with both hybrid shape and hybrid color. Breeding among the first generation yielded second-generation plants in which genes for each characteristic were equally likely to be either dominant or recessive. What is the probability $P[Y]$ that a second-generation pea plant has yellow seeds? What is the probability $P[R]$ that a second-generation plant has round peas? Are R and Y independent events? How

many visibly different kinds of pea plants would Mendel observe in the second generation? What are the probabilities of each of these kinds?

1.6.11 For independent events A and B , prove that

- (a) A and B^c are independent.
- (b) A^c and B are independent.
- (c) A^c and B^c are independent.

1.6.12 Use a Venn diagram in which the event areas are proportional to their probabilities to illustrate three events A , B , and C that are independent.

1.6.13 Use a Venn diagram in which event areas are in proportion to their probabilities to illustrate events A , B , and C that are pairwise independent but not independent.

1.7.1 Following Quiz 1.3, use MATLAB, but not the `randi` function, to generate a vector $\mathbf{T}$ of 200 independent test scores such that all scores between 51 and 100 are equally likely.

scalar X , or a vector or matrix $\mathbf{X}$. In addition, we use $X_{i,j}$ to denote the i, j th element. Thus, $\mathbf{X}$ and $\mathbf{x}$ (in a MATLAB code fragment) may both refer to the same variable.

Quiz 2.5

The flip of a thick coin yields heads with probability 0.4, tails with probability 0.5, or lands on its edge with probability 0.1. Simulate 100 thick coin flips. Your output should be a 3×1 vector $\mathbf{X}$ such that X_1 , X_2 , and X_3 are the number of occurrences of heads, tails, and edge.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

2.1.1 ● Suppose you flip a coin twice. On any flip, the coin comes up heads with probability $1/4$. Use H_i and T_i to denote the result of flip i .

- What is the probability, $P[H_1|H_2]$, that the first flip is heads given that the second flip is heads?
- What is the probability that the first flip is heads and the second flip is tails?

2.1.2 ● For Example 2.2, suppose $P[G_1] = 1/2$, $P[G_2|G_1] = 3/4$, and $P[G_2|R_1] = 1/4$. Find $P[G_2]$, $P[G_2|G_1]$, and $P[G_1|G_2]$.

2.1.3 ● At the end of regulation time, a basketball team is trailing by one point and a player goes to the line for two free throws. If the player makes exactly one free throw, the game goes into overtime. The probability that the first free throw is good is $1/2$. However, if the first attempt is good, the player relaxes and the second attempt is good with probability $3/4$. However, if the player misses the first attempt, the added pressure reduces the success probability to $1/4$. What is the probability that the game goes into overtime?

2.1.4 ● You have two biased coins. Coin A comes up heads with probability $1/4$. Coin B comes up heads with probability $3/4$. However, you are not sure which is which, so you choose a coin randomly and you flip it. If the flip is heads, you guess that the flipped coin is B ; otherwise, you guess that the flipped coin is A . What is the probability $P[C]$ that your guess is correct?

2.1.5 ■ Suppose that for the general population, 1 in 5000 people carries the human immunodeficiency virus (HIV). A test for the presence of HIV yields either a positive (+) or negative (−) response. Suppose the test gives the correct answer 99% of the time. What is $P[-|H]$, the conditional probability that a person tests negative given that the person does have the HIV virus? What is $P[H|+]$, the conditional probability that a randomly chosen person has the HIV virus given that the person tests positive?

2.1.6 ■ A machine produces photo detectors in pairs. Tests show that the first photo detector is acceptable with probability $3/5$. When the first photo detector is acceptable, the second photo detector is acceptable with probability $4/5$. If the first photo detector is defective, the second photo detector is acceptable with probability $2/5$.

- Find the probability that exactly one photo detector of a pair is acceptable.
- Find the probability that both photo detectors in a pair are defective.

2.1.7 ■ You have two biased coins. Coin A comes up heads with probability $1/4$. Coin B comes up heads with probability $3/4$. However, you are not sure which is which so you flip each coin once, choosing the first coin randomly. Use H_i and T_i to denote the result of flip i . Let A_1 be the event that coin A was flipped first. Let B_1 be the event that coin B was flipped first. What is $P[H_1H_2]$?

Are H_1 and H_2 independent? Explain your answer.

2.1.8 ■ A particular birth defect of the heart is rare; a newborn infant will have the defect D with probability $P[D] = 10^{-4}$. In the general exam of a newborn, a particular heart arrhythmia A occurs with probability 0.99 in infants with the defect. However, the arrhythmia also appears with probability 0.1 in infants without the defect. When the arrhythmia is present, a lab test for the defect is performed. The result of the lab test is either positive (event T^+) or negative (event T^-). In a newborn with the defect, the lab test is positive with probability $p = 0.999$ independent from test to test. In a newborn without the defect, the lab test is negative with probability $p = 0.999$. If the arrhythmia is present and the test is positive, then heart surgery (event H) is performed.

- Given the arrhythmia A is present, what is the probability the infant has the defect D ?
- Given that an infant has the defect, what is the probability $P[H|D]$ that heart surgery is performed?
- Given that the infant does not have the defect, what is the probability $q = P[H|D^c]$ that an unnecessary heart surgery is performed?
- Find the probability $P[H]$ that an infant has heart surgery performed for the arrhythmia.
- Given that heart surgery is performed, what is the probability that the newborn does *not* have the defect?

2.1.9 ■ Suppose Dagwood (Blondie's husband) wants to eat a sandwich but needs to go on a diet. Dagwood decides to let the flip of a coin determine whether he eats. Using an unbiased coin, Dagwood will postpone the diet (and go directly to the refrigerator) if either (a) he flips heads on his first flip or (b) he flips tails on the first flip but then proceeds to get two heads out of the next three flips. Note that the first flip is *not*

counted in the attempt to win two of three and that Dagwood never performs any unnecessary flips. Let H_i be the event that Dagwood flips heads on try i . Let T_i be the event that tails occurs on flip i .

- Draw the tree for this experiment. Label the probabilities of all outcomes.
- What are $P[H_3]$ and $P[T_3]$?
- Let D be the event that Dagwood must diet. What is $P[D]$? What is $P[H_1|D]$?
- Are H_3 and H_2 independent events?

2.1.10 ■ The quality of each pair of photo detectors produced by the machine in Problem 2.1.6 is independent of the quality of every other pair of detectors.

- What is the probability of finding no good detectors in a collection of n pairs produced by the machine?
- How many pairs of detectors must the machine produce to reach a probability of 0.99 that there will be at least one acceptable photo detector?

2.1.11 ■ In Steven Strogatz's New York Times blog <http://opinionator.blogs.nytimes.com/2010/04/25/chances-are/?ref=opinion>, the following problem was posed to highlight the confusing character of conditional probabilities.

Before going on vacation for a week, you ask your spacey friend to water your ailing plant. Without water, the plant has a 90 percent chance of dying. Even with proper watering, it has a 20 percent chance of dying. And the probability that your friend will forget to water it is 30 percent. (a) What's the chance that your plant will survive the week? (b) If it's dead when you return, what's the chance that your friend forgot to water it? (c) If your friend forgot to water it, what's the chance it'll be dead when you return?

Solve parts (a), (b) and (c) of this problem.

2.1.12 ■ Each time a fisherman casts his line, a fish is caught with probability p , independent of whether a fish is caught on any other cast of the line. The fisherman will fish all day until a fish is caught and

then he will quit and go home. Let C_i denote the event that on cast i the fisherman catches a fish. Draw the tree for this experiment and find $P[C_1]$, $P[C_2]$, and $P[C_n]$ as functions of p .

2.2.1 ● On each turn of the knob, a gumball machine is equally likely to dispense a red, yellow, green or blue gumball, independent from turn to turn. After eight turns, what is the probability $P[R_2Y_2G_2B_2]$ that you have received 2 red, 2 yellow, 2 green and 2 blue gumballs?

2.2.2 ● A Starburst candy package contains 12 individual candy pieces. Each piece is equally likely to be berry, orange, lemon, or cherry, independent of all other pieces.

- What is the probability that a Starburst package has only berry or cherry pieces and zero orange or lemon pieces?
- What is the probability that a Starburst package has no cherry pieces?
- What is the probability $P[F_1]$ that all twelve pieces of your Starburst are the same flavor?

2.2.3 ● Your Starburst candy has 12 pieces, three pieces of each of four flavors: berry, lemon, orange, and cherry, arranged in a random order in the pack. You draw the first three pieces from the pack.

- What is the probability they are all the same flavor?
- What is the probability they are all different flavors?

2.2.4 ■ Your Starburst candy has 12 pieces, three pieces of each of four flavors: berry, lemon, orange, and cherry, arranged in a random order in the pack. You draw the first four pieces from the pack.

- What is the probability $P[F_1]$ they are all the same flavor?
- What is the probability $P[F_4]$ they are all different flavors?
- What is the probability $P[F_2]$ that your Starburst has exactly two pieces of each of two different flavors?

2.2.5 ● In a game of rummy, you are dealt a seven-card hand.

- What is the probability $P[R_7]$ that your hand has only red cards?
- What is the probability $P[F]$ that your hand has only face cards?
- What is the probability $P[R_7F]$ that your hand has only red face cards? (The face cards are jack, queen, and king.)

2.2.6 ■ In a game of poker, you are dealt a five-card hand.

- What is the probability $P[R_5]$ that your hand has only red cards?
- What is the probability of a “full house” with three-of-a-kind and two-of-a-kind?

2.2.7 ● Consider a binary code with 5 bits (0 or 1) in each code word. An example of a code word is 01010. How many different code words are there? How many code words have exactly three 0's?

2.2.8 ● Consider a language containing four letters: A, B, C, D . How many three-letter words can you form in this language? How many four-letter words can you form if each letter appears only once in each word?

2.2.9 ● On an American League baseball team with 15 field players and 10 pitchers, the manager selects a starting lineup with 8 field players, 1 pitcher, and 1 designated hitter. The lineup specifies the players for these positions and the positions in a batting order for the 8 field players and designated hitter. If the designated hitter must be chosen among all the field players, how many possible starting lineups are there?

2.2.10 ■ Suppose that in Problem 2.2.9, the designated hitter can be chosen from among all the players. How many possible starting lineups are there?

2.2.11 ● At a casino, the only game is numberless roulette. On a spin of the wheel, the ball lands in a space with color red (r), green (g), or black (b). The wheel has 19 red spaces, 19 green spaces and 2 black spaces.

- (a) In 40 spins of the wheel, find the probability of the event

$$A = \{19 \text{ reds, } 19 \text{ greens, and } 2 \text{ blacks}\}.$$

- (b) In 40 spins of the wheel, find the probability of $G_{19} = \{19 \text{ greens}\}$.
- (c) The only bets allowed are red and green. Given that you randomly choose to bet red or green, what is the probability p that your bet is a winner?

2.2.12 ■ A basketball team has three pure centers, four pure forwards, four pure guards, and one swingman who can play either guard or forward. A pure position player can play only the designated position. If the coach must start a lineup with one center, two forwards, and two guards, how many possible lineups can the coach choose?

2.2.13 ♦ An instant lottery ticket consists of a collection of boxes covered with gray wax. For a subset of the boxes, the gray wax hides a special mark. If a player scratches off the correct number of the marked boxes (and no boxes without the mark), then that ticket is a winner. Design an instant lottery game in which a player scratches five boxes and the probability that a ticket is a winner is approximately 0.01.

2.3.1 ● Consider a binary code with 5 bits (0 or 1) in each code word. An example of a code word is 01010. In each code word, a bit is a zero with probability 0.8, independent of any other bit.

- (a) What is the probability of the code word 00111?
- (b) What is the probability that a code word contains exactly three ones?

2.3.2 ● The Boston Celtics have won 16 NBA championships over approximately 50 years. Thus it may seem reasonable to assume that in a given year the Celtics win the title with probability $p = 16/50 = 0.32$, independent of any other year. Given such a model, what would be the probability

of the Celtics winning eight straight championships beginning in 1959? Also, what would be the probability of the Celtics winning the title in 10 out of 11 years, starting in 1959? Given your answers, do you trust this simple probability model?

2.3.3 ● Suppose each day that you drive to work a traffic light that you encounter is either green with probability $7/16$, red with probability $7/16$, or yellow with probability $1/8$, independent of the status of the light on any other day. If over the course of five days, G , Y , and R denote the number of times the light is found to be green, yellow, or red, respectively, what is the probability that $P[G = 2, Y = 1, R = 2]$? Also, what is the probability $P[G = R]$?

2.3.4 ■ In a game between two equal teams, the home team wins with probability $p > 1/2$. In a best of three playoff series, a team with the home advantage has a game at home, followed by a game away, followed by a home game if necessary. The series is over as soon as one team wins two games. What is $P[H]$, the probability that the team with the home advantage wins the series? Is the home advantage increased by playing a three-game series rather than a one-game playoff? That is, is it true that $P[H] \geq p$ for all $p \geq 1/2$?

2.3.5 ♦ A collection of field goal kickers are divided into groups 1 and 2. Group i has $3i$ kickers. On any kick, a kicker from group i will kick a field goal with probability $1/(i+1)$, independent of the outcome of any other kicks.

- (a) A kicker is selected at random from among all the kickers and attempts one field goal. Let K be the event that a field goal is kicked. Find $P[K]$.
- (b) Two kickers are selected at random; K_j is the event that kicker j kicks a field goal. Are K_1 and K_2 independent?
- (c) A kicker is selected at random and attempts 10 field goals. Let M be the number of misses. Find $P[M = 5]$.

2.4.1 ■ A particular operation has six components. Each component has a failure probability q , independent of any other component. A successful operation requires both of the following conditions:

- Components 1, 2, and 3 all work, or component 4 works.
- Component 5 or component 6 works.

Draw a block diagram for this operation similar to those of Figure 2.2 on page 53. Derive a formula for the probability $P[W]$ that the operation is successful.

2.4.2 ■ We wish to modify the cellular telephone coding system in Example 2.21 in order to reduce the number of errors. In particular, if there are two or three zeroes in the received sequence of 5 bits, we will say that a deletion (event D) occurs. Otherwise, if at least 4 zeroes are received, the receiver decides a zero was sent, or if at least 4 ones are received, the receiver decides a one was sent. We say that an error occurs if i was sent and the receiver decides $j \neq i$ was sent. For this modified protocol, what is the probability $P[E]$ of an error? What is the probability $P[D]$ of a deletion?

2.4.3 ■ Suppose a 10-digit phone number is transmitted by a cellular phone using four binary symbols for each digit, using the model of binary symbol errors and deletions given in Problem 2.4.2. Let C denote the number of bits sent correctly, D the number of deletions, and E the number of errors. Find $P[C = c, D = d, E = e]$ for all c , d , and e .

2.4.4 ♦ Consider the device in Problem 2.4.1. Suppose we can replace any one component with an ultrareliable component that has a failure probability of $q/2 = 0.05$. Which component should we replace?

2.5.1 ■ Build a MATLAB simulation of 50 trials of the experiment of Example 2.3. Your output should be a pair of 50×1 vectors $\mathbf{C}$ and $\mathbf{H}$. For the i th trial, H_i will

record whether it was heads ($H_i = 1$) or tails ($H_i = 0$), and $C_i \in \{1, 2\}$ will record which coin was picked.

2.5.2 ■ Following Quiz 2.3, suppose the communication link has different error probabilities for transmitting 0 and 1. When a 1 is sent, it is received as a 0 with probability 0.01. When a 0 is sent, it is received as a 1 with probability 0.03. Each bit in a packet is still equally likely to be a 0 or 1. Packets have been coded such that if five or fewer bits are received in error, then the packet can be decoded. Simulate the transmission of 100 packets, each containing 100 bits. Count the number of packets decoded correctly.

2.5.3 ■ For a failure probability $q = 0.2$, simulate 100 trials of the six-component test of Problem 2.4.1. How many devices were found to work? Perform 10 repetitions of the 100 trials. What do you learn from 10 repetitions of 100 trials compared to a simulated experiment with 100 trials?

2.5.4 ♦ Write a MATLAB function

`N=countequal(G,T)`

that duplicates the action of `hist(G,T)` in Example 2.26. Hint: Use `ndgrid`.

2.5.5 ♦ In this problem, we use a MATLAB simulation to “solve” Problem 2.4.4. Recall that a particular operation has six components. Each component has a failure probability q independent of any other component. The operation is successful if both

- Components 1, 2, and 3 all work, or component 4 works.
- Component 5 or component 6 works.

With $q = 0.2$, simulate the replacement of a component with an ultrareliable component. For each replacement of a regular component, perform 100 trials. Are 100 trials sufficient to decide which component should be replaced?

of samples $x(1), x(2), \dots, x(n)$ of a random variable X will converge to $E[X]$ as n becomes large. For a discrete uniform $(0, 10)$ random variable X , use MATLAB to examine this convergence.

- (a) For 100 sample values of X , plot the sequence $m_1, m_2, \dots, m_{100}$. Repeat this experiment five times, plotting all five m_n curves on common axes.
- (b) Repeat part (a) for 1000 sample values of X .

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

3.2.1 ● The random variable N has PMF

$$P_N(n) = \begin{cases} c(1/2)^n & n = 0, 1, 2, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[N \leq 1]$?

3.2.2 ● The random variable V has PMF

$$P_V(v) = \begin{cases} cv^2 & v = 1, 2, 3, 4, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the value of the constant c .
- (b) Find $P[V \in \{u^2 | u = 1, 2, 3, \dots\}]$.
- (c) Find the probability that V is even.
- (d) Find $P[V > 2]$.

3.2.3 ● The random variable X has PMF

$$P_X(x) = \begin{cases} c/x & x = 2, 4, 8, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[X = 4]$?
- (c) What is $P[X < 4]$?
- (d) What is $P[3 \leq X \leq 9]$?

3.2.4 ● In each at-bat in a baseball game, mighty Casey swings at every pitch. The

result is either a home run (with probability $q = 0.05$) or a strike. Of course, three strikes and Casey is out.

- (a) What is the probability $P[H]$ that Casey hits a home run?
- (b) For one at-bat, what is the PMF of N , the number of times Casey swings his bat?

3.2.5 ■ A tablet computer transmits a file over a wi-fi link to an access point. Depending on the size of the file, it is transmitted as N packets where N has PMF

$$P_N(n) = \begin{cases} c/n & n = 1, 2, 3, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the constant c .
- (b) What is the probability that N is odd?
- (c) Each packet is received correctly with probability p , and the file is received correctly if all N packets are received correctly. Find $P[C]$, the probability that the file is received correctly.

3.2.6 ■ In college basketball, when a player is fouled while not in the act of shooting and the opposing team is “in the penalty,” the player is awarded a “1 and 1.” In the 1 and 1, the player is awarded one free throw, and if that free throw goes in the player is awarded a second free throw. Find the PMF of Y , the number of points scored in

a 1 and 1 given that any free throw goes in with probability p , independent of any other free throw.

3.2.7 ■ You roll a 6-sided die repeatedly. Starting with roll $i = 1$, let R_i denote the result of roll i . If $R_i > i$, then you will roll again; otherwise you stop. Let N denote the number of rolls.

- (a) What is $P[N > 3]$?
- (b) Find the PMF of N .

3.2.8 ■ You are manager of a ticket agency that sells concert tickets. You assume that people will call three times in an attempt to buy tickets and then give up. You want to make sure that you are able to serve at least 95% of the people who want tickets. Let p be the probability that a caller gets through to your ticket agency. What is the minimum value of p necessary to meet your goal?

3.2.9 ■ In the ticket agency of Problem 3.2.8, each telephone ticket agent is available to receive a call with probability 0.2. If all agents are busy when someone calls, the caller hears a busy signal. What is the minimum number of agents that you have to hire to meet your goal of serving 95% of the customers who want tickets?

3.2.10 ■ Suppose when a baseball player gets a hit, a single is twice as likely as a double, which is twice as likely as a triple, which is twice as likely as a home run. Also, the player's batting average, i.e., the probability the player gets a hit, is 0.300. Let B denote the number of bases touched safely during an at-bat. For example, $B = 0$ when the player makes an out, $B = 1$ on a single, and so on. What is the PMF of B ?

3.2.11 ♦ When someone presses SEND on a cellular phone, the phone attempts to set up a call by transmitting a SETUP message to a nearby base station. The phone waits for a response, and if none arrives within 0.5 seconds it tries again. If it doesn't get a response after $n = 6$ tries, the phone stops transmitting messages and generates a busy signal.

- (a) Draw a tree diagram that describes the call setup procedure.
- (b) If all transmissions are independent and the probability is p that a SETUP message will get through, what is the PMF of K , the number of messages transmitted in a call attempt?
- (c) What is the probability that the phone will generate a busy signal?
- (d) As manager of a cellular phone system, you want the probability of a busy signal to be less than 0.02. If $p = 0.9$, what is the minimum value of n necessary to achieve your goal?

3.3.1 ● In a package of M&Ms, Y , the number of yellow M&Ms, is uniformly distributed between 5 and 15.

- (a) What is the PMF of Y ?
- (b) What is $P[Y < 10]$?
- (c) What is $P[Y > 12]$?
- (d) What is $P[8 \leq Y \leq 12]$?

3.3.2 ● In a bag of 25 M&Ms, each piece is equally likely to be red, green, orange, blue, or brown, independent of the color of any other piece. Find the PMF of R , the number of red pieces. What is the probability a bag has no red M&Ms?

3.3.3 ● When a conventional paging system transmits a message, the probability that the message will be received by the pager it is sent to is p . To be confident that a message is received at least once, a system transmits the message n times.

- (a) Assuming all transmissions are independent, what is the PMF of K , the number of times the pager receives the same message?
- (b) Assume $p = 0.8$. What is the minimum value of n that produces a probability of 0.95 of receiving the message at least once?

3.3.4 ● You roll a pair of fair dice until you roll "doubles" (i.e., both dice are the same). What is the expected number, $E[N]$, of rolls?

3.3.5 ● When you go fishing, you attach m hooks to your line. Each time you cast your line, each hook will be swallowed by a fish with probability h , independent of whether any other hook is swallowed. What is the PMF of K , the number of fish that are hooked on a single cast of the line?

3.3.6 ● Any time a child throws a Frisbee, the child's dog catches the Frisbee with probability p , independent of whether the Frisbee is caught on any previous throw. When the dog catches the Frisbee, it runs away with the Frisbee, never to be seen again. The child continues to throw the Frisbee until the dog catches it. Let X denote the number of times the Frisbee is thrown.

- What is the PMF $P_X(x)$?
- If $p = 0.2$, what is the probability that the child will throw the Frisbee more than four times?

3.3.7 ● When a two-way paging system transmits a message, the probability that the message will be received by the pager it is sent to is p . When the pager receives the message, it transmits an acknowledgment signal (ACK) to the paging system. If the paging system does not receive the ACK, it sends the message again.

- What is the PMF of N , the number of times the system sends the same message?
- The paging company wants to limit the number of times it has to send the same message. It has a goal of $P[N \leq 3] \geq 0.95$. What is the minimum value of p necessary to achieve the goal?

3.3.8 ● The number of bytes B in an HTML file is the geometric $(2.5 \cdot 10^{-5})$ random variable. What is the probability $P[B > 500,000]$ that a file has over 500,000 bytes?

3.3.9 ●

- Starting on day 1, you buy one lottery ticket each day. Each ticket is a winner with probability 0.1. Find the PMF of

K , the number of tickets you buy up to and including your fifth winning ticket.

- L is the number of flips of a fair coin up to and including the 33rd occurrence of tails. What is the PMF of L ?
- Starting on day 1, you buy one lottery ticket each day. Each ticket is a winner with probability 0.01. Let M equal the number of tickets you buy up to and including your first winning ticket. What is the PMF of M ?

3.3.10 ● The number of buses that arrive at a bus stop in T minutes is a Poisson random variable B with expected value $T/5$.

- What is the PMF of B , the number of buses that arrive in T minutes?
- What is the probability that in a two-minute interval, three buses will arrive?
- What is the probability of no buses arriving in a 10-minute interval?
- How much time should you allow so that with probability 0.99 at least one bus arrives?

3.3.11 ● In a wireless automatic meter-reading system, a base station sends out a wake-up signal to nearby electric meters. On hearing the wake-up signal, a meter transmits a message indicating the electric usage. Each message is repeated eight times.

- If a single transmission of a message is successful with probability p , what is the PMF of N , the number of successful message transmissions?
- I is an indicator random variable such that $I = 1$ if at least one message is transmitted successfully; otherwise $I = 0$. Find the PMF of I .

3.3.12 ● A Zipf $(n, \alpha = 1)$ random variable X has PMF

$$P_X(x) = \begin{cases} c(n)/x & x = 1, 2, \dots, n, \\ 0 & \text{otherwise.} \end{cases}$$

The constant $c(n)$ is set so that $\sum_{x=1}^n P_X(x) = 1$. Calculate $c(n)$ for $n = 1, 2, \dots, 6$.

3.3.13 ■ In a bag of 64 “holiday season” M&Ms, each M&M is equally likely to be red or green, independent of any other M&M in the bag.

- If you randomly grab four M&Ms, what is the probability $P[E]$ that you grab an equal number of red and green M&Ms?
- What is the PMF of G , the number of green M&Ms in the bag?
- You begin eating randomly chosen M&Ms one by one. Let R equal the number of red M&Ms you eat before you eat your first green M&M. What is the PMF of R ?

3.3.14 ■ A radio station gives a pair of concert tickets to the sixth caller who knows the birthday of the performer. For each person who calls, the probability is 0.75 of knowing the performer’s birthday. All calls are independent.

- What is the PMF of L , the number of calls necessary to find the winner?
- What is the probability of finding the winner on the tenth call?
- What is the probability that the station will need nine or more calls to find a winner?

3.3.15 ■ In a packet voice communications system, a source transmits packets containing digitized speech to a receiver. Because transmission errors occasionally occur, an acknowledgment (ACK) or a negative acknowledgment (NAK) is transmitted back to the source to indicate the status of each received packet. When the transmitter gets a NAK, the packet is retransmitted. Voice packets are delay sensitive, and a packet can be transmitted a maximum of d times. If a packet transmission is an independent Bernoulli trial with success probability p , what is the PMF of T , the number of times a packet is transmitted?

3.3.16 ♦ At Newark airport, your jet joins a line as the tenth jet waiting for takeoff. At Newark, takeoffs and landings are synchronized to the minute. In each one-minute

interval, an arriving jet lands with probability $p = 2/3$, independent of an arriving jet in any other minute. Such an arriving jet blocks any waiting jet from taking off in that one-minute interval. However, if there is no arrival, then the waiting jet at the head of the line takes off. Each take-off requires exactly one minute.

- Let L_1 denote the number of jets that land before the jet at the front of the line takes off. Find the PMF $P_{L_1}(l)$.
- Let W denote the number of minutes you wait until your jet takes off. Find $P[W = 10]$. (Note that if no jets land for ten minutes, then one waiting jet will take off each minute and $W = 10$.)
- What is the PMF of W ?

3.3.17 ♦ Suppose each day (starting on day 1) you buy one lottery ticket with probability $1/2$; otherwise, you buy no tickets. A ticket is a winner with probability p independent of the outcome of all other tickets. Let N_i be the event that on day i you do *not* buy a ticket. Let W_i be the event that on day i , you buy a winning ticket. Let L_i be the event that on day i you buy a losing ticket.

- What are $P[W_{33}]$, $P[L_{87}]$, and $P[N_{99}]$?
- Let K be the number of the day on which you buy your first lottery ticket. Find the PMF $P_K(k)$.
- Find the PMF of R , the number of losing lottery tickets you have purchased in m days.
- Let D be the number of the day on which you buy your j th losing ticket. What is $P_D(d)$? Hint: If you buy your j th losing ticket on day d , how many losers did you have after $d - 1$ days?

3.3.18 ♦ The Sixers and the Celtics play a best out of five playoff series. The series ends as soon as one of the teams has won three games. Assume that either team is equally likely to win any game independently of any other game played. Find

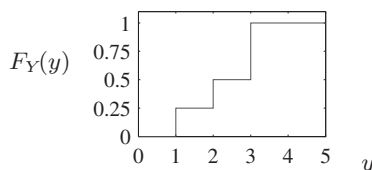
- The PMF $P_N(n)$ for the total number N of games played in the series;

- (b) The PMF $P_W(w)$ for the number W of Celtics wins in the series;
- (c) The PMF $P_L(l)$ for the number L of Celtics losses in the series.

3.3.19 ♦ For a binomial random variable K representing the number of successes in n trials, $\sum_{k=0}^n P_K(k) = 1$. Use this fact to prove the binomial theorem for any $a > 0$ and $b > 0$. That is, show that

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

3.4.1 ● Discrete random variable Y has the CDF $F_Y(y)$ as shown:



Use the CDF to find the following probabilities:

- (a) $P[Y < 1]$ and $P[Y \leq 1]$
- (b) $P[Y > 2]$ and $P[Y \geq 2]$
- (c) $P[Y = 3]$ and $P[Y > 3]$
- (d) $P_Y(y)$

3.4.2 ● The random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ 0.2 & -1 \leq x < 0, \\ 0.7 & 0 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

- (a) Draw a graph of the CDF.
- (b) Write $P_X(x)$, the PMF of X . Be sure to write the value of $P_X(x)$ for all x from $-\infty$ to ∞ .

3.4.3 ● The random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < -3, \\ 0.4 & -3 \leq x < 5, \\ 0.8 & 5 \leq x < 7, \\ 1 & x \geq 7. \end{cases}$$

- (a) Draw a graph of the CDF.
- (b) Write $P_X(x)$, the PMF of X .

3.4.4 ● Following Example 3.22, show that a geometric (p) random variable K has CDF

$$F_K(k) = \begin{cases} 0 & k < 1, \\ 1 - (1 - p)^{\lfloor k \rfloor} & k \geq 1. \end{cases}$$

3.4.5 ■ At the One Top Pizza Shop, a pizza sold has mushrooms with probability $p = 2/3$. On a day in which 100 pizzas are sold, let N equal the number of pizzas sold before the first pizza with mushrooms is sold. What is the PMF of N ? What is the CDF of N ?

3.4.6 ■ In Problem 3.2.10, find and sketch the CDF of B , the number of bases touched safely during an at-bat.

3.4.7 ■ In Problem 3.2.6, find and sketch the CDF of Y , the number of points scored in a 1 and 1 for $p = 1/4$, $p = 1/2$, and $p = 3/4$.

3.4.8 ■ In Problem 3.2.11, find and sketch the CDF of N , the number of attempts made by the cellular phone for $p = 1/2$.

3.5.1 ● Let X have the uniform PMF

$$P_X(x) = \begin{cases} 0.01 & x = 1, 2, \dots, 100, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find a mode x_{mod} of X . If the mode is not unique, find the set X_{mod} of all modes of X .
- (b) Find a median x_{med} of X . If the median is not unique, find the set X_{med} of all numbers x that are medians of X .

3.5.2 ● It costs 20 cents to receive a photo and 30 cents to send a photo from a cell-phone. C is the cost of one photo (either sent or received). The probability of receiving a photo is 0.6. The probability sending a photo is 0.4.

- (a) Find $P_C(c)$, the PMF of C .

(b) What is $E[C]$, the expected value of C ?

3.5.3 ●

- (a) The number of trains J that arrive at the station in time t minutes is a Poisson random variable with $E[J] = t$. Find t such that $P[J > 0] = 0.9$.
- (b) The number of buses K that arrive at the station in one hour is a Poisson random variable with $E[K] = 10$. Find $P[K = 10]$.
- (c) In a 1 ms interval, the number of hits L on a Web server is a Poisson random variable with expected value $E[L] = 2$ hits. What is $P[L \leq 1]$?

3.5.4 ● You simultaneously flip a pair of fair coins. Your friend gives you one dollar if both coins come up heads. You repeat this ten times and your friend gives you X dollars. Find $E[X]$, the expected number of dollars you receive. What is the probability that you do “worse than average”?

3.5.5 ● A packet received by your smartphone is error-free with probability 0.95, independent of any other packet.

- (a) Out of 10 packets received, let X equal the number of packets received with errors. What is the PMF of X ?
- (b) In one hour, your smartphone receives 12,000 packets. Let X equal the number of packets received with errors. What is $E[X]$?

3.5.6 ● Find the expected value of the random variable Y in Problem 3.4.1.

3.5.7 ● Find the expected value of the random variable X in Problem 3.4.2.

3.5.8 ● Find the expected value of the random variable X in Problem 3.4.3.

3.5.9 ● Use Definition 3.13 to calculate the expected value of a binomial $(4, 1/2)$ random variable X .

3.5.10 ● X is the discrete uniform $(1, 5)$ random variable.

- (a) What is $P[X = E[X]]$?

(b) What is $P[X > E[X]]$?

3.5.11 ● K is the geometric $(1/11)$ random variable.

- (a) What is $P[K = E[K]]$?
- (b) What is $P[K > E[K]]$?
- (c) What is $P[K < E[K]]$?

3.5.12 ● At a casino, people line up to pay \$20 each to be a contestant in the following game: The contestant flips a fair coin repeatedly. If she flips heads 20 times in a row, she walks away with $R = 20$ million dollars; otherwise she walks away with $R = 0$ dollars.

- (a) Find the PMF of R , the reward earned by the contestant.
- (b) The casino counts “losing contestants” who fail to win the 20 million dollar prize. Let L equal the number of losing contestants before the first winning contestant. What is the PMF of L ?
- (c) Why does the casino offer this game?

3.5.13 ● Give examples of practical applications of probability theory that can be modeled by the following PMFs. In each case, state an experiment, the sample space, the range of the random variable, the PMF of the random variable, and the expected value:

- (a) Bernoulli
- (b) Binomial
- (c) Pascal
- (d) Poisson

Make up your own examples. (Don’t copy examples from the text.)

3.5.14 ■ Find $P[K < E[K]]$ when

- (a) K is geometric $(1/3)$.
- (b) K is binomial $(6, 1/2)$.
- (c) K is Poisson (3) .
- (d) K is discrete uniform $(0, 6)$.

3.5.15 ■ Suppose you go to a casino with exactly \$63. At this casino, the only game is roulette and the only bets allowed are red and green. The payoff for a winning bet

is the amount of the bet. In addition, the wheel is fair so that $P[\text{red}] = P[\text{green}] = 1/2$. You have the following strategy: First, you bet \$1. If you win the bet, you quit and leave the casino with \$64. If you lose, you then bet \$2. If you win, you quit and go home. If you lose, you bet \$4. In fact, whenever you lose, you double your bet until either you win a bet or you lose all of your money. However, as soon as you win a bet, you quit and go home. Let Y equal the amount of money that you take home. Find $P_Y(y)$ and $E[Y]$. Would you like to play this game every day?

3.5.16 ♦ In a TV game show, there are three identical-looking suitcases. The first suitcase has 3 dollars, the second has 30 dollars and the third has 300 dollars. You start the game by randomly choosing a suitcase. *Between the two unchosen suitcases*, the game show host opens the suitcase with more money. The host then asks you if you want to keep your suitcase or switch to the other remaining suitcase. After you make your decision, you open your suitcase and keep the D dollars inside. Should you switch suitcases? To answer this question, solve the following subproblems and use the following notation:

- C_i is the event that you first choose the suitcase with i dollars.
- O_i denotes the event that the host opens a suitcase with i dollars.

In addition, you may wish to go back and review the Monty Hall problem in Example 2.4.

- (a) Suppose you never switch; you always stick with your original choice. Use a tree diagram to find the PMF $P_D(d)$ and expected value $E[D]$.
- (b) Suppose you always switch. Use a tree diagram to find the PMF $P_D(d)$ and expected value $E[D]$.
- (c) Perhaps your rule for switching should depend on how many dollars are in the suitcase that the host opens? What is the optimal strategy to maximize

$E[D]$? Hint: Consider making a random decision; if the host opens a suitcase with i dollars, let α_i denote the probability that you switch.

3.5.17 ♦ You are a contestant on a TV game show; there are four identical-looking suitcases containing \$100, \$200, \$400, and \$800. You start the game by randomly choosing a suitcase. *Among the three unchosen suitcases*, the game show host opens the suitcase that holds the median amount of money. (For example, if the unopened suitcases contain \$100, \$400 and \$800, the host opens the \$400 suitcase.) The host then asks you if you want to keep your suitcase or switch one of the other remaining suitcases. For your analysis, use the following notation for events:

- C_i is the event that you choose a suitcase with i dollars.
- O_i denotes the event that the host opens a suitcase with i dollars.
- R is the reward in dollars that you keep.

- (a) You refuse the host's offer and open the suitcase you first chose. Find the PMF of R and the expected value $E[R]$.
- (b) You always switch and randomly choose one of the two remaining suitcases with equal probability. You receive the R dollars in this chosen suitcase. Sketch a tree diagram for this experiment, and find the PMF and expected value of R .
- (c) Can you do better than either always switching or always staying with your original choice? Explain.

3.5.18 ♦ You are a contestant on a TV game show; there are four identical-looking suitcases containing \$200, \$400, \$800, and \$1600. You start the game by randomly choosing a suitcase. *Among the three unchosen suitcases*, the game show host opens the suitcase that holds the least money. The host then asks you if you want to keep

your suitcase or switch one of the other remaining suitcases. For the following analysis, use the following notation for events:

- C_i is the event that you choose a suitcase with i dollars.
 - O_i denotes the event that the host opens a suitcase with i dollars.
 - R is the reward in dollars that you keep.
- (a) You refuse the host's offer and open the suitcase you first chose. Find the PMF of R and the expected value $E[R]$.
- (b) You switch and randomly choose one of the two remaining suitcases. You receive the R dollars in this chosen suitcase. Sketch a tree diagram for this experiment, and find the PMF and expected value of R .

3.5.19 ♦ Let binomial random variable X_n denote the number of successes in n Bernoulli trials with success probability p . Prove that $E[X_n] = np$. Hint: Use the fact that $\sum_{x=0}^{n-1} P_{X_{n-1}}(x) = 1$.

3.5.20 ♦ Prove that if X is a nonnegative integer-valued random variable, then

$$E[X] = \sum_{k=0}^{\infty} P[X > k].$$

3.5.21 ♦ At the gym, a weightlifter can bench press a maximum of 100 kg. For a mass of m kg, ($1 \leq m \leq 100$), the maximum number of repetitions she can complete is R , a geometric random variable with expected value $100/m$.

- (a) In terms of the mass m , what is the PMF of R ?
- (b) When she performs one repetition, she lifts the m kg mass a height $h = 5/9.8$ meters and thus does work $w = mgh = 5m$ Joules. For R repetitions, she does $W = 5mR$ Joules of work. What is the expected work $E[W]$ that she will complete?
- (c) A friend offers to pay her 1000 dollars if she can perform 1000 Joules of

weightlifting work. What mass m in the range $1 \leq m \leq 100$ should she use to maximize her probability of winning the money? For the best choice of m , what is the probability that she wins the money?

3.5.22 ♦ At the gym, a weightlifter can bench press a maximum of 200 kg. For a mass of m kg, ($1 \leq m \leq 200$), the maximum number of repetitions she can complete is R , a geometric random variable with expected value $200/m$.

- (a) In terms of the mass m , what is the PMF of R ?
- (b) When she performs one repetition, she lifts the m kg mass a height $h = 4/9.8$ meters and thus does work $w = mgh = 4m$ Joules. For R repetitions, she does $W = 4mR$ Joules of work. What is the expected work $E[W]$ that she will complete?
- (c) A friend offers to pay her 1000 dollars if she can perform 1000 Joules of weightlifting work. What mass m in the range $1 \leq m \leq 200$ should she use to maximize her probability of winning the money?

3.6.1 ● Given the random variable Y in Problem 3.4.1, let $U = g(Y) = Y^2$.

- (a) Find $P_U(u)$.
- (b) Find $F_U(u)$.
- (c) Find $E[U]$.

3.6.2 ● Given the random variable X in Problem 3.4.2, let $V = g(X) = |X|$.

- (a) Find $P_V(v)$.
- (b) Find $F_V(v)$.
- (c) Find $E[V]$.

3.6.3 ● Given the random variable X in Problem 3.4.3, let $W = g(X) = -X$.

- (a) Find $P_W(w)$.
- (b) Find $F_W(w)$.
- (c) Find $E[W]$.

3.6.4 ● At a discount brokerage, a stock purchase or sale worth less than \$10,000 incurs a brokerage fee of 1% of the value of the transaction. A transaction worth more than \$10,000 incurs a fee of \$100 plus 0.5% of the amount exceeding \$10,000. Note that for a fraction of a cent, the brokerage always charges the customer a full penny. You wish to buy 100 shares of a stock whose price D in dollars has PMF

$$P_D(d) = \begin{cases} 1/3 & d = 99.75, 100, 100.25, \\ 0 & \text{otherwise.} \end{cases}$$

What is the PMF of C , the cost of buying the stock (including the brokerage fee)?

3.6.5 ■ A source transmits data packets to a receiver over a radio link. The receiver uses error detection to identify packets that have been corrupted by radio noise. When a packet is received error free, the receiver sends an acknowledgment (ACK) back to the source. When the receiver gets a packet with errors, a negative acknowledgment (NAK) message is sent back to the source. Each time the source receives a NAK, the packet is retransmitted. We assume that each packet transmission is independently corrupted by errors with probability q .

- Find the PMF of X , the number of times that a packet is transmitted by the source.
- Suppose each packet takes 1 millisecond to transmit and that the source waits an additional millisecond to receive the acknowledgment message (ACK or NAK) before retransmitting. Let T equal the time required until the packet is successfully received. What is the relationship between T and X ? What is the PMF of T ?

3.6.6 ■ Suppose that a cellular phone costs \$20 per month with 30 minutes of use included and that each additional minute of use costs \$0.50. If the number of minutes you use in a month is a geometric random variable M with expected value of

$E[M] = 1/p = 30$ minutes, what is the PMF of C , the cost of the phone for one month?

3.6.7 ■ A professor tries to count the number of students attending lecture. For each student in the audience, the professor either counts the student properly (with probability p) or overlooks (and does not count) the student with probability $1 - p$. The exact number of attending students is 70.

- The number of students counted by the professor is a random variable N . What is the PMF of N ?
- Let $U = 70 - N$ denote the number of uncounted students. What is the PMF of U ?
- What is the probability that the undercount U is 2 or more?
- For what value of p does $E[U] = 2$?

3.6.8 ■ A forgetful professor tries to count the M&Ms in a package; however, the professor often loses his place and double counts an M&M. For each M&M in the package, the professor counts the M&M and then, with probability p counts the M&M again. The exact number of M&Ms in the pack is 20.

- Find the PMF of R , the number of double-counted M&Ms.
- Find the PMF of N , the number of M&Ms counted by the professor.

3.7.1 ● Starting on day $n = 1$, you buy one lottery ticket each day. Each ticket costs 1 dollar and is independently a winner that can be cashed for 5 dollars with probability 0.1; otherwise the ticket is worthless. Let X_n equal your net profit after n days. What is $E[X_n]$?

3.7.2 ● For random variable T in Quiz 3.6, first find the expected value $E[T]$ using Theorem 3.10. Next, find $E[T]$ using Definition 3.13.

3.7.3 ● In a certain lottery game, the chance of getting a winning ticket is exactly one in a thousand. Suppose a person buys one ticket each day (except on the leap year day February 29) over a period of fifty years.

What is the expected number $E[T]$ of winning tickets in fifty years? If each winning ticket is worth \$1000, what is the expected amount $E[R]$ collected on these winning tickets? Lastly, if each ticket costs \$2, what is your expected net profit $E[Q]$?

3.7.4 ● Suppose an NBA basketball player shooting an uncontested 2-point shot will make the basket with probability 0.6. However, if you foul the shooter, the shot will be missed, but two free throws will be awarded. Each free throw is an independent Bernoulli trial with success probability p . Based on the expected number of points the shooter will score, for what values of p may it be desirable to foul the shooter?

3.7.5 ● It can take up to four days after you call for service to get your computer repaired. The computer company charges for repairs according to how long you have to wait. The number of days D until the service technician arrives and the service charge C , in dollars, are described by

d	1	2	3	4
$P_D(d)$	0.2	0.4	0.3	0.1

and

$$C = \begin{cases} 90 & \text{for 1-day service,} \\ 70 & \text{for 2-day service,} \\ 40 & \text{for 3-day service,} \\ 40 & \text{for 4-day service.} \end{cases}$$

- What is the expected waiting time $\mu_D = E[D]$?
- What is the expected deviation $E[D - \mu_D]$?
- Express C as a function of D .
- What is the expected value $E[C]$?

3.7.6 ■ True or False: For any random variable X , $E[1/X] = 1/E[X]$.

3.7.7 ■ For the cellular phone in Problem 3.6.6, express the monthly cost C as a function of M , the number of minutes used. What is the expected monthly cost $E[C]$?

3.7.8 ■ A new cellular phone billing plan costs \$15 per month plus \$1 for each minute of use. If the number of minutes you use the phone in a month is a geometric random variable with expected value $1/p$, what is the expected monthly cost $E[C]$ of the phone? For what values of p is this billing plan preferable to the billing plan of Problem 3.6.6 and Problem 3.7.7?

3.7.9 ♦ A particular circuit works if all 10 of its component devices work. Each circuit is tested before leaving the factory. Each working circuit can be sold for k dollars, but each nonworking circuit is worthless and must be thrown away. Each circuit can be built with either ordinary devices or ultrareliable devices. An ordinary device has a failure probability of $q = 0.1$ and costs \$1. An ultrareliable device has a failure probability of $q/2$ but costs \$3. Assuming device failures are independent, should you build your circuit with ordinary devices or ultrareliable devices in order to maximize your expected profit $E[R]$? Keep in mind that your answer will depend on k .

3.7.10 ♦♦ In the New Jersey state lottery, each \$1 ticket has six randomly marked numbers out of $1, \dots, 46$. A ticket is a winner if the six marked numbers match six numbers drawn at random at the end of a week. For each ticket sold, 50 cents is added to the pot for the winners. If there are k winning tickets, the pot is divided equally among the k winners. Suppose you bought a winning ticket in a week in which $2n$ tickets are sold and the pot is n dollars.

- What is the probability q that a random ticket will be a winner?
- Find the PMF of K_n , the number of other (besides your own) winning tickets.
- What is the expected value of W_n , the prize for your winning ticket?

3.7.11 ♦♦ If there is no winner for the lottery described in Problem 3.7.10, then the pot is carried over to the next week. Suppose that in a given week, an r dollar pot is carried over from the previous week and

2n tickets sold. Answer the following questions.

- What is the probability q that a random ticket will be a winner?
- If you own one of the $2n$ tickets sold, what is the expected value of V , the value (i.e., the amount you win) of that ticket? Is it ever possible that $E[V] > 1$?
- Suppose that in the instant before the ticket sales are stopped, you are given the opportunity to buy one of each possible ticket. For what values (if any) of n and r should you do it?

3.8.1 In an experiment to monitor two packets, the PMF of N , the number of video packets, is

n	0	1	2
$P_N(n)$	0.2	0.7	0.1

Find $E[N]$, $E[N^2]$, $\text{Var}[N]$, and σ_N .

3.8.2 Find the variance of the random variable Y in Problem 3.4.1.

3.8.3 Find the variance of the random variable X in Problem 3.4.2.

3.8.4 Find the variance of the random variable X in Problem 3.4.3.

3.8.5 Let X have the binomial PMF

$$P_X(x) = \binom{4}{x} (1/2)^4.$$

- Find the standard deviation of X .
- What is $P[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X]$, the probability that X is within one standard deviation of the expected value?

3.8.6 X is the binomial $(5, 0.5)$ random variable.

- Find the standard deviation of X .
- Find $P[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X]$, the probability that X is within one standard deviation of the expected value.

3.8.7 Show that the variance of $Y = aX + b$ is $\text{Var}[Y] = a^2 \text{Var}[X]$.

3.8.8 Given a random variable X with expected value μ_X and variance σ_X^2 , find the expected value and variance of

$$Y = \frac{X - \mu_X}{\sigma_X}.$$

3.8.9 In real-time packet data transmission, the time between successfully received packets is called the *interarrival time*, and randomness in packet interarrival times is called *jitter*. Jitter is undesirable. One measure of jitter is the standard deviation of the packet interarrival time. From Problem 3.6.5, calculate the jitter σ_T . How large must the successful transmission probability q be to ensure that the jitter is less than 2 milliseconds?

3.8.10 Random variable K has a Poisson (α) distribution. Derive the properties $E[K] = \text{Var}[K] = \alpha$. Hint: $E[K^2] = E[K(K-1)] + E[K]$.

3.8.11 For the delay D in Problem 3.7.5, what is the standard deviation σ_D of the waiting time?

3.9.1 Let X be the binomial $(100, 1/2)$ random variable. Let E_2 denote the event that X is a perfect square. Calculate $P[E_2]$.

3.9.2 Write a MATLAB function `x=shipweight8(m)` that produces m random sample values of the package weight X with PMF given in Example 3.27.

3.9.3 Use the `unique` function to write a MATLAB script `shipcostpmf.m` that outputs the pair of vectors `sy` and `py` representing the PMF $P_Y(y)$ of the shipping cost Y in Example 3.27.

3.9.4 For $m = 10$, $m = 100$, and $m = 1000$, use MATLAB to find the average cost of sending m packages using the model of Example 3.27. Your program input should have the number of trials m as the input. The output should be $\bar{Y} = \frac{1}{m} \sum_{i=1}^m Y_i$, where Y_i is the cost of the i th package. As m becomes large, what do you observe?

3.9.5 ■ The Zipf $(n, \alpha = 1)$ random variable X introduced in Problem 3.3.12 is often used to model the “popularity” of a collection of n objects. For example, a Web server can deliver one of n Web pages. The pages are numbered such that the page 1 is the most requested page, page 2 is the second most requested page, and so on. If page k is requested, then $X = k$.

To reduce external network traffic, an ISP gateway caches copies of the k most popular pages. Calculate, as a function of n for $1 \leq n \leq 1000$, how large k must be to ensure that the cache can deliver a page with probability 0.75.

3.9.6 ■ Generate n independent samples of the Poisson (5) random variable Y . For each $y \in S_Y$, let $n(y)$ denote the number of times that y was observed. Thus $\sum_{y \in S_Y} n(y) = n$ and the relative frequency of y is $R(y) = n(y)/n$. Compare the relative frequency of y against $P_Y(y)$ by plotting $R(y)$ and $P_Y(y)$ on the same graph as functions of y for $n = 100$, $n = 1000$ and

$n = 10,000$. How large should n be to have reasonable agreement?

3.9.7 ■ Test the convergence of Theorem 3.8. For $\alpha = 10$, plot the PMF of K_n for $(n, p) = (10, 1)$, $(n, p) = (100, 0.1)$, and $(n, p) = (1000, 0.01)$ and compare each result with the Poisson (α) PMF.

3.9.8 ■ Use the result of Problem 3.4.4 and the Random Sample Algorithm on Page 102 to write a MATLAB function `k=geometricrv(p,m)` that generates m samples of a geometric (p) random variable.

3.9.9 ♦ Find n^* , the smallest value of n for which the function `poissonpmf(n,n)` shown in Example 3.37 reports an error. What is the source of the error? Write a function `bigpoissonpmf(alpha,n)` that calculates `poissonpmf(n,n)` for values of n much larger than n^* . Hint: For a Poisson (α) random variable K ,

$$P_K(k) = \exp \left(-\alpha + k \ln(\alpha) - \sum_{j=1}^k \ln(j) \right).$$

Quiz 4.8

Write a MATLAB function `t=t2rv(m)` that generates m samples of a random variable with the PDF $f_{T|T>2}(t)$ as given in Example 7.10.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

4.2.1 ● The cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0 & x < -1, \\ (x+1)/2 & -1 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

- (a) What is $P[X > 1/2]$?
- (b) What is $P[-1/2 < X \leq 3/4]$?
- (c) What is $P[|X| \leq 1/2]$?
- (d) What is the value of a such that $P[X \leq a] = 0.8$?

4.2.2 ● The CDF of the continuous random variable V is

$$F_V(v) = \begin{cases} 0 & v < -5, \\ c(v+5)^2 & -5 \leq v < 7, \\ 1 & v \geq 7. \end{cases}$$

- (a) What is c ?
- (b) What is $P[V > 4]$?
- (c) What is $P[-3 < V \leq 0]$?
- (d) What is the value of a such that $P[V > a] = 2/3$?

4.2.3 ● In this problem, we verify that $\lim_{n \rightarrow \infty} \lceil nx \rceil / n = x$.

- (a) Verify that $nx \leq \lceil nx \rceil \leq nx + 1$.
- (b) Use part (a) to show

$$\lim_{n \rightarrow \infty} \lceil nx \rceil / n = x.$$

- (c) Use a similar argument to show that $\lim_{n \rightarrow \infty} \lfloor nx \rfloor / n = x$.

4.2.4 ● The CDF of random variable W is

$$F_W(w) = \begin{cases} 0 & w < -5, \\ \frac{w+5}{8} & -5 \leq w < -3, \\ \frac{1}{4} & -3 \leq w < 3, \\ \frac{1}{4} + \frac{3(w-3)}{8} & 3 \leq w < 5, \\ 1 & w \geq 5. \end{cases}$$

- (a) What is $P[W \leq 4]$?
- (b) What is $P[-2 < W \leq 2]$?
- (c) What is $P[W > 0]$?
- (d) What is the value of a such that $P[W \leq a] = 1/2$?

4.3.1 ● The random variable X has probability density function

$$f_X(x) = \begin{cases} cx & 0 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

Use the PDF to find

- (a) the constant c ,
- (b) $P[0 \leq X \leq 1]$,
- (c) $P[-1/2 \leq X \leq 1/2]$,
- (d) the CDF $F_X(x)$.

4.3.2 ● The cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0 & x < -1, \\ (x+1)/2 & -1 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

Find the PDF $f_X(x)$ of X .

4.3.3 ● Find the PDF $f_U(u)$ of the random variable U in Problem 4.2.4.

4.3.4 ■ For a constant parameter $a > 0$, a Rayleigh random variable X has PDF

$$f_X(x) = \begin{cases} a^2 x e^{-a^2 x^2/2} & x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

What is the CDF of X ?

4.3.5 ♦ Random variable X has a PDF of the form $f_X(x) = \frac{1}{2}f_1(x) + \frac{1}{2}f_2(x)$, where

$$f_1(x) = \begin{cases} c_1 & 0 \leq x \leq 2, \\ 0 & \text{otherwise,} \end{cases}$$

$$f_2(x) = \begin{cases} c_2 e^{-x} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

What conditions must c_1 and c_2 satisfy so that $f_X(x)$ is a valid PDF?

4.3.6 ♦♦ For constants a and b , random variable X has PDF

$$f_X(x) = \begin{cases} ax^2 + bx & 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

What conditions on a and b are necessary and sufficient to guarantee that $f_X(x)$ is a valid PDF?

4.4.1 ● Random variable X has PDF

$$f_X(x) = \begin{cases} 1/4 & -1 \leq x \leq 3, \\ 0 & \text{otherwise.} \end{cases}$$

Define the random variable Y by $Y = h(X) = X^2$.

- Find $E[X]$ and $\text{Var}[X]$.
- Find $h(E[X])$ and $E[h(X)]$.
- Find $E[Y]$ and $\text{Var}[Y]$.

4.4.2 ● Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 1/8 & 1 \leq x \leq 9, \\ 0 & \text{otherwise.} \end{cases}$$

Let $Y = h(X) = 1/\sqrt{X}$.

- Find $E[X]$ and $\text{Var}[X]$.
- Find $h(E[X])$ and $E[h(X)]$.
- Find $E[Y]$ and $\text{Var}[Y]$.

4.4.3 ● Random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < 0, \\ x/2 & 0 \leq x \leq 2, \\ 1 & x > 2. \end{cases}$$

- What is $E[X]$?
- What is $\text{Var}[X]$?

4.4.4 ● The probability density function of random variable Y is

$$f_Y(y) = \begin{cases} y/2 & 0 \leq y < 2, \\ 0 & \text{otherwise.} \end{cases}$$

What are $E[Y]$ and $\text{Var}[Y]$?

4.4.5 ● The cumulative distribution function of the random variable Y is

$$F_Y(y) = \begin{cases} 0 & y < -1, \\ (y+1)/2 & -1 \leq y \leq 1, \\ 1 & y > 1. \end{cases}$$

What are $E[Y]$ and $\text{Var}[Y]$?

4.4.6 ■ The cumulative distribution function of random variable V is

$$F_V(v) = \begin{cases} 0 & v < -5, \\ (v+5)^2/144 & -5 \leq v < 7, \\ 1 & v \geq 7. \end{cases}$$

- What are $E[V]$ and $\text{Var}[V]$?
- What is $E[V^3]$?

4.4.7 ■ The cumulative distribution function of random variable U is

$$F_U(u) = \begin{cases} 0 & u < -5, \\ \frac{u+5}{8} & -5 \leq u < -3, \\ \frac{1}{4} & -3 \leq u < 3, \\ \frac{3u-7}{8} & 3 \leq u < 5, \\ 1 & u \geq 5. \end{cases}$$

- What are $E[U]$ and $\text{Var}[U]$?
- What is $E[2^U]$?

4.4.8 ■ X is a Pareto (α, μ) random variable, as defined in Appendix A. What is the largest value of n for which the n th moment $E[X^n]$ exists? For all feasible values of n , find $E[X^n]$.

4.5.1 ● Y is a continuous uniform $(1, 5)$ random variable.

- (a) What is $P[Y > E[Y]]$?
- (b) What is $P[Y \leq \text{Var}[Y]]$?

4.5.2 ● The current Y across a $1 \text{ k}\Omega$ resistor is a continuous uniform $(-10, 10)$ random variable. Find $P[|Y| < 3]$.

4.5.3 ● Radars detect flying objects by measuring the power reflected from them. The reflected power of an aircraft can be modeled as a random variable Y with PDF

$$f_Y(y) = \begin{cases} \frac{1}{P_0} e^{-y/P_0} & y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where $P_0 > 0$ is some constant. The aircraft is correctly identified by the radar if the reflected power of the aircraft is larger than its average value. What is the probability $P[C]$ that an aircraft is correctly identified?

4.5.4 ● Y is an exponential random variable with variance $\text{Var}[Y] = 25$.

- (a) What is the PDF of Y ?
- (b) What is $E[Y^2]$?
- (c) What is $P[Y > 5]$?

4.5.5 ● The time delay Y (in milliseconds) that your computer needs to connect to an access point is an exponential random variable.

- (a) Find $P[Y > E[Y]]$.
- (b) Find $P[Y > 2E[Y]]$.

4.5.6 ● X is an Erlang (n, λ) random variable with parameter $\lambda = 1/3$ and expected value $E[X] = 15$.

- (a) What is the value of the parameter n ?
- (b) What is the PDF of X ?
- (c) What is $\text{Var}[X]$?

4.5.7 ● Y is an Erlang $(n = 2, \lambda = 2)$ random variable.

- (a) What is $E[Y]$?
- (b) What is $\text{Var}[Y]$?
- (c) What is $P[0.5 \leq Y < 1.5]$?

4.5.8 ■ U is a zero mean continuous uniform random variable. What is $P[U^2 \leq \text{Var}[U]]$?

4.5.9 ■ U is a continuous uniform random variable such that $E[U] = 10$ and $P[U > 12] = 1/4$. What is $P[U < 9]$?

4.5.10 ■ X is a continuous uniform $(-5, 5)$ random variable.

- (a) What is the PDF $f_X(x)$?
- (b) What is the CDF $F_X(x)$?
- (c) What is $E[X]$?
- (d) What is $E[X^5]$?
- (e) What is $E[e^X]$?

4.5.11 ■ X is a continuous uniform $(-a, a)$ random variable. Find $P[|X| \leq \text{Var}[X]]$.

4.5.12 ■ X is a uniform random variable with expected value $\mu_X = 7$ and variance $\text{Var}[X] = 3$. What is the PDF of X ?

4.5.13 ■ The probability density function of random variable X is

$$f_X(x) = \begin{cases} (1/2)e^{-x/2} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $P[1 \leq X \leq 2]$?
- (b) What is $F_X(x)$, the cumulative distribution function of X ?
- (c) What is $E[X]$, the expected value of X ?
- (d) What is $\text{Var}[X]$, the variance of X ?

4.5.14 ■ Verify parts (b) and (c) of Theorem 4.6 by directly calculating the expected value and variance of a uniform random variable with parameters $a < b$.

4.5.15 ■ Long-distance calling plan A offers flat-rate service at 10 cents per minute. Calling plan B charges 99 cents for every call under 20 minutes; for calls over 20 minutes, the charge is 99 cents for the first 20 minutes plus 10 cents for every additional

minute. (Note that these plans measure your call duration exactly, without rounding to the next minute or even second.) If your long-distance calls have exponential distribution with expected value τ minutes, which plan offers a lower expected cost per call?

4.5.16 ■ In this problem we verify that an Erlang (n, λ) PDF integrates to 1. Let the integral of the n th order Erlang PDF be denoted by

$$I_n = \int_0^\infty \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n-1)!} dx.$$

First, show directly that the Erlang PDF with $n = 1$ integrates to 1 by verifying that $I_1 = 1$. Second, use integration by parts (Appendix B, Math Fact B.10) to show that $I_n = I_{n-1}$.

4.5.17 ■ Calculate the k th moment $E[X^k]$ of an Erlang (n, λ) random variable X . Use your result to verify Theorem 4.10. Hint: Remember that the Erlang $(n + k, \lambda)$ PDF integrates to 1.

4.5.18 ♦ In this problem, we outline the proof of Theorem 4.11.

- (a) Let X_n denote an Erlang (n, λ) random variable. Use the definition of the Erlang PDF to show that for any $x \geq 0$,

$$F_{X_n}(x) = \int_0^x \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!} dt.$$

- (b) Apply integration by parts (see Appendix B, Math Fact B.10) to this integral to show that for $x \geq 0$,

$$F_{X_n}(x) = F_{X_{n-1}}(x) - \frac{(\lambda x)^{n-1} e^{-\lambda x}}{(n-1)!}.$$

- (c) Use the fact that $F_{X_1}(x) = 1 - e^{-\lambda x}$ for $x \geq 0$ to verify the claim of Theorem 4.11.

4.5.19 ♦ Prove by induction that an exponential random variable X with expected

value $1/\lambda$ has n th moment

$$E[X^n] = \frac{n!}{\lambda^n}.$$

Hint: Use integration by parts (Appendix B, Math Fact B.10).

4.5.20 ♦♦ This problem outlines the steps needed to show that a nonnegative continuous random variable X has expected value

$$E[X] = \int_0^\infty [1 - F_X(x)] dx.$$

- (a) For any $r \geq 0$, show that

$$r P[X > r] \leq \int_r^\infty x f_X(x) dx.$$

- (b) Use part (a) to argue that if $E[X] < \infty$, then

$$\lim_{r \rightarrow \infty} r P[X > r] = 0.$$

- (c) Now use integration by parts (Appendix B, Math Fact B.10) to evaluate

$$\int_0^\infty [1 - F_X(x)] dx.$$

4.6.1 ● The peak temperature T , as measured in degrees Fahrenheit, on a July day in New Jersey is the Gaussian $(85, 10)$ random variable. What is $P[T > 100]$, $P[T < 60]$, and $P[70 \leq T \leq 100]$?

4.6.2 ● What is the PDF of Z , the standard normal random variable?

4.6.3 ● Find each probability.

- (a) V is a Gaussian $(\mu = 0, \sigma = 2)$ random variable. Find $P[V > 4]$.
- (b) W is a Gaussian $(\mu = 2, \sigma = 5)$ random variable. What is $P[W \leq 2]$?
- (c) For a Gaussian $(\mu, \sigma = 2)$ random variable X , find $P[X \leq \mu + 1]$.
- (d) Y is a Gaussian $(\mu = 50, \sigma = 10)$ random variable. Calculate $P[Y > 65]$.

4.6.4 ● In each of the following cases, Y is a Gaussian random variable. Find the expected value $\mu = E[Y]$.

- (a) Y has standard deviation $\sigma = 10$ and $P[Y \leq 10] = 0.933$.
- (b) Y has standard deviation $\sigma = 10$ and $P[Y \leq 0] = 0.067$.
- (c) Y has standard deviation σ and $P[Y \leq 10] = 0.977$. (Find μ as a function of σ .)
- (d) $P[Y > 5] = 1/2$.

4.6.5 ● Your internal body temperature T in degrees Fahrenheit is a Gaussian ($\mu = 98.6, \sigma = 0.4$) random variable. In terms of the $\Phi(\cdot)$ function, find $P[T > 100]$. Does this model seem reasonable?

4.6.6 ● The temperature T in this thermostatically controlled lecture hall is a Gaussian random variable with expected value $\mu = 68$ degrees Fahrenheit. In addition, $P[T < 66] = 0.1587$. What is the variance of T ?

4.6.7 ● X is a Gaussian random variable with $E[X] = 0$ and $P[|X| \leq 10] = 0.1$. What is the standard deviation σ_X ?

4.6.8 ● A function commonly used in communications textbooks for the tail probabilities of Gaussian random variables is the complementary error function, defined as

$$\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^\infty e^{-x^2} dx.$$

Show that

$$Q(z) = \frac{1}{2} \operatorname{erfc}\left(\frac{z}{\sqrt{2}}\right).$$

4.6.9 ■ The peak temperature T , in degrees Fahrenheit, on a July day in Antarctica is a Gaussian random variable with a variance of 225. With probability $1/2$, the temperature T exceeds -75 degrees. What is $P[T > 0]$? What is $P[T < -100]$?

4.6.10 ■ A professor pays 25 cents for each blackboard error made in lecture to the student who points out the error. In a career

of n years filled with blackboard errors, the total amount in dollars paid can be approximated by a Gaussian random variable Y_n with expected value $40n$ and variance $100n$. What is the probability that Y_{20} exceeds 1000? How many years n must the professor teach in order that $P[Y_n > 1000] > 0.99$?

4.6.11 ■ Suppose that out of 100 million men in the United States, 23,000 are at least 7 feet tall. Suppose that the heights of U.S. men are independent Gaussian random variables with a expected value of $5'10''$. Let N equal the number of men who are at least $7'6''$ tall.

- (a) Calculate σ_X , the standard deviation of the height of U.S. men.
- (b) In terms of the $\Phi(\cdot)$ function, what is the probability that a randomly chosen man is at least 8 feet tall?
- (c) What is the probability that no man alive in the United States today is at least $7'6''$ tall?
- (d) What is $E[N]$?

4.6.12 ■ In this problem, we verify that for $x \geq 0$,

$$\Phi(x) = \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{x}{\sqrt{2}}\right).$$

- (a) Let Y have a Gaussian $(0, 1/\sqrt{2})$ distribution and show that

$$F_Y(y) = \int_{-\infty}^y f_Y(u) du = \frac{1}{2} + \operatorname{erf}(y).$$

- (b) Observe that $Z = \sqrt{2}Y$ is Gaussian $(0, 1)$ and show that

$$\Phi(z) = F_Z(z) = F_Y\left(\frac{z}{\sqrt{2}}\right).$$

4.6.13 ■ This problem outlines the steps needed to show that the Gaussian PDF integrates to unity. For a Gaussian (μ, σ) random variable W , we will show that

$$I = \int_{-\infty}^{\infty} f_W(w) dw = 1.$$

- (a) Use the substitution $x = (w - \mu)/\sigma$ to show that

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} dx.$$

- (b) Show that

$$I^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy.$$

- (c) Change to polar coordinates to show that $I^2 = 1$.

4.6.14 ♦ At time $t = 0$, the price of a stock is a constant k dollars. At time $t > 0$ the price of a stock is a Gaussian random variable X with $E[X] = k$ and $\text{Var}[X] = t$. At time t , a *Call Option at Strike k* has value $V = (X - k)^+$, where the operator $(\cdot)^+$ is defined as $(z)^+ = \max(z, 0)$.

- (a) Find the expected value $E[V]$.
- (b) Suppose you can buy the call option for d dollars at time $t = 0$. At time t , you can sell the call for V dollars and earn a profit (or loss perhaps) of $R = V - d$ dollars. Let d_0 denote the value of d such that $P[R > 0] = 1/2$. Your strategy is that you buy the option if $d \leq d_0$ so that your probability of a profit is $P[R > 0] \geq 1/2$. Find d_0 .
- (c) Let d_1 denote the value of d such that $E[R] = 0.01 \times d$. Now your strategy is to buy the option if $d \leq d_1$ so that your expected return is at least one percent of the option cost. Find d_1 .
- (d) Are the strategies “Buy the option if $d \leq d_0$ ” and “Buy the option if $d \leq d_1$ ” reasonable strategies?

4.6.15 ♦ In mobile radio communications, the radio channel can vary randomly. In particular, in communicating with a fixed transmitter power over a “Rayleigh fading” channel, the receiver signal-to-noise ratio Y is an exponential random variable with expected value γ . Moreover, when $Y = y$, the probability of an error in decoding a transmitted bit is $P_e(y) = Q(\sqrt{2y})$ where $Q(\cdot)$ is the standard normal complementary CDF.

The average probability of bit error, also known as the bit error rate or BER, is

$$\bar{P}_e = E[P_e(Y)] = \int_{-\infty}^{\infty} Q(\sqrt{2y}) f_Y(y) dy.$$

Find a simple formula for the BER $\bar{P}_e$ as a function of the average SNR γ .

4.6.16 ♦♦ At time $t = 0$, the price of a stock is a constant k dollars. At some future time $t > 0$, the price X of the stock is a uniform $(k - t, k + t)$ random variable. At this time t , a *Put Option at Strike k* (which is the right to sell the stock at price k) has value $(k - X)^+$ dollars where the operator $(\cdot)^+$ is defined as $(z)^+ = \max(z, 0)$. Similarly a *Call Option at Strike k* (the right to buy the stock at price k) at time t has value $(X - k)^+$.

- (a) At time 0, you sell the put and receive d dollars. At time t , you purchase the put for $(k - X)^+$ dollars to cancel your position. Your gain is

$$R = g_p(X) = d - (k - X)^+.$$

Find the central moments $E[R]$ and $\text{Var}[R]$.

- (b) In a short straddle, you sell the put for d dollars and you also sell the call for d dollars. At a future time $t > 0$, you purchase the put for $(k - X)^+$ dollars and the call for $(X - k)^+$ dollars to cancel both positions. Your gain on the put is $g_p(X) = d - (k - X)^+$ dollars and your gain on the call is $g_c(X) = d - (X - k)^+$ dollars. Your net gain is

$$R' = g_p(X) + g_c(X).$$

Find the expected value $E[R']$ and variance $\text{Var}[R']$.

- (c) Explain why selling the straddle might be attractive compared to selling just the put or just the call.

4.6.17 ♦♦ Continuing Problem 4.6.16, suppose you sell the straddle at time $t = 0$ and liquidate your position at time t , generating a profit (or perhaps a loss) R' . Find the

PDF $f_{R'}(r)$ of R' . Suppose d is sufficiently large that $E[R'] > 0$. Would you be interested in selling the short straddle? Are you getting something, namely $E[R']$ dollars, for nothing?

4.7.1 ● Let X be a random variable with CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ x/3 + 1/3 & -1 \leq x < 0, \\ x/3 + 2/3 & 0 \leq x < 1, \\ 1 & 1 \leq x. \end{cases}$$

Sketch the CDF and find

- (a) $P[X < -1]$ and $P[X \leq -1]$,
- (b) $P[X < 0]$ and $P[X \leq 0]$,
- (c) $P[0 < X \leq 1]$ and $P[0 \leq X \leq 1]$.

4.7.2 ● Let X be a random variable with CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ x/4 + 1/2 & -1 \leq x < 1, \\ 1 & 1 \leq x. \end{cases}$$

Sketch the CDF and find

- (a) $P[X < -1]$ and $P[X \leq -1]$.
- (b) $P[X < 0]$ and $P[X \leq 0]$.
- (c) $P[X > 1]$ and $P[X \geq 1]$.

4.7.3 ● For random variable X of Problem 4.7.2, find $f_X(x)$, $E[X]$, and $\text{Var}[X]$.

4.7.4 ● X is Bernoulli random variable with expected value p . What is the PDF $f_X(x)$?

4.7.5 ● X is a geometric random variable with expected value $1/p$. What is the PDF $f_X(x)$?

4.7.6 ■ When you make a phone call, the line is busy with probability 0.2 and no one answers with probability 0.3. The random variable X describes the conversation time (in minutes) of a phone call that is answered. X is an exponential random variable with $E[X] = 3$ minutes. Let the random variable W denote the conversation time (in seconds) of all calls ($W = 0$ when the line is busy or there is no answer.)

- (a) What is $F_W(w)$?
- (b) What is $f_W(w)$?
- (c) What are $E[W]$ and $\text{Var}[W]$?

4.7.7 ■ For 80% of lectures, Professor X arrives on time and starts lecturing with delay $T = 0$. When Professor X is late, the starting time delay T is uniformly distributed between 0 and 300 seconds. Find the CDF and PDF of T .

4.7.8 ♦ With probability 0.7, the toss of an Olympic shot-putter travels $D = 60 + X$ feet, where X is an exponential random variable with expected value $\mu = 10$. Otherwise, with probability 0.3, a foul is committed by stepping outside of the shot-put circle and we say $D = 0$. What are the CDF and PDF of random variable D ?

4.7.9 ♦ For 70% of lectures, Professor Y arrives on time. When Professor Y is late, the arrival time delay is a continuous random variable uniformly distributed from 0 to 10 minutes. Yet, as soon as Professor Y is 5 minutes late, all the students get up and leave. (It is unknown if Professor Y still conducts the lecture.) If a lecture starts when Professor Y arrives and always ends 80 minutes after the scheduled starting time, what is the PDF of T , the length of time that the students observe a lecture.

4.8.1 ● Write a function `y=quiz31rv(m)` that produces m samples of random variable Y defined in Quiz 4.2.

4.8.2 ● For the Gaussian $(0, 1)$ complementary CDF $Q(z)$, a useful numerical approximation for $z \geq 0$ is

$$\hat{Q}(z) = \left(\sum_{n=1}^5 a_n t^n \right) e^{-z^2/2},$$

where

$$\begin{aligned} t &= \frac{1}{1 + 0.231641888z} & a_1 &= 0.127414796 \\ a_2 &= -0.142248368 & a_3 &= 0.7107068705 \\ a_4 &= -0.7265760135 & a_5 &= 0.5307027145. \end{aligned}$$

To compare this approximation to $Q(z)$, use MATLAB to graph

$$e(z) = \frac{Q(z) - \hat{Q}(z)}{Q(z)}.$$

4.8.3 ■ Use `exponentialrv.m` and Theorem 4.9 and to write a MATLAB function `k=georv(p,m)` that generates m samples of a geometric (p) random variable K . Compare the resulting algorithm to the technique employed in Problem 3.9.8 for `geometricrv(p,m)`.

4.8.4 ■ Applying Equation (4.14) with x replaced by $i\Delta$ and dx replaced by Δ , we obtain

$$P[i\Delta < X \leq i\Delta + \Delta] = f_X(i\Delta) \Delta.$$

If we generate a large number n of samples of random variable X , let n_i denote the number of occurrences of the event

$$\{i\Delta < X \leq (i+1)\Delta\}.$$

We would expect that

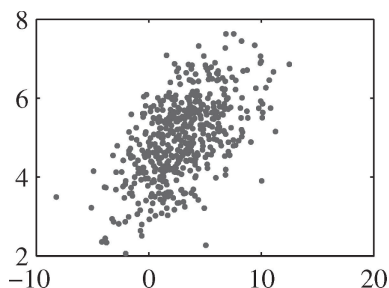
$$\lim_{n \rightarrow \infty} \frac{n_i}{n} = f_X(i\Delta) \Delta,$$

or equivalently,

$$\lim_{n \rightarrow \infty} \frac{n_i}{n\Delta} = f_X(i\Delta).$$

Use MATLAB to confirm this with $\Delta = 0.01$ for

- (a) an exponential ($\lambda = 1$) random variable X and for $i = 0, \dots, 500$,
 - (b) a Gaussian $(3, 1)$ random variable X and for $i = 0, \dots, 600$.
-



The sample output of `gauss2rv` shown here is produced with the commands

```
>> xy=gauss2rv(3,3,5,1,0.5,500);
>> plot(xy(1,:),xy(2,:),'.');

```

We observe that the center of the cloud is $(\mu_X, \mu_Y) = (3, 5)$. In addition, we note that the X and Y axes are scaled differently because $\sigma_X = 3$ and $\sigma_Y = 1$.

We observe that this example with $\rho_{X,Y} = 0.5$ shows random variables that are less correlated than the examples in Figure 5.5 with $|\rho| = 0.9$.

We note that bivariate Gaussian random variables are a special case of n -dimensional Gaussian random vectors, which are introduced in Chapter 8. Based on linear algebra techniques, Chapter 8 introduces the `gaussvector` function to generate samples of Gaussian random vectors that generalizes `gauss2rv` to n dimensions.

Beyond bivariate Gaussian pairs, there exist a variety of techniques for generating sample values of pairs of continuous random variables of specific types. A basic approach is to generate X based on the marginal PDF $f_X(x)$ and then generate Y using a conditional probability model that depends on the value of X . Conditional probability models and MATLAB techniques that employ these models are the subject of Chapter 7.

Problems

Difficulty: ● Easy ■ Moderate ◆ Difficult ◆◆ Experts Only

5.1.1 ● Random variables X and Y have the joint CDF

$$F_{X,Y}(x,y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}) & x \geq 0; \\ & y \geq 0, \\ 0 & \text{ow.} \end{cases}$$

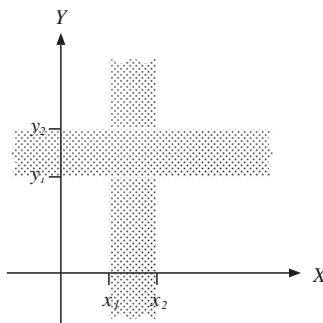
- (a) What is $P[X \leq 2, Y \leq 3]$?
- (b) What is the marginal CDF, $F_X(x)$?
- (c) What is the marginal CDF, $F_Y(y)$?

5.1.2 ● Express the following extreme values of $F_{X,Y}(x,y)$ in terms of the marginal cumulative distribution functions $F_X(x)$ and $F_Y(y)$.

- (a) $F_{X,Y}(x, -\infty)$
- (b) $F_{X,Y}(x, \infty)$
- (c) $F_{X,Y}(-\infty, \infty)$
- (d) $F_{X,Y}(-\infty, y)$

(e) $F_{X,Y}(\infty, y)$

5.1.3 ■ For continuous random variables X, Y with joint CDF $F_{X,Y}(x,y)$ and marginal CDFs $F_X(x)$ and $F_Y(y)$, find $P[x_1 \leq X < x_2 \cup y_1 \leq Y < y_2]$. This is the probability of the shaded “cross” region in the following diagram.



5.1.4 ■ Random variables X and Y have CDF $F_X(x)$ and $F_Y(y)$. Is $F(x, y) = F_X(x)F_Y(y)$ a valid CDF? Explain your answer.

5.1.5 ♦ In this problem, we prove Theorem 5.2.

- (a) Sketch the following events on the X, Y plane:

$$\begin{aligned} A &= \{X \leq x_1, y_1 < Y \leq y_2\}, \\ B &= \{x_1 < X \leq x_2, Y \leq y_1\}, \\ C &= \{x_1 < X \leq x_2, y_1 < Y \leq y_2\}. \end{aligned}$$

- (b) Express the probability of the events A , B , and $A \cup B \cup C$ in terms of the joint CDF $F_{X,Y}(x, y)$.
- (c) Use the observation that events A , B , and C are mutually exclusive to prove Theorem 5.2.

5.1.6 ♦ Can the following function be the joint CDF of random variables X and Y ?

$$F(x, y) = \begin{cases} 1 - e^{-(x+y)} & x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

5.2.1 ● Random variables X and Y have the joint PMF

$$P_{X,Y}(x, y) = \begin{cases} cxy & x = 1, 2, 4; \quad y = 1, 3, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[Y < X]$?
- (c) What is $P[Y > X]$?
- (d) What is $P[Y = X]$?
- (e) What is $P[Y = 3]$?

5.2.2 ● Random variables X and Y have the joint PMF

$$P_{X,Y}(x, y) = \begin{cases} c|x+y| & x = -2, 0, 2; \\ & y = -1, 0, 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?

- (b) What is $P[Y < X]$?
- (c) What is $P[Y > X]$?
- (d) What is $P[Y = X]$?
- (e) What is $P[X < 1]$?

5.2.3 ● Test two integrated circuits. In each test, the probability of rejecting the circuit is p , independent of the other test. Let X be the number of rejects (either 0 or 1) in the first test and let Y be the number of rejects in the second test. Find the joint PMF $P_{X,Y}(x, y)$.

5.2.4 ■ For two independent flips of a fair coin, let X equal the total number of tails and let Y equal the number of heads on the last flip. Find the joint PMF $P_{X,Y}(x, y)$.

5.2.5 ■ In Figure 5.2, the axes of the figures are labeled X and Y because the figures depict possible values of the random variables X and Y . However, the figure at the end of Example 5.3 depicts $P_{X,Y}(x, y)$ on axes labeled with lowercase x and y . Should those axes be labeled with the uppercase X and Y ? Hint: Reasonable arguments can be made for both views.

5.2.6 ■ As a generalization of Example 5.3, consider a test of n circuits such that each circuit is acceptable with probability p , independent of the outcome of any other test. Show that the joint PMF of X , the number of acceptable circuits, and Y , the number of acceptable circuits found before observing the first reject, is

$$\begin{aligned} P_{X,Y}(x, y) &= \begin{cases} \binom{n-y-1}{x-y} p^x (1-p)^{n-x} & 0 \leq y \leq x < n, \\ p^n & x = y = n, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Hint: For $0 \leq y \leq x < n$, show that

$$\{X = x, Y = y\} = A \cap B \cap C,$$

where

- A : The first y tests are acceptable.
 B : Test $y + 1$ is a rejection.
 C : The remaining $n - y - 1$ tests yield $x - y$ acceptable circuits

5.2.7 ■ With two minutes left in a five-minute overtime, the score is 0–0 in a Rutgers soccer match versus Villanova. (Note that the overtime is NOT *sudden-death*.) In the next-to-last minute of the game, either (1) Rutgers scores a goal with probability $p = 0.2$, (2) Villanova scores with probability $p = 0.2$, or (3) neither team scores with probability $1 - 2p = 0.6$. If neither team scores in the next-to-last minute, then in the final minute, either (1) Rutgers scores a goal with probability $q = 0.3$, (2) Villanova scores with probability $q = 0.3$, or (3) neither team scores with probability $1 - 2q = 0.4$. However, if a team scores in the next-to-last minute, the trailing team goes for broke so that in the last minute, either (1) the leading team scores with probability 0.5, or (2) the trailing team scores with probability 0.5. For the final two minutes of overtime:

- Sketch a probability tree and construct a table for $P_{R,V}(r, v)$, the joint PMF of R , the number of Rutgers goals scored, and V , the number of Villanova goals scored.
- What is the probability $P[T]$ that the overtime ends in a tie?
- What is the PMF of R , the number of goals scored by Rutgers?
- What is the PMF of G , the total number of goals scored?

5.2.8 ♦ Each test of an integrated circuit produces an acceptable circuit with probability p , independent of the outcome of the test of any other circuit. In testing n circuits, let K denote the number of circuits rejected and let X denote the number of acceptable circuits (either 0 or 1) in the last test. Find the joint PMF $P_{K,X}(k, x)$.

5.2.9 ♦ Each test of an integrated circuit produces an acceptable circuit with probability p , independent of the outcome of the test of any other circuit. In testing n circuits, let K denote the number of circuits rejected and let X denote the number of acceptable circuits that appear before the

first reject is found. Find the joint PMF $P_{K,X}(k, x)$.

5.3.1 ● Given the random variables X and Y in Problem 5.2.1, find

- The marginal PMFs $P_X(x)$ and $P_Y(y)$,
- The expected values $E[X]$ and $E[Y]$,
- The standard deviations σ_X and σ_Y .

5.3.2 ● Given the random variables X and Y in Problem 5.2.2, find

- The marginal PMFs $P_X(x)$ and $P_Y(y)$,
- The expected values $E[X]$ and $E[Y]$,
- The standard deviations σ_X and σ_Y .

5.3.3 ● For $n = 0, 1, \dots$ and $0 \leq k \leq 100$, the joint PMF of random variables N and K is

$$P_{N,K}(n, k) = \frac{100^n e^{-100}}{n!} \binom{100}{k} p^k (1-p)^{100-k}.$$

Otherwise, $P_{N,K}(n, k) = 0$. Find the marginal PMFs $P_N(n)$ and $P_K(k)$.

5.3.4 ■ Random variables X and Y have joint PMF

$$P_{X,Y}(x, y) = \begin{cases} 1/21 & x = 0, 1, 2, 3, 4, 5; \\ & y = 0, 1, \dots, x, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PMFs $P_X(x)$ and $P_Y(y)$ and the expected values $E[X]$ and $E[Y]$.

5.3.5 ■ Random variables N and K have the joint PMF

$$P_{N,K}(n, k) = \begin{cases} \frac{(1-p)^{n-1} p}{n} & k=1, \dots, n; \\ & n=1, 2, \dots \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PMFs $P_N(n)$ and $P_K(k)$.

5.3.6 ♦ Random variables N and K have the joint PMF

$$P_{N,K}(n, k) = \begin{cases} \frac{100^n e^{-100}}{(n+1)!} & k=0, 1, \dots, n; \\ & n=0, 1, \dots \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PMF $P_N(n)$. Show that the marginal PMF $P_K(k)$ satisfies $P_K(k) = P[N > k]/100$.

5.4.1 ● Random variables X and Y have the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} c & x \geq 0, y \geq 0, x + y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- What is the value of the constant c ?
- What is $P[X \leq Y]$?
- What is $P[X + Y \leq 1/2]$?

5.4.2 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} cxy^2 & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- Find the constant c .
- Find $P[X > Y]$ and $P[Y < X^2]$.
- Find $P[\min(X, Y) \leq 1/2]$.
- Find $P[\max(X, Y) \leq 3/4]$.

5.4.3 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6e^{-(2x+3y)} & x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- Find $P[X > Y]$ and $P[X + Y \leq 1]$.
- Find $P[\min(X, Y) \geq 1]$.
- Find $P[\max(X, Y) \leq 1]$.

5.4.4 ♦ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 8xy & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Following the method of Example 5.8, find the joint CDF $F_{X,Y}(x, y)$.

5.5.1 ● Random variables X and Y have the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 1/2 & -1 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Sketch the region of nonzero probability and answer the following questions.

- What is $P[X > 0]$?
- What is $f_X(x)$?
- What is $E[X]$?

5.5.2 ● Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} cx & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- Find the constant c .
- Find the marginal PDF $f_X(x)$.
- Are X and Y independent? Justify your answer.

5.5.3 ● X and Y are random variables with the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & x + y \leq 1, x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- What is the marginal PDF $f_X(x)$?
- What is the marginal PDF $f_Y(y)$?

5.5.4 ■ Over the circle $X^2 + Y^2 \leq r^2$, random variables X and Y have the uniform PDF

$$f_{X,Y}(x, y) = \begin{cases} 1/(\pi r^2) & x^2 + y^2 \leq r^2, \\ 0 & \text{otherwise.} \end{cases}$$

- What is the marginal PDF $f_X(x)$?
- What is the marginal PDF $f_Y(y)$?

5.5.5 ■ X and Y are random variables with the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 5x^2/2 & -1 \leq x \leq 1, \\ 0 & 0 \leq y \leq x^2 \\ \text{otherwise.} \end{cases}$$

- What is the marginal PDF $f_X(x)$?
- What is the marginal PDF $f_Y(y)$?

5.5.6 ■ Over the circle $X^2 + Y^2 \leq r^2$, random variables X and Y have the PDF

$$f_{X,Y}(x,y) = \begin{cases} 2|xy|/r^4 & x^2 + y^2 \leq r^2, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the marginal PDF $f_X(x)$?
 (b) What is the marginal PDF $f_Y(y)$?

5.5.7 ■ For a random variable X , let $Y = aX + b$. Show that if $a > 0$ then $\rho_{X,Y} = 1$. Also show that if $a < 0$, then $\rho_{X,Y} = -1$.

5.5.8 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} (x+y)/3 & 0 \leq x \leq 1; \\ & 0 \leq y \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.
 (b) What are $E[X]$ and $\text{Var}[X]$?
 (c) What are $E[Y]$ and $\text{Var}[Y]$?

5.5.9 ♦ Random variables X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} cy & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Draw the region of nonzero probability.
 (b) What is the value of the constant c ?
 (c) What is $F_X(x)$?
 (d) What is $F_Y(y)$?
 (e) What is $P[Y \leq X/2]$?

5.6.1 ● An ice cream company needs to order ingredients from its suppliers. Depending on the size of the order, the weight of the shipment can be either

- 1 kg for a small order,
 2 kg for a big order.

The company has three different suppliers. The vanilla supplier is 20 miles away. The chocolate supplier is 100 miles away. The

strawberry supplier is 300 miles away. An experiment consists of monitoring an order and observing W , the weight of the order, and D , the distance the shipment must be sent. The following probability model describes the experiment:

	van.	choc.	straw.
small	0.2	0.2	0.2
big	0.1	0.2	0.1

- (a) What is the joint PMF $P_{W,D}(w,d)$ of the weight and the distance?
 (b) Find the expected shipping distance $E[D]$.
 (c) Are W and D independent?

5.6.2 ● A company receives shipments from two factories. Depending on the size of the order, a shipment can be in

- 1 box for a small order,
 2 boxes for a medium order,
 3 boxes for a large order.

The company has two different suppliers. Factory Q is 60 miles from the company. Factory R is 180 miles from the company.

An experiment consists of monitoring a shipment and observing B , the number of boxes, and M , the number of miles the shipment travels. The following probability model describes the experiment:

	Factory Q	Factory R
small order	0.3	0.2
medium order	0.1	0.2
large order	0.1	0.1

- (a) Find $P_{B,M}(b,m)$, the joint PMF of the number of boxes and the distance.
 (b) What is $E[B]$, the expected number of boxes?
 (c) Are B and M independent?

5.6.3 ● Observe 100 independent flips of a fair coin. Let X equal the number of heads in the first 75 flips. Let Y equal the number of heads in the remaining 25 flips. Find $P_X(x)$ and $P_Y(y)$. Are X and Y independent? Find $P_{X,Y}(x,y)$.

5.6.4 ● Observe independent flips of a fair coin until heads occurs twice. Let X_1 equal the number of flips up to and including the

first H . Let X_2 equal the number of additional flips up to and including the second H . What are $P_{X_1}(x_1)$ and $P_{X_2}(x_2)$. Are X_1 and X_2 independent? Find $P_{X_1, X_2}(x_1, x_2)$.

5.6.5 ● X is the continuous uniform $(0, 2)$ random variable. Y has the continuous uniform $(0, 5)$ PDF, independent of X . What is the joint PDF $f_{X,Y}(x, y)$?

5.6.6 ● X_1 and X_2 are independent random variables such that X_i has PDF

$$f_{X_i}(x) = \begin{cases} \lambda_i e^{-\lambda_i x} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

What is $P[X_2 < X_1]$?

5.6.7 ■ In terms of a positive constant k , random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} k + 3x^2 & -1/2 \leq x \leq 1/2, \\ 0 & \text{otherwise.} \end{cases}$$

- What is k ?
- What is the marginal PDF of X ?
- What is the marginal PDF of Y ?
- Are X and Y independent?

5.6.8 ■ X_1 and X_2 are independent, identically distributed random variables with PDF

$$f_X(x) = \begin{cases} x/2 & 0 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

- Find the CDF, $F_X(x)$.
- What is $P[X_1 \leq 1, X_2 \leq 1]$, the probability that X_1 and X_2 are both less than or equal to 1?
- Let $W = \max(X_1, X_2)$. What is $F_W(1)$, the CDF of W evaluated at $w = 1$?
- Find the CDF $F_W(w)$.

5.6.9 ♦ Prove that random variables X and Y are independent if and only if

$$F_{X,Y}(x, y) = F_X(x) F_Y(y).$$

5.7.1 ● Continuing Problem 5.6.1, the price per kilogram for shipping the order is one cent per mile. C cents is the shipping cost of one order. What is $E[C]$?

5.7.2 ● Continuing Problem 5.6.2, the price per mile of shipping each box is one cent per mile the box travels. C cents is the price of one shipment. What is $E[C]$, the expected price of one shipment?

5.7.3 ● A random ECE sophomore has height X (rounded to the nearest foot) and GPA Y (rounded to the nearest integer). These random variables have joint PMF

$P_{X,Y}(x, y)$	$y = 1$	$y = 2$	$y = 3$	$y = 4$
$x = 5$	0.05	0.1	0.2	0.05
$x = 6$	0.1	0.1	0.3	0.1

Find $E[X + Y]$ and $\text{Var}[X + Y]$.

5.7.4 ● X and Y are independent, identically distributed random variables with PMF

$$P_X(k) = P_Y(k) = \begin{cases} 3/4 & k = 0, \\ 1/4 & k = 20, \\ 0 & \text{otherwise.} \end{cases}$$

Find the following quantities:

$$E[X], \quad \text{Var}[X], \\ E[X + Y], \quad \text{Var}[X + Y], \quad E[XY2^{XY}].$$

5.7.5 ● X and Y are random variables with $E[X] = E[Y] = 0$ and $\text{Var}[X] = 1$, $\text{Var}[Y] = 4$ and correlation coefficient $\rho = 1/2$. Find $\text{Var}[X + Y]$.

5.7.6 ● X and Y are random variables such that X has expected value $\mu_X = 0$ and standard deviation $\sigma_X = 3$ while Y has expected value $\mu_Y = 1$ and standard deviation $\sigma_Y = 4$. In addition, X and Y have covariance $\text{Cov}[X, Y] = -3$. Find the expected value and variance of $W = 2X + 2Y$.

5.7.7 ● Observe independent flips of a fair coin until heads occurs twice. Let X_1 equal the number of flips up to and including the first H . Let X_2 equal the number of additional flips up to and including the sec-

and H . Let $Y = X_1 - X_2$. Find $E[Y]$ and $\text{Var}[Y]$. Hint: Don't try to find $P_Y(y)$.

5.7.8 ● X_1 and X_2 are independent identically distributed random variables with expected value $E[X]$ and variance $\text{Var}[X]$.

- (a) What is $E[X_1 - X_2]$?
- (b) What is $\text{Var}[X_1 - X_2]$?

5.7.9 ■ X and Y are identically distributed random variables with $E[X] = E[Y] = 0$ and covariance $\text{Cov}[X, Y] = 3$ and correlation coefficient $\rho_{X,Y} = 1/2$. For nonzero constants a and b , $U = aX$ and $V = bY$.

- (a) Find $\text{Cov}[U, V]$.
- (b) Find the correlation coefficient $\rho_{U,V}$.
- (c) Let $W = U + V$. For what values of a and b are X and W uncorrelated?

5.7.10 ■ True or False: For identically distributed random variables Y_1 and Y_2 with $E[Y_1] = E[Y_2] = 0$, $\text{Var}[Y_1 + Y_2] \geq \text{Var}[Y_1]$.

5.7.11 ■ X and Y are random variables with $E[X] = E[Y] = 0$ such that X has standard deviation $\sigma_X = 2$ while Y has standard deviation $\sigma_Y = 4$.

- (a) For $V = X - Y$, what are the smallest and largest possible values of $\text{Var}[V]$?
- (b) For $W = X - 2Y$, what are the smallest and largest possible values of $\text{Var}[W]$?

5.7.12 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 4xy & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What are $E[X]$ and $\text{Var}[X]$?
- (b) What are $E[Y]$ and $\text{Var}[Y]$?
- (c) What is $\text{Cov}[X, Y]$?
- (d) What is $E[X + Y]$?
- (e) What is $\text{Var}[X + Y]$?

5.7.13 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 5x^2/2 & -1 \leq x \leq 1; \\ & 0 \leq y \leq x^2, \\ 0 & \text{otherwise.} \end{cases}$$

Answer the following questions.

- (a) What are $E[X]$ and $\text{Var}[X]$?
- (b) What are $E[Y]$ and $\text{Var}[Y]$?
- (c) What is $\text{Cov}[X, Y]$?
- (d) What is $E[X + Y]$?
- (e) What is $\text{Var}[X + Y]$?

5.7.14 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What are $E[X]$ and $\text{Var}[X]$?
- (b) What are $E[Y]$ and $\text{Var}[Y]$?
- (c) What is $\text{Cov}[X, Y]$?
- (d) What is $E[X + Y]$?
- (e) What is $\text{Var}[X + Y]$?

5.7.15 ■ A transmitter sends a signal X and a receiver makes the observation $Y = X + Z$, where Z is a receiver noise that is independent of X and $E[X] = E[Z] = 0$. Since the average power of the signal is $E[X^2]$ and the average power of the noise is $E[Z^2]$, a quality measure for the received signal is the signal-to-noise ratio

$$\Gamma = \frac{E[X^2]}{E[Z^2]}.$$

How is Γ related to the correlation coefficient $\rho_{X,Y}$?

5.8.1 ● X and Z are independent random variables with $E[X] = E[Z] = 0$ and variance $\text{Var}[X] = 1$ and $\text{Var}[Z] = 16$. Let $Y = X + Z$. Find the correlation coefficient ρ of X and Y . Are X and Y independent?

5.8.2 ● For the random variables X and Y in Problem 5.2.1, find

- (a) The expected value of $W = Y/X$,
- (b) The correlation, $r_{X,Y} = E[XY]$,
- (c) The covariance, $\text{Cov}[X, Y]$,

- (d) The correlation coefficient, $\rho_{X,Y}$,
 (e) The variance of $X + Y$, $\text{Var}[X + Y]$.

(Refer to the results of Problem 5.3.1 to answer some of these questions.)

5.8.3 For the random variables X and Y in Problem 5.2.2 find

- (a) The expected value of $W = 2^{XY}$,
 (b) The correlation, $r_{X,Y} = E[XY]$,
 (c) The covariance, $\text{Cov}[X, Y]$,
 (d) The correlation coefficient, $\rho_{X,Y}$,
 (e) The variance of $X + Y$, $\text{Var}[X + Y]$.

(Refer to the results of Problem 5.3.2 to answer some of these questions.)

5.8.4 Let H and B be the random variables in Quiz 5.3. Find $r_{H,B}$ and $\text{Cov}[H, B]$.

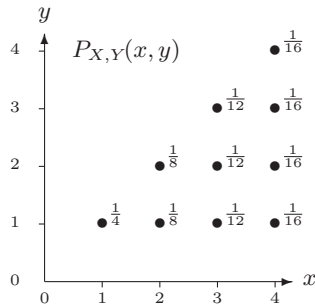
5.8.5 X and Y are independent random variables with PDFs

$$f_X(x) = \begin{cases} \frac{1}{3}e^{-x/3} & x \geq 0, \\ 0 & \text{otherwise,} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{2}e^{-y/2} & y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the correlation $r_{X,Y}$.
 (b) Find the covariance $\text{Cov}[X, Y]$.

5.8.6 The random variables X and Y have joint PMF



Find

- (a) The expected values $E[X]$ and $E[Y]$,
 (b) The variances $\text{Var}[X]$ and $\text{Var}[Y]$,

- (c) The correlation, $r_{X,Y} = E[XY]$,
 (d) The covariance, $\text{Cov}[X, Y]$,
 (e) The correlation coefficient, $\rho_{X,Y}$.

5.8.7 For X and Y with PMF $P_{X,Y}(x, y)$ given in Problem 5.8.6, let $W = \min(X, Y)$ and $V = \max(X, Y)$. Find

- (a) The expected values, $E[W]$ and $E[V]$,
 (b) The variances, $\text{Var}[W]$ and $\text{Var}[V]$,
 (c) The correlation, $r_{W,V}$,
 (d) The covariance, $\text{Cov}[W, V]$,
 (e) The correlation coefficient, $\rho_{W,V}$.

5.8.8 Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 1/2 & -1 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find $r_{X,Y}$ and $E[e^{X+Y}]$.

5.8.9 This problem outlines a proof of Theorem 5.13.

- (a) Show that

$$\hat{X} - E[\hat{X}] = a(X - E[X]),$$

$$\hat{Y} - E[\hat{Y}] = c(Y - E[Y]).$$

- (b) Use part (a) to show that

$$\text{Cov}[\hat{X}, \hat{Y}] = ac \text{Cov}[X, Y].$$

- (c) Show that $\text{Var}[\hat{X}] = a^2 \text{Var}[X]$ and $\text{Var}[\hat{Y}] = c^2 \text{Var}[Y]$.

- (d) Combine parts (b) and (c) to relate $\rho_{\hat{X}, \hat{Y}}$ and $\rho_{X,Y}$.

5.8.10 Random variables N and K have the joint PMF

$$P_{N,K}(n, k) = \begin{cases} (1-p)^{n-1}p/n & k = 1, \dots, n; \\ & n = 1, 2, \dots, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PMF $P_N(n)$ and the expected values $E[N]$, $\text{Var}[N]$, $E[N^2]$, $E[K]$, $\text{Var}[K]$, $E[N + K]$, $r_{N,K}$, $\text{Cov}[N, K]$.

5.9.1 Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = ce^{-(x^2/8)-(y^2/18)}.$$

What is the constant c ? Are X and Y independent?

5.9.2 X is the Gaussian ($\mu = 1, \sigma = 2$) random variable. Y is the Gaussian ($\mu = 2, \sigma = 4$) random variable. X and Y are independent.

- (a) What is the PDF of $V = X + Y$?
- (b) What is the PDF of $W = 3X + 2Y$?

5.9.3 TRUE OR FALSE: X_1 and X_2 are bivariate Gaussian random variables. For any constant y , there exists a constant a such that $P[X_1 + aX_2 \leq y] = 1/2$.

5.9.4 X_1 and X_2 are identically distributed Gaussian $(0, 1)$ random variables. Moreover, they are jointly Gaussian. Under what conditions are X_1 , X_2 and $X_1 + X_2$ identically distributed?

5.9.5 Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = ce^{-(2x^2-4xy+4y^2)}.$$

- (a) What are $E[X]$ and $E[Y]$?
- (b) Find the correlation coefficient $\rho_{X,Y}$.
- (c) What are $\text{Var}[X]$ and $\text{Var}[Y]$?
- (d) What is the constant c ?
- (e) Are X and Y independent?

5.9.6 An archer shoots an arrow at a circular target of radius 50 cm. The arrow pierces the target at a random position (X, Y) , measured in centimeters from the center of the disk at position $(X, Y) = (0, 0)$. The bullseye is a solid black circle of radius 2 cm, at the center of the target. Calculate the probability $P[B]$ of the event that the archer hits the bullseye under each of the following models:

- (a) X and Y are iid continuous uniform $(-50, 50)$ random variables.

(b) The PDF $f_{X,Y}(x, y)$ is uniform over the 50 cm circular target.

(c) X and Y are iid Gaussian ($\mu = 0, \sigma = 10$) random variables.

5.9.7 A person's white blood cell (WBC) count W (measured in thousands of cells per microliter of blood) and body temperature T (in degrees Celsius) can be modeled as bivariate Gaussian random variables such that W is Gaussian $(7, 2)$ and T is Gaussian $(37, 1)$. To determine whether a person is sick, first the person's temperature T is measured. If $T > 38$, then the person's WBC count is measured. If $W > 10$, the person is declared ill (event I).

- (a) Suppose W and T are uncorrelated. What is $P[I]$? Hint: Draw a tree diagram for the experiment.
- (b) Now suppose W and T have correlation coefficient $\rho_{W,T} = 1/\sqrt{2}$. Find the conditional probability $P[I|T = t]$ that a person is declared ill given that the person's temperature is $T = t$.

5.9.8 Suppose your grade in a probability course depends on your exam scores X_1 and X_2 . The professor, a fan of probability, releases exam scores in a normalized fashion such that X_1 and X_2 are iid Gaussian ($\mu = 0, \sigma = \sqrt{2}$) random variables. Your semester average is $X = 0.5(X_1 + X_2)$.

- (a) You earn an A grade if $X > 1$. What is $P[A]$?
- (b) To improve his SIRS (Student Instructional Rating Service) score, the professor decides he should award more A 's. Now you get an A if $\max(X_1, X_2) > 1$. What is $P[A]$ now?
- (c) The professor found out he is unpopular at ratemyprofessor.com and decides to award an A if either $X > 1$ or $\max(X_1, X_2) > 1$. Now what is $P[A]$?
- (d) Under criticism of grade inflation from the department chair, the professor adopts a new policy. An A is awarded if $\max(X_1, X_2) > 1$ and $\min(X_1, X_2) > 0$. Now what is $P[A]$?

5.9.9 ♦ Your course grade depends on two test scores: X_1 and X_2 . Your score X_i on test i is Gaussian ($\mu = 74, \sigma = 16$) random variable, independent of any other test score.

- (a) With equal weighting, grades are determined by $Y = X_1/2 + X_2/2$. You earn an A if $Y \geq 90$. What is $P[A] = P[Y \geq 90]$?
- (b) A student asks the professor to choose a weight factor w , $0 \leq w \leq 1$, such that

$$Y = wX_1 + (1 - w)X_2.$$

Find $P[A]$ as a function of the weight w . What value or values of w maximize $P[A] = P[Y \geq 90]$?

- (c) A different student proposes that the better exam is the one that should count and that grades should be based on $M = \max(X_1, X_2)$. In a fit of generosity, the professor agrees! Now what is $P[A] = P[M \geq 90]$?
- (d) How generous was the professor? In a class of 100 students, what is the expected increase in the number of A's awarded?

5.9.10 ♦ Under what conditions on the constants a , b , c , and d is

$$f(x, y) = de^{-(a^2x^2 + bxy + c^2y^2)}$$

a joint Gaussian PDF?

5.9.11 ♦ Show that the joint Gaussian PDF $f_{X,Y}(x, y)$ given by Definition 5.10 satisfies

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dx \, dy = 1.$$

Hint: Use Equation (5.68) and the result of Problem 4.6.13.

5.9.12 ♦ Random variables X_1 and X_2 are independent identical Gaussian $(0, 1)$ random variables. Let

$$Y_1 = X_1 \operatorname{sgn}(X_2), \quad Y_2 = X_2 \operatorname{sgn}(X_1),$$

where

$$\operatorname{sgn}(x) = \begin{cases} 1 & x > 0, \\ -1 & x \leq 0. \end{cases}$$

- (a) Find the CDF $F_{Y_1}(y_1)$ in terms of the $\Phi(\cdot)$ function.
- (b) Show that Y_1 and Y_2 are both Gaussian random variables.
- (c) Are Y_1 and Y_2 bivariate Gaussian random variables?

5.10.1 ● Every laptop returned to a repair center is classified according its needed repairs: (1) LCD screen, (2) motherboard, (3) keyboard, or (4) other. A random broken laptop needs a type i repair with probability $p_i = 2^{4-i}/15$. Let N_i equal the number of type i broken laptops returned on a day in which four laptops are returned.

- (a) Find the joint PMF of N_1, N_2, N_3, N_4 .
- (b) What is the probability that two laptops require LCD repairs?
- (c) What is the probability that more laptops require motherboard repairs than keyboard repairs?

5.10.2 ● When ordering a personal computer, a customer can add the following features to the basic configuration: (1) additional memory, (2) flat panel display, (3) professional software, and (4) wireless modem. A random computer order has feature i with probability $p_i = 2^{-i}$ independent of other features. In an hour in which three computers are ordered, let N_i equal the number of computers with feature i .

- (a) Find the joint PMF

$$P_{N_1, N_2, N_3, N_4}(n_1, n_2, n_3, n_4).$$

- (b) What is the probability of selling a computer with no additional features?
- (c) What is the probability of selling a computer with at least three additional features?

5.10.3 The random variables $X_1, \dots, X_n$ have the joint PDF

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \begin{cases} 1 & 0 \leq x_i \leq 1; \\ & i = 1, \dots, n, \\ 0 & \text{otherwise.} \end{cases}$$

Find

- (a) The joint CDF, $F_{X_1, \dots, X_n}(x_1, \dots, x_n)$,
- (b) $P[\min(X_1, X_2, X_3) \leq 3/4]$.

5.10.4 Are N_1, N_2, N_3, N_4 in Problem 5.10.1 independent?

5.10.5 In a compressed data file of 10,000 bytes, each byte is equally likely to be any one of 256 possible characters $b_0, \dots, b_{255}$ independent of any other byte. If N_i is the number of times b_i appears in the file, find the joint PMF of $N_0, \dots, N_{255}$. Also, what is the joint PMF of N_0 and N_1 ?

5.10.6 In Example 5.22, we derived the joint PMF of the the number of pages in each of four downloads:

$$P_{X,Y,Z}(x, y, z) = \begin{pmatrix} 4 \\ x, y, z \end{pmatrix} \frac{1}{3^x} \frac{1}{2^y} \frac{1}{6^z}.$$

- (a) In a group of four downloads, what is the PMF of the number of 3-page documents?
- (b) In a group of four downloads, what is the expected number of 3-page documents?
- (c) Given that there are two 3-page documents in a group of four, what is the joint PMF of the number of 1-page documents and the number of 2-page documents?
- (d) Given that there are two 3-page documents in a group of four, what is the expected number of 1-page documents?
- (e) In a group of four downloads, what is the joint PMF of the number of 1-page documents and the number of 2-page documents?

5.10.7 X_1, X_2, X_3 are iid exponential (λ) random variables. Find:

- (a) the PDF of $V = \min(X_1, X_2, X_3)$,
- (b) the PDF of $W = \max(X_1, X_2, X_3)$.

5.10.8 In a race of 10 sailboats, the finishing times of all boats are iid Gaussian random variables with expected value 35 minutes and standard deviation 5 minutes.

- (a) What is the probability that the winning boat will finish the race in less than 25 minutes?
- (b) What is the probability that the last boat will cross the finish line in more than 50 minutes?
- (c) Given this model, what is the probability that a boat will finish before it starts (negative finishing time)?

5.10.9 Random variables $X_1, X_2, \dots, X_n$ are iid; each X_j has CDF $F_X(x)$ and PDF $f_X(x)$. Consider

$$L_n = \min(X_1, \dots, X_n) \\ U_n = \max(X_1, \dots, X_n).$$

In terms of $F_X(x)$ and/or $f_X(x)$:

- (a) Find the CDF $F_{U_n}(u)$.
- (b) Find the CDF $F_{L_n}(l)$.
- (c) Find the joint CDF $F_{L_n, U_n}(l, u)$.

5.10.10 Suppose you have n suitcases and suitcase i holds X_i dollars where $X_1, X_2, \dots, X_n$ are iid continuous uniform $(0, m)$ random variables. (Think of a number like one million for the symbol m .) Unfortunately, you don't know X_i until you open suitcase i .

Suppose you can open the suitcases one by one, starting with suitcase n and going down to suitcase 1. After opening suitcase i , you can either accept or reject X_i dollars. If you accept suitcase i , the game ends. If you reject, then you get to choose only from the still unopened suitcases.

What should you do? Perhaps it is not so obvious? In fact, you can decide before the game on a policy, a set of rules to follow. We will specify a policy by a vector $(\tau_1, \dots, \tau_n)$ of threshold parameters.

- After opening suitcase i , you accept the amount X_i if $X_i \geq \tau_i$.
 - Otherwise, you reject suitcase i and open suitcase $i - 1$.
 - If you have rejected suitcases n down through 2, then you must accept the amount X_1 in suitcase 1. Thus the threshold $\tau_1 = 0$ since you never reject the amount in the last suitcase.
- (a) Suppose you reject suitcases n through $i + 1$, but then you accept suitcase i . Find $E[X_i | X_i \geq \tau_i]$.
- (b) Let W_k denote your reward given that there are k unopened suitcases remaining. What is $E[W_1]$?
- (c) As a function of τ_k , find a recursive relationship for $E[W_k]$ in terms of τ_k and $E[W_{k-1}]$.
- (d) For $n = 4$ suitcases, find the policy $(\tau_1^*, \dots, \tau_4^*)$, that maximizes $E[W_4]$.

5.10.11 ♦♦ Given the set $\{U_1, \dots, U_n\}$ of iid uniform $(0, T)$ random variables, we define

$$X_k = \text{small}_k(U_1, \dots, U_n)$$

as the k th “smallest” element of the set. That is, X_1 is the minimum element, X_2 is the second smallest, and so on, up to X_n , which is the maximum element of $\{U_1, \dots, U_n\}$. Note that $X_1, \dots, X_n$ are known as the *order statistics* of $U_1, \dots, U_n$. Prove that

$$\begin{aligned} f_{X_1, \dots, X_n}(x_1, \dots, x_n) \\ = \begin{cases} n!/T^n & 0 \leq x_1 < \dots < x_n \leq T, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

5.11.1 ● For random variables X and Y in Example 5.26, use MATLAB to generate a list of the form

$$\begin{array}{ccc} x_1 & y_1 & P_{X,Y}(x_1, y_1) \\ x_2 & y_2 & P_{X,Y}(x_2, y_2) \\ \vdots & \vdots & \vdots \end{array}$$

that includes all possible pairs (x, y) .

5.11.2 ● For random variables X and Y in Example 5.26, use MATLAB to calculate $E[X]$, $E[Y]$, the correlation $E[XY]$, and the covariance $\text{Cov}[X, Y]$.

5.11.3 ● You generate random variable $W = \mathbf{W}$ by typing `W=sum(4*randn(1,2))` in a MATLAB Command window. What is $\text{Var}[W]$?

5.11.4 ● Write `trianglecdfplot.m`, a script that graphs $F_{X,Y}(x, y)$ of Figure 5.4.

5.11.5 ■ Problem 5.2.6 extended Example 5.3 to a test of n circuits and identified the joint PDF of X , the number of acceptable circuits, and Y , the number of successful tests before the first reject. Write a MATLAB function

$$[\mathbf{SX}, \mathbf{SY}, \mathbf{PXY}] = \text{circuits}(n, p)$$

that generates the sample space grid for the n circuit test. Check your answer against Equation (5.11) for the $p = 0.9$ and $n = 2$ case. For $p = 0.9$ and $n = 50$, calculate the correlation coefficient $\rho_{X,Y}$.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

6.1.1 ● Random variables X and Y have joint PMF

$$P_{X,Y}(x,y) = \begin{cases} |x+y|/14 & x = -2, 0, 2; \\ & y = -1, 0, 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the PMF of $W = X - Y$.

6.1.2 ● For random variables X and Y in Problem 6.1.1, find the PMF of $W = X + 2Y$.

6.1.3 ● N is a binomial ($n = 100, p = 0.4$) random variable. M is a binomial ($n = 50, p = 0.4$) random variable. Given that M and N are independent, what is the PMF of $L = M + N$?

6.1.4 ● Let X and Y be discrete random variables with joint PMF $P_{X,Y}(x,y)$ that is zero except when x and y are integers. Let $W = X + Y$ and show that the PMF of W satisfies

$$P_W(w) = \sum_{x=-\infty}^{\infty} P_{X,Y}(x, w-x).$$

6.1.5 ■ Let X and Y be discrete random variables with joint PMF

$$P_{X,Y}(x,y) = \begin{cases} 0.01 & x = 1, 2, \dots, 10, \\ & y = 1, 2, \dots, 10, \\ 0 & \text{otherwise.} \end{cases}$$

What is the PMF of $W = \min(X, Y)$?

6.1.6 ■ For random variables X and Y in Problem 6.1.5, what is the PMF of $V = \max(X, Y)$?

6.2.1 ● The voltage X across a $1\ \Omega$ resistor is a uniform random variable with parameters 0 and 1. The instantaneous power is $Y = X^2$. Find the CDF $F_Y(y)$ and the PDF $f_Y(y)$ of Y .

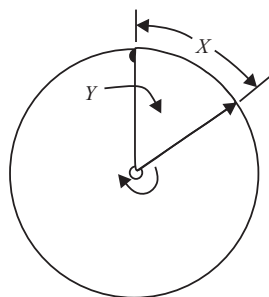
6.2.2 ● X is the Gaussian $(0, 1)$ random variable. Find the CDF of $Y = |X|$ and its expected value $E[Y]$.

6.2.3 ■ In a 50 km Tour de France time trial, a rider's time T , measured in minutes, is the continuous uniform $(60, 75)$ random variable. Let $V = 3000/T$ denote the rider's speed over the course in km/hr. Find the PDF of V .

6.2.4 ● In the presence of a headwind of normalized intensity W , your speed on your bike is $V = g(W) = 20 - 10W^{1/3}$ mi/hr. The wind intensity W is the continuous uniform $(-1, 1)$ random variable. (Note: If W is negative, then the headwind is actually a tailwind.) Find the PDF $f_V(v)$.

6.2.5 ● If X has an exponential (λ) PDF, what is the PDF of $W = X^2$?

6.2.6 ■ Let X denote the position of the pointer after a spin on a wheel of circumference 1. For that same spin, let Y denote the area within the arc defined by the stopping position of the pointer:



(a) What is the relationship between X and Y ?

(b) What is $F_Y(y)$?

(c) What is $f_Y(y)$?

(d) What is $E[Y]$?

6.2.7 ■ U is the uniform $(0, 1)$ random variable and $X = -\ln(1 - U)$.

(a) What is $F_X(x)$?

(b) What is $f_X(x)$?

(c) What is $E[X]$?

6.2.8 ■ X is the uniform $(0, 1)$ random variable. Find a function $g(x)$ such that the PDF of $Y = g(X)$ is

$$f_Y(y) = \begin{cases} 3y^2 & 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

6.2.9 ■ An amplifier circuit has power consumption Y that grows nonlinearly with the input signal voltage X . When the input signal is X volts, the instantaneous power consumed by the amplifier is $Y = 20 + 15X^2$ Watts. The input signal X is the continuous uniform $(-1, 1)$ random variable. Find the PDF $f_Y(y)$.

6.2.10 ■ Use Theorem 6.2 to prove Theorem 6.3.

6.2.11 ■ For the uniform $(0, 1)$ random variable U , find the CDF and PDF of $Y = a + (b - a)U$ with $a < b$. Show that Y is the uniform (a, b) random variable.

6.2.12 ■ Theorem 6.5 required the inverse CDF $F^{-1}(u)$ to exist for $0 < u < 1$. Why was it *not* necessary that $F^{-1}(u)$ exist at either $u = 0$ or $u = 1$?

6.2.13 ♦ X is a continuous random variable. $Y = aX + b$, where $a, b \neq 0$. Prove that

$$f_Y(y) = \frac{f_X((y - b)/a)}{|a|}.$$

Hint: Consider the cases $a < 0$ and $a > 0$ separately.

6.2.14 ♦ Let continuous random variable X have a CDF $F(x)$ such that $F^{-1}(u)$ exists for all u in $[0, 1]$. Show that $U = F(X)$ is the uniform $(0, 1)$ random variable. Hint: U is a random variable such that when $X = x'$, $U = F(x')$. That is, we evaluate the CDF of X at the observed value of X .

6.3.1 ■ X has CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ x/3 + 1/3 & -1 \leq x < 0, \\ x/3 + 2/3 & 0 \leq x < 1, \\ 1 & 1 \leq x. \end{cases}$$

$Y = g(X)$ where

$$g(X) = \begin{cases} 0 & X < 0, \\ 100 & X \geq 0. \end{cases}$$

- What is $F_Y(y)$?
- What is $f_Y(y)$?
- What is $E[Y]$?

6.3.2 ■ In a 50 km cycling time trial, a rider's exact time T , measured in minutes, is the continuous uniform $(50, 60)$ random variable. However, a rider's recorded time R in seconds is obtained by rounding *up* T to next whole second. That is, if T is 50 minutes, 27.001 seconds, then $R = 3028$ seconds. On the other hand, if T is exactly 50 minutes 27 seconds, then $R = 3027$. What is the PMF of R ?

6.3.3 ■ The voltage V at the output of a microphone is the continuous uniform $(-1, 1)$ random variable. The microphone voltage is processed by a clipping rectifier with output

$$L = \begin{cases} |V| & |V| \leq 0.5, \\ 0.5 & \text{otherwise.} \end{cases}$$

- What is $P[L = 0.5]$?
- What is $F_L(l)$?
- What is $E[L]$?

6.3.4 ■ U is the uniform random variable with parameters 0 and 2. The random variable W is the output of the clipper:

$$W = g(U) = \begin{cases} U & U \leq 1, \\ 1 & U > 1. \end{cases}$$

Find the CDF $F_W(w)$, the PDF $f_W(w)$, and the expected value $E[W]$.

6.3.5 ■ X is a random variable with CDF by $F_X(x)$. Let $Y = g(X)$ where

$$g(x) = \begin{cases} 10 & x < 0, \\ -10 & x \geq 0. \end{cases}$$

Express $F_Y(y)$ in terms of $F_X(x)$.

6.3.6 ■ Suppose that a cellular phone costs \$30 per month with 300 minutes of use included and that each additional minute of use costs \$0.50. The number of minutes you use the phone in a month is an exponential random variable T with expected value $E[T] = 200$ minutes. The telephone company charges you for exactly how many minutes you use without any rounding of fractional minutes. Let C denote the cost in dollars of one month of service.

- What is $P[C = 30]$?
- What is the PDF of C ?
- What is $E[C]$?

6.3.7 ■ The input voltage to a rectifier is the continuous uniform $(0, 1)$ random variable U . The rectifier output is a random variable W defined by

$$W = g(U) = \begin{cases} 0 & U < 0, \\ U & U \geq 0. \end{cases}$$

Find the CDF $F_W(w)$ and the expected value $E[W]$.

6.3.8 ■ Random variable X has PDF

$$f_X(x) = \begin{cases} x/2 & 0 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

X is processed by a clipping circuit with output

$$Y = \begin{cases} 0.5 & X \leq 1, \\ X & X > 1. \end{cases}$$

- What is $P[Y = 0.5]$?
- Find the CDF $F_Y(y)$.

6.3.9 ■ Given an input voltage V , the output voltage of a half-wave rectifier is given

$$W = \begin{cases} 0 & V \leq 0, \\ V & 0 < V < 10, \\ 10 & V \geq 10. \end{cases}$$

Suppose the input V is the continuous uniform $(-15, 15)$ random variable. Find the PDF of W .

6.3.10 ■ The current X across a resistor is the continuous uniform $(-2, 2)$ random variable. The power dissipated in the resistor is $Y = 9X^2$ Watts.

- Find the CDF and PDF of Y .
- A power measurement circuit is range-limited so that its output is

$$W = \begin{cases} Y & Y < 16, \\ 16 & \text{otherwise.} \end{cases}$$

Find the PDF of W .

6.3.11 ♦ A defective voltmeter measures small voltages as zero. In particular, when the input voltage is V , the measured voltage is

$$W = \begin{cases} 0 & |V| < 0.6, \\ V & \text{otherwise.} \end{cases}$$

If V is the continuous uniform $(-5, 5)$ random variable, what is the PDF of W ?

6.3.12 ♦ X is the continuous uniform $(-3, 3)$ random variable. When X is passed through a limiter, the output is the discrete random variable

$$\hat{X} = g(X) = \begin{cases} -c & X < 0 \\ c & X \geq 0 \end{cases}$$

where c is an unspecified positive constant.

- What is the PMF $P_{\hat{X}}(x)$ of $\hat{X}$?
- When the limiter input is X , the distortion D between the input X and the limiter output $\hat{X}$ is

$$D = d(X) = (X - g(X))^2.$$

In terms of c , find the expected distortion $E[D] = E[d(X)]$. What value of c minimizes $E[D]$?

- (c) Y is a Gaussian random variable with the same expected value and variance as X . What is the PDF of Y ?
- (d) Suppose Y is passed through the limiter yielding the output $\hat{Y} = g(Y)$. The distortion D between the input Y and the limiter output $\hat{Y}$ is

$$D = d(Y) = (Y - g(Y))^2.$$

In terms of c , find the expected distortion $E[D] = E[d(Y)]$. What value of c minimizes $E[D]$?

6.3.13 ♦♦ In this problem we prove a generalization of Theorem 6.5. Given a random variable X with CDF $F_X(x)$, define

$$\tilde{F}(u) = \min \{x | F_X(x) \geq u\}.$$

This problem proves that for a continuous uniform $(0, 1)$ random variable U , $\hat{X} = \tilde{F}(U)$ has CDF $F_{\hat{X}}(x) = F_X(x)$.

- (a) Show that when $F_X(x)$ is a continuous, strictly increasing function (i.e., X is not mixed, $F_X(x)$ has no jump discontinuities, and $F_X(x)$ has no “flat” intervals (a, b) where $F_X(x) = c$ for $a \leq x \leq b$), then $\tilde{F}(u) = F_X^{-1}(u)$ for $0 < u < 1$.
- (b) Show that if $F_X(x)$ has a jump at $x = x_0$, then $\tilde{F}(u) = x_0$ for all u in the interval

$$F_X(x_0^-) \leq u \leq F_X(x_0^+).$$

- (c) Prove that $\hat{X} = \tilde{F}(U)$ has CDF $F_{\hat{X}}(x) = F_X(x)$.

6.4.1 ● Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6xy^2 & 0 \leq x, y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $V = \max(X, Y)$. Find the CDF and PDF of V .

6.4.2 ● For random variables X and Y in Problem 6.4.1, find the CDF and PDF of $W = \min(X, Y)$.

6.4.3 ■ X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & x \geq 0, y \geq 0, x + y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Are X and Y independent?
- (b) Let $U = \min(X, Y)$. Find the CDF and PDF of U .
- (c) Let $V = \max(X, Y)$. Find the CDF and PDF of V .

6.4.4 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} x + y & 0 \leq x, y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $W = \max(X, Y)$.

- (a) What is S_W , the range of W ?
- (b) Find $F_W(w)$ and $f_W(w)$.

6.4.5 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6y & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $W = Y - X$.

- (a) What is S_W , the range of W ?
- (b) Find $F_W(w)$ and $f_W(w)$.

6.4.6 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $W = Y/X$.

- (a) What is S_W , the range of W ?
- (b) Find $F_W(w)$, $f_W(w)$, and $E[W]$.

6.4.7 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $W = X/Y$.

- (a) What is S_W , the range of W ?
- (b) Find $F_W(w)$, $f_W(w)$, and $E[W]$.

6.4.8 ■ In a simple model of a cellular telephone system, a portable telephone is equally likely to be found anywhere in a circular cell of radius 4 km. (See Problem 5.5.4.) Find the CDF $F_R(r)$ and PDF $f_R(r)$ of R , the distance (in km) between the telephone and the base station at the center of the cell.

6.4.9 ■ X and Y are independent identically distributed Gaussian $(0, 1)$ random variables. Find the CDF of $W = X^2 + Y^2$.

6.4.10 ■ X is the exponential (2) random variable and Z is the Bernoulli $(1/2)$ random variable that is independent of X . Find the PDF of $Y = ZX$.

6.4.11 ♦ X is the Gaussian $(0, 1)$ random variable and Z , independent of X , has PMF

$$P_Z(z) = \begin{cases} 1-p & z = -1, \\ p & z = 1. \end{cases}$$

Find the PDF of $Y = ZX$.

6.4.12 ♦ You are waiting on the platform of the first stop of a Manhattan subway line. You could ride either a local or express train to your destination, which is the last stop on the line. The waiting time X for the next express train is the exponential random variable with $E[X] = 10$ minutes. The waiting time Y for the next local train is the exponential random variable with $E[Y] = 5$ minutes. Although the arrival times X and Y of the trains are random and independent, the trains' travel times are deterministic; the local train travels from first stop to last stop in exactly 15 minutes while the express travels from first to last stop in exactly 5 minutes.

- (a) What is the joint PDF $f_{X,Y}(x, y)$?
- (b) Find $P[L]$ that the local train arrives first at the platform?
- (c) Suppose you board the first train that arrives. Find the PDF of your waiting time $W = \min(X, Y)$.

- (d) The time until the first train (express or local) reaches final stop is $T = \min(X + 5, Y + 15)$. Find $f_T(t)$.
- (e) Suppose the local train does arrive first at your platform. Should you board the local train? Justify your answer. (There may be more than one correct answer.)

6.4.13 ♦ For a constant $a > 0$, random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 1/a^2 & 0 \leq x, y \leq a, \\ 0 & \text{otherwise.} \end{cases}$$

Find the CDF and PDF of random variable

$$W = \max\left(\frac{X}{Y}, \frac{Y}{X}\right).$$

Hint: Is it possible to observe $W < 1$?

6.4.14 ♦ The joint PDF of X and Y is

$$f_{X,Y}(x, y) = \begin{cases} \lambda^2 e^{-\lambda y} & 0 \leq x < y, \\ 0 & \text{otherwise.} \end{cases}$$

What is the PDF of $W = Y - X$?

6.4.15 ♦♦ Consider random variables X , Y , and W from Problem 6.4.14.

- (a) Are W and X independent?
- (b) Are W and Y independent?

6.4.16 ♦♦ X and Y are independent random variables with CDFs $F_X(x)$ and $F_Y(y)$. Let $U = \min(X, Y)$ and $V = \max(X, Y)$.

- (a) What is $F_{U,V}(u, v)$?
- (b) What is $f_{U,V}(u, v)$?

Hint: To find the joint CDF, let $A = \{U \leq u\}$ and $B = \{V \leq v\}$ and note that $P[AB] = P[B] - P[A^c B]$.

6.5.1 ● Let X and Y be independent discrete random variables such that $P_X(k) = P_Y(k) = 0$ for all non-integer k . Show that the PMF of $W = X + Y$ satisfies

$$P_W(w) = \sum_{k=-\infty}^{\infty} P_X(k) P_Y(w - k).$$

6.5.2 ● X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & x \geq 0, y \geq 0, x+y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the PDF of $W = X + Y$.

6.5.3 ■ Find the PDF of $W = X + Y$ when X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

6.5.4 ■ Find the PDF of $W = X + Y$ when X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 1 & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

6.5.5 ■ Random variables X and Y are independent exponential random variables with expected values $E[X] = 1/\lambda$ and $E[Y] = 1/\mu$. If $\mu \neq \lambda$, what is the PDF of $W = X + Y$? If $\mu = \lambda$, what is $f_W(w)$?

6.5.6 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 8xy & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

What is the PDF of $W = X + Y$?

6.5.7 ■ Continuous random variables X and Y have joint PDF $f_{X,Y}(x,y)$. Show that $W = X - Y$ has PDF

$$f_W(w) = \int_{-\infty}^{\infty} f_{X,Y}(y+w, y) dy.$$

Use a variable substitution to show

$$f_W(w) = \int_{-\infty}^{\infty} f_{X,Y}(x, x-w) dx.$$

6.5.8 ♦ In this problem we show directly that the sum of independent Poisson random variables is Poisson. Let J and K be independent Poisson random variables with

expected values α and β , respectively, and show that $N = J + K$ is a Poisson random variable with expected value $\alpha + \beta$. Hint: Show that

$$P_N(n) = \sum_{m=0}^n P_K(m) P_J(n-m),$$

and then simplify the summation by extracting the sum of a binomial PMF over all possible values.

6.6.1 ■ Use `icdfrv.m` to write a function `w=wrw1(m)` that generates m samples of random variable W from Problem 4.2.4. Note that $F_W^{-1}(u)$ does not exist for $u = 1/4$; however, you must define a function `icdfw(u)` that returns a value for `icdfw(0.25)`. Does it matter what value you return for `u=0.25`?

6.6.2 ■ Write a MATLAB function `u=urw(m)` that generates m samples of random variable U defined in Problem 4.4.7.

6.6.3 ■ For random variable W of Example 6.10, we can generate random samples in two different ways:

1. Generate samples of X and Y and calculate $W = Y/X$.
2. Find the CDF $F_W(w)$ and generate samples using Theorem 6.5.

Write MATLAB functions `w=wrw1(m)` and `w=wrw2(m)` to implement these methods. Does one method run much faster? If so, why? (Use `cputime` to make comparisons.)

6.6.4 ♦ Write a function `y=deltarv(m)` that returns m samples of the random variable X with PDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ (x+1)/4 & -1 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

Since $F_X^{-1}(u)$ is not defined for $1/2 \leq u < 1$, use the result of Problem 6.3.13.

Example 7.22

Write a function `xy = xytrianglererv(m)` that generates m sample pairs (X, Y) in Example 7.16.

In Example 7.16, we found that

$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1, \\ 0 & \text{otherwise,} \end{cases} \quad f_{Y|X}(y|x) = \begin{cases} 1/x & 0 \leq y \leq x, \\ 0 & \text{otherwise.} \end{cases} \quad (7.73)$$

```
function xy = xytrianglererv(m);
x=sqrt(rand(m,1));
y=x.*rand(m,1);
xy=[x y];
```

For $0 \leq x \leq 1$, we have that $F_X(x) = x^2$. Using Theorem 6.5 to generate sample values of X , we define $u = F_X(x) = x^2$. Then, for $0 < u < 1$, $x = \sqrt{u}$. By Theorem 6.5, if U is uniform $(0, 1)$, then $\sqrt{U}$ has PDF $f_X(x)$. Next, we observe that

given $X = x_i$, Y is the uniform $(0, x_i)$ random variable. Given another uniform $(0, 1)$ random variable U_i , Theorem 6.3(a) states that $Y_i = x_i U_i$ is the uniform $(0, x_i)$ random variable. We implement these ideas in the function `xytrianglererv.m`.

Quiz 7.7

For random variables X and Y with joint PMF $P_{X,Y}(x, y)$ given in Example 7.11, write a MATLAB function `xy=dtrianglererv(m)` that generates m sample pairs.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

7.1.1 ● Random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < -3, \\ 0.4 & -3 \leq x < 5, \\ 0.8 & 5 \leq x < 7, \\ 1 & x \geq 7. \end{cases}$$

Find the conditional CDF $F_{X|X>0}(x)$ and PMF $P_{X|X>0}(x)$.

7.1.2 ● X is the discrete uniform $(0, 5)$ random variable. What is $E[X|X \geq E[X]]$?

7.1.3 ● X has PMF

$$P_X(x) = \binom{4}{x} (1/2)^4.$$

Find $P_{X|B}(x)$ where $B = \{X \neq 0\}$.

7.1.4 ● In a youth basketball league, a player is fouled in the act of shooting a layup. There is a probability $q = 0.2$ that the layup is good, scoring 2 points. If the layup is good, the player is also awarded 1 free throw, giving the player a chance at a three-point play. If the layup is missed, then (because of the foul) the player is still awarded one point automatically and is also awarded one free throw, enabling a chance to score two points in total. The player makes a free throw with probability $p = 1/2$.

- What is the PMF of X , the number of points scored by the player?
- Find the conditional PMF $P_{X|T}(x)$ of X given event T that the free throw is good.

7.1.5 ■ Every day you consider going jogging. Before each mile, including the first, you will quit with probability q , independent of the number of miles you have already run. However, you are sufficiently decisive that you never run a fraction of a mile. Also, we say you have run a marathon whenever you run at least 26 miles.

- Let M equal the number of miles that you run on an arbitrary day. Find the PMF $P_M(m)$.
- Let r be the probability that you run a marathon on an arbitrary day. Find r .
- Let J be the number of days in one year (not a leap year) in which you run a marathon. Find the PMF $P_J(j)$. This answer may be expressed in terms of r found in part (b).
- Define $K = M - 26$. Let A be the event that you have run a marathon. Find $P_{K|A}(k)$.

7.1.6 ■ A random ECE student has height X in inches given by the PDF

$$f_X(x) = \frac{4e^{-(x-70)^2/8} + e^{-(x-65)^2/8}}{5\sqrt{8\pi}}.$$

- Sketch $f_X(x)$ over the interval $60 \leq x \leq 75$. (For purposes of sketching, note that $\sqrt{8\pi} \approx 5$.)
- Find the probability that a random ECE student is less than 5 feet 8 inches tall.
- Use conditional PDFs to explain why $f_X(x)$ might be a reasonable model for ECE students.

7.1.7 ■ A test for diabetes is a measurement X of a person's blood sugar level following an overnight fast. For a healthy person, a blood sugar level X in the range of 70–110 mg/dl is considered normal. When a measurement X is used as a test for diabetes, the result is called positive (event T^+) if $X \geq 140$; the test is negative (event T^-) if $X \leq 110$, and the test is ambiguous (event T^0) if $110 < X < 140$.

Given that a person is healthy (event H), a blood sugar measurement X is the Gaussian $(90, 20)$ random variable. Given that a person has diabetes, (event D), X is the Gaussian $(60, 40)$ random variable. A randomly chosen person is healthy with probability $P[H] = 0.9$ or has diabetes with probability $P[D] = 0.1$.

- What is the conditional PDF $f_{X|H}(x)$?
- Calculate the conditional probabilities $P[T^+|H]$, and $P[T^-|H]$.
- Find $P[H|T^-]$, the conditional probability that a person is healthy given the event of a negative test.
- When a person has an ambiguous test result (T^0), the test is repeated, possibly many times, until either a positive T^+ or negative T^- result is obtained. Let N denote the number of times the test is given. Assuming that for a given person the result of each test is independent of the result of all other tests, find the conditional PMF of N given event H that a person is healthy. Note that $N = 1$ if the person has a positive T^+ or negative T^- result on the first test.

7.1.8 ■ For the quantizer of Example 7.6, the difference $Z = X - Y$ is the quantization error or quantization “noise.” As in Example 7.6, assume that X has a uniform $(-r/2, r/2)$ PDF.

- Given event B_i that $Y = y_i = \Delta/2 + i\Delta$ and X is in the i th quantization interval, find the conditional PDF of Z .
- Show that Z is a uniform random variable. Find the PDF, the expected value, and the variance of Z .

7.1.9 ♦ For the quantizer of Example 7.6, we showed in Problem 7.1.8 that the quantization noise Z is a uniform random variable. If X is not uniform, show that Z is nonuniform by calculating the PDF of Z for a simple example.

7.2.1 ● X is the binomial $(5, 1/2)$ random variable. Find $P_{X|B}(x)$, where the condi-

tion $B = \{X \geq \mu_X\}$. What are $E[X|B]$ and $\text{Var}[X|B]$?

7.2.2 ● Random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ 0.2 & -1 \leq x < 0, \\ 0.7 & 0 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

Given $B = \{|X| > 0\}$, find $P_{X|B}(x)$. What are $E[X|B]$ and $\text{Var}[X|B]$?

7.2.3 ● X is the continuous uniform $(-5, 5)$ random variable. Given the event $B = \{|X| \leq 3\}$, find the

- conditional PDF, $f_{X|B}(x)$,
- conditional expected value, $E[X|B]$,
- conditional variance, $\text{Var}[X|B]$.

7.2.4 ● Y is the exponential (0.2) random variable. Given $A = \{Y < 2\}$, find:

- $f_{Y|A}(y)$,
- $E[Y|A]$.

7.2.5 ● For the experiment of spinning the pointer three times and observing the maximum pointer position, Example 4.5, find the conditional PDF given the event R that the maximum position is on the right side of the circle. What are the conditional expected value and the conditional variance?

7.2.6 ■ The number of pages X in a document has PMF

$$P_X(x) = \begin{cases} 0.15 & x = 1, 2, 3, 4, \\ 0.1 & x = 5, 6, 7, 8, \\ 0 & \text{otherwise.} \end{cases}$$

A firm sends all documents with an even number of pages to printer A and all documents with an odd number of pages to printer B .

- Find the conditional PMF of the length X of a document, given the document was sent to A . What are the conditional expected length and standard deviation?

- Find the conditional PMF of the length X of a document, given the document was sent to B and had no more than six pages. What are the conditional expected length and standard deviation?

7.2.7 ■ Select integrated circuits, test them in sequence until you find the first failure, and then stop. Let N be the number of tests. All tests are independent, with probability of failure $p = 0.1$. Consider the condition $B = \{N \geq 20\}$.

- Find the PMF $P_N(n)$.
- Find $P_{N|B}(n)$, the conditional PMF of N given that there have been 20 consecutive tests without a failure.
- What is $E[N|B]$, the expected number of tests given that there have been 20 consecutive tests without a failure?

7.2.8 ■ W is the Gaussian $(0, 4)$ random variable. Given the event $C = \{W > 0\}$, find the conditional PDF, $f_{W|C}(w)$, the conditional expected value, $E[W|C]$, and the conditional variance, $\text{Var}[W|C]$.

7.2.9 ■ The time between telephone calls at a telephone switch is the exponential random variable T with expected value 0.01.

- What is $E[T|T > 0.02]$, the conditional expected value of T ?
- What is $\text{Var}[T|T > 0.02]$, the conditional variance of T ?

7.2.10 ■ As the final rider in the final 60 km time trial of the Tour de France, Roy must finish in time $T \leq 1$ hour to win the Tour. He has the choice of bike made of (1) carbon fiber or (2) titanium. On the carbon fiber bike, his speed V over the course is the continuous uniform random variable with $E[V] = 58$ km/hr and $\text{Var}[V] = 12$. On the titanium bike, V is the exponential random variable with $E[V] = 60$ km/hr.

- Roy chooses his bike to maximize $P[W]$, the probability he wins the Tour. Which bike does Roy choose and what is $P[W]$?
- Suppose instead that Roy flips a fair coin to choose his bike. What is $P[W]$?

7.2.11 ■ For the distance D of a shot-put toss in Problem 4.7.8, find the conditional PDFs $f_{D|D>0}(d)$ and $f_{D|D\leq 70}(d)$.

7.3.1 ● X and Y are independent identical discrete uniform $(1, 10)$ random variables. Let A denote the event that $\min(X, Y) > 5$. Find the conditional PMF $P_{X,Y|A}(x, y)$.

7.3.2 ● Continuing Problem 7.3.1, let B denote the event that $\max(X, Y) \leq 5$. Find the conditional PMF $P_{X,Y|B}(x, y)$.

7.3.3 ● Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6e^{-(2x+3y)} & x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Let A be the event that $X + Y \leq 1$. Find the conditional PDF $f_{X,Y|A}(x, y)$.

7.3.4 ■ N and K have joint PMF

$$P_{N,K}(n, k) = \begin{cases} \frac{(1-p)^{n-1}p}{n} & n=1, 2, \dots, \\ 0 & k=1, \dots, n, \\ & \text{otherwise.} \end{cases}$$

Let B denote the event that $N \geq 10$.

- Find the conditional PMFs $P_{N|B}(n)$ and $P_{N,K|B}(n, k)$. Which should you find first?
- Find the conditional expected values $E[N|B]$, $E[K|B]$, $E[N + K|B]$, $\text{Var}[N|B]$, $\text{Var}[K|B]$, $E[NK|B]$.

7.3.5 ■ X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} (x+y)/3 & 0 \leq x \leq 1; \\ & 0 \leq y \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

Let $A = \{Y \leq 1\}$.

- What is $P[A]$?
- Find $f_{X,Y|A}(x, y)$.
- Find $f_{X|A}(x)$ and $f_{Y|A}(y)$.

7.3.6 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} (4x + 2y)/3 & 0 \leq x \leq 1; \\ & 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $A = \{Y \leq 1/2\}$.

- What is $P[A]$?
- Find $f_{X,Y|A}(x, y)$.
- Find $f_{X|A}(x)$, and $f_{Y|A}(y)$.

7.3.7 ■ A study examined whether there was correlation between how much football a person watched and how bald the person was. The time T watching football was measured on a 0, 1, 2 scale such that $T = 0$ if a person never watched football, $T = 1$ if a person watched football occasionally, and $T = 2$ if a person watched a lot of football. Similarly, baldness B was measured on the same scale: $B = 0$ for a person with a full head of hair, $B = 1$ for a person with thinning hair, and $B = 2$ for a person who has not much hair at all. The experiment was to learn B and T for a randomly chosen person, equally likely to be a man (event M) or a woman (event W). The study found that given a person was a man (event M), random variables B and T were conditionally independent. Similarly, given that a person was a woman (event W), B and T were conditionally independent. Moreover, B and T had conditional joint PMFs

b	0	1	2
$P_{B M}(b)$	0.2	0.3	0.5
t	0	1	2
$P_{T M}(t)$	0.2	0.2	0.6
b	0	1	2
$P_{B W}(b)$	0.6	0.3	0.1
t	0	1	2
$P_{T W}(t)$	0.6	0.2	0.2

- Find the conditional PMF $P_{B,T|W}(b, t)$ of B and T given that a person is a woman.
- Find the conditional PMF $P_{B,T|M}(b, t)$ of B and T given that a person is a man.
- Find the joint PMF $P_{B,T}(b, t)$.
- Find the covariance of B and T . Are B and T independent?

7.3.8 ♦ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 5x^2/2 & -1 \leq x \leq 1; \\ & 0 \leq y \leq x^2, \\ 0 & \text{otherwise.} \end{cases}$$

Let $A = \{Y \leq 1/4\}$.

- Find the conditional PDF $f_{X,Y|A}(x,y)$.
- Find $f_{Y|A}(y)$ and $E[Y|A]$.
- Find $f_{X|A}(x)$ and $E[X|A]$.

7.3.9 ♦ X and Y are independent random variables with PDFs

$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$f_Y(y) = \begin{cases} 3y^2 & 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $A = \{X > Y\}$.

- What are $E[X]$ and $E[Y]$?
- What are $E[X|A]$ and $E[Y|A]$?

7.4.1 ● Given $X = x$,

- Y_1 is Gaussian with conditional expected value x and conditional variance 1.
- Y_2 is Gaussian with conditional expected value x and conditional variance x^2 .

Find the conditional PDFs $f_{Y_1|X}(y_1|x)$ and $f_{Y_2|X}(y_2|x)$.

7.4.2 ● X is the continuous uniform $(0,1)$ random variable. Given $X = x$, Y has a continuous uniform $(0,x)$ PDF. What is the joint PDF $f_{X,Y}(x,y)$? Sketch the region of the X,Y plane for which $f_{X,Y}(x,y) > 0$.

7.4.3 ● X is the continuous uniform $(0,1)$ random variable. Given $X = x$, Y is conditionally a continuous uniform $(0,1+x)$ random variable. What is the joint PDF $f_{X,Y}(x,y)$ of X and Y ?

7.4.4 ● Z is a Gaussian $(0,1)$ noise random variable that is independent of X , and

$Y = X + Z$ is a noisy observation of X . What is the conditional PDF $f_{Y|X}(y|x)$?

7.4.5 ● A business trip is equally likely to take 2, 3, or 4 days. After a d -day trip, the change in the traveler's weight, measured as an integer number of pounds, is a uniform $(-d,d)$ random variable. For one such trip, denote the number of days by D and the change in weight by W . Find the joint PMF $P_{D,W}(d,w)$.

7.4.6 ● X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} (4x+2y)/3 & 0 \leq x \leq 1; \\ & 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- For which values of y is $f_{X|Y}(x|y)$ defined? What is $f_{X|Y}(x|y)$?
- For which values of x is $f_{Y|X}(y|x)$ defined? What is $f_{Y|X}(y|x)$?

7.4.7 ● A student's final exam grade depends on how close the student sits to the center of the classroom during lectures. If a student sits r feet from the center of the room, the grade is a Gaussian random variable with expected value $80 - r$ and standard deviation r . If r is a sample value of random variable R , and X is the exam grade, what is $f_{X|R}(x|r)$?

7.4.8 ■ $Y = ZX$ where X is the Gaussian $(0,1)$ random variable and Z , independent of X , has PMF

$$P_Z(z) = \begin{cases} 1-p & z = -1, \\ p & z = 1. \end{cases}$$

True or False:

- Y and Z are independent.
- Y and X are independent.

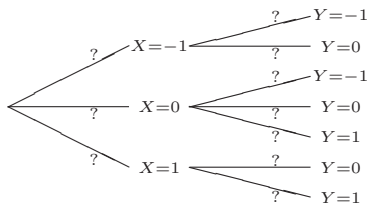
7.4.9 ■ At the One Top Pizza Shop, mushrooms are the only topping. Curiously, a pizza sold before noon has mushrooms with probability $p = 1/3$ while a pizza sold after noon never has mushrooms. Also, a pizza is equally likely to be sold before noon as

after noon. On a day in which 100 pizzas are sold, let N equal the number of pizzas sold before noon and let M equal the number of mushroom pizzas sold during the day. What is the joint PMF $P_{M,N}(m,n)$? Are M and N independent? Hint: Find the conditional PMF of M given N .

7.4.10 ■ Random variables X and Y have the joint PMF in the following table.

$P_{X,Y}(x,y)$	$y = -1$	$y = 0$	$y = 1$
$x = -1$	3/16	1/16	0
$x = 0$	1/6	1/6	1/6
$x = 1$	0	1/8	1/8

- Are X and Y independent?
- The experiment from which X and Y are derived is performed sequentially. First, X is found, then Y is found. In this context, label the conditional branch probabilities of the following tree:



7.4.11 ■ Flip a coin twice. On each flip, the probability of heads equals p . Let X_i equal the number of heads (either 0 or 1) on flip i . Let $W = X_1 - X_2$ and $Y = X_1 + X_2$. Find $P_{W,Y}(w,y)$, $P_{W|Y}(w|y)$, and $P_{Y|W}(y|w)$.

7.4.12 ■ Show that

$$\begin{aligned} \lim_{\Delta \rightarrow 0} \mathbb{P}[x_1 < X \leq x_2 | y < Y \leq y + \Delta] \\ = \int_{x_1}^{x_2} f_{X|Y}(x|y) \, dx. \end{aligned}$$

Hint: $\mathbb{P}[x_1 < X \leq x_2, y < Y \leq y + \Delta]$ can be written as an integral of $f_{X,Y}(x,y)$.

7.4.13 ♦ Packets arriving at an Internet router are either voice packets (v) or data packets (d). Each packet is a voice packet

with probability p , independent of any other packet. Observe packets at the Internet router until you see two voice packets. Let M equal the number of packets up to and including the first voice packet. Let N equal the number of packets observed up to and including the second voice packet. Find the conditional PMFs $P_{M|N}(m|n)$ and $P_{N|M}(n|m)$. Interpret your results.

7.4.14 ♦♦ Suppose you arrive at a bus stop at time 0, and at the end of each minute, with probability p , a bus arrives, or with probability $1 - p$, no bus arrives. Whenever a bus arrives, you board that bus with probability q and depart. Let T equal the number of minutes you stand at a bus stop. Let N be the number of buses that arrive while you wait at the bus stop.

- Identify the set of points (n,t) for which $\mathbb{P}[N = n, T = t] > 0$.
- Find $P_{N,T}(n,t)$.
- Find the marginal PMFs $P_N(n)$ and $P_T(t)$.
- Find the conditional PMFs $P_{N|T}(n|t)$ and $P_{T|N}(t|n)$.

7.4.15 ♦♦ Each millisecond at an Internet router, a packet independently arrives with probability p . Each packet is either a data packet (d) with probability q or a video packet (v). Each data packet belongs to an email with probability r . Let N equal the number of milliseconds required to observe the first 100 email packets. Let T equal the number of milliseconds you observe the router waiting for the first email packet. Find the marginal PMF $P_T(t)$ and the conditional PMF $P_{N|T}(n|t)$. Lastly, find the conditional PMF $P_{T|N}(t|n)$.

7.5.1 ● X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & 0 \leq y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the PDF $f_Y(y)$, the conditional PDF $f_{X|Y}(x|y)$, and the conditional expected value $\mathbb{E}[X|Y = y]$.

7.5.2 Let random variables X and Y have joint PDF $f_{X,Y}(x, y)$ given in Problem 7.5.1. Find the PDF $f_X(x)$, the conditional PDF $f_{Y|X}(y|x)$, and the conditional expected value $E[Y|X = x]$.

7.5.3 The probability model for random variable A is

$$P_A(a) = \begin{cases} 1/3 & a = -1, \\ 2/3 & a = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The conditional probability model for random variable B given A is:

$$P_{B|A}(b|1) = \begin{cases} 1/3 & b = 0, \\ 2/3 & b = 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$P_{B|A}(b|1) = \begin{cases} 1/2 & b = 0, \\ 1/2 & b = 1, \\ 0 & \text{otherwise.} \end{cases}$$

- What is the probability model for random variables A and B ? Write the joint PMF $P_{A,B}(a, b)$ as a table.
- If $A = 1$, what is the conditional expected value $E[B|A = 1]$?
- If $B = 1$, what is the conditional PMF $P_{A|B}(a|1)$?
- If $B = 1$, what is the conditional variance $\text{Var}[A|B = 1]$ of A ?
- What is the covariance $\text{Cov}[A, B]$?

7.5.4 For random variables A and B given in Problem 7.5.3, let $U = E[B|A]$. Find the PMF $P_U(u)$. What is $E[U] = E[E[B|A]]$?

7.5.5 Random variables N and K have the joint PMF

$$P_{N,K}(n, k) = \begin{cases} \frac{100^n e^{-100}}{(n+1)!} & n=0, 1, \dots; \\ & k=0, 1, \dots, n, \\ 0 & \text{otherwise.} \end{cases}$$

- Find the marginal PMF $P_N(n)$ and the conditional PMF $P_{K|N}(k|n)$.
- Find the conditional expected value $E[K|N = n]$.

- Express the random variable $E[K|N]$ as a function of N and use the iterated expectation to find $E[K]$.

7.5.6 Random variables X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 1/2 & -1 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- What is $f_Y(y)$?
- What is $f_{X|Y}(x|y)$?
- What is $E[X|Y = y]$?

7.5.7 Over the circle $X^2 + Y^2 \leq r^2$, random variables X and Y have the uniform PDF

$$f_{X,Y}(x, y) = \begin{cases} 1/(\pi r^2) & x^2 + y^2 \leq r^2, \\ 0 & \text{otherwise.} \end{cases}$$

- What is $f_{Y|X}(y|x)$?
- What is $E[Y|X = x]$?

7.5.8 (Continuation of Problem 4.6.14) At time $t = 0$, the price of a stock is a constant k dollars. At time $t > 0$ the price of a stock is a Gaussian random variable X with $E[X] = k$ and $\text{Var}[X] = t$. At time t , a *Call Option at Strike k* has value

$$V = (X - k)^+,$$

where the operator $(\cdot)^+$ is defined as $(z)^+ = \max(z, 0)$. Suppose that at the start of each $t = 30$ day month, you can buy the call option at strike k at a price D that is a random variable that fluctuates every month. You decide to buy the call only if the price D is no more than a threshold d^* . What value of the threshold d^* maximizes the expected return $E[R]$?

7.5.9 In a weekly lottery, each \$1 ticket sold adds 50 cents to the jackpot that starts at \$1 million before any tickets are sold. The jackpot is announced each morning to encourage people to play. On the morning of the i th day before the drawing, the current value of the jackpot J_i is announced.

On that day, the number of tickets sold, N_i , is a Poisson random variable with expected value J_i . Thus, six days before the drawing, the morning jackpot starts at \$1 million and N_6 tickets are sold that day. On the day of the drawing, the announced jackpot is J_0 dollars and N_0 tickets are sold before the evening drawing. What are the expected value and variance of J , the value of the jackpot the instant before the drawing? Hint: Use conditional expectations.

7.6.1 ● You wish to measure random variable X with expected value $E[X] = 1$ and variance $\text{Var}[X] = 1$, but your measurement procedure yields the noisy observation $Y = X + Z$, where Z is the Gaussian $(0, 2)$ noise that is independent of X .

- Find the conditional PDF $f_{Z|X}(z|x)$ of Z given $X = x$.
- Find the conditional PDF $f_{Y|X}(y|2)$ of Y given $X = 2$. Hint: Given $X = x$, $Y = x + Z$.

7.6.2 ■ X and Y are jointly Gaussian random variables with $E[X] = E[Y] = 0$ and $\text{Var}[X] = \text{Var}[Y] = 1$. Furthermore, $E[Y|X] = X/2$. Find $f_{X,Y}(x, y)$.

7.6.3 ■ A study of bicycle riders found that a male cyclist's speed X (in miles per hour over a 100-mile "century" ride) and weight Y (kg) could be modeled by a bivariate Gaussian PDF $f_{X,Y}(x, y)$ with parameters $\mu_X = 20$, $\sigma_X = 2$, $\mu_Y = 75$, $\sigma_Y = 5$ and $\rho_{X,Y} = -0.6$. In addition, a female cyclist's speed X' and weight Y' could be modeled by a bivariate Gaussian PDF $f_{X',Y'}(x', y')$ with parameters $\mu_{X'} = 15$, $\sigma_{X'} = 2$, $\mu_{Y'} = 50$, $\sigma_{Y'} = 5$ and $\rho_{X',Y'} = -0.6$. For men and women, the negative correlation of speed and weight reflects the common wisdom that fast cyclists are thin. As it happens, cycling is much more popular among men than women; in a mixed group of cyclists, a cyclist is a male with probability $p = 0.80$.

You suspect it's OK to ignore the differences between men and women since for both groups, weight and speed are negatively correlated, with $\rho = -0.6$. To convince yourself this is OK, you decide to study the speed $\hat{X}$ and weight $\hat{Y}$ of a cyclist randomly chosen from a large mixed group of male and female cyclists. How are $\hat{X}$ and $\hat{Y}$ correlated? Explain your answer.

7.6.4 ♦ Let X_1 and X_2 have a bivariate Gaussian PDF with correlation coefficient ρ_{12} such that each X_i is a Gaussian (μ_i, σ_i) random variable. Show that $Y = X_1 X_2$ has variance

$$\begin{aligned} \text{Var}[Y] &= \sigma_1^2 \sigma_2^2 (1 + \rho_{12}^2) \\ &\quad + \sigma_1^2 \mu_2^2 + \mu_1^2 \sigma_2^2 - \mu_1^2 \mu_2^2. \end{aligned}$$

Hints: Look ahead to Problem 9.2.4 and also use the iterated expectation to find

$$E[X_1^2 X_2^2] = E[E[X_1^2 X_2^2 | X_2]].$$

7.6.5 ♦ Use the iterated expectation for a proof of Theorem 5.19 without integrals.

7.7.1 ● For the modem receiver voltage X with PDF given in Example 7.8, use MATLAB to plot the PDF and CDF of random variable X . Write a MATLAB function `x=modemrv(m)` that produces `m` samples of the modem voltage X .

7.7.2 ■ For the quantizer of Example 7.6, we showed in Problem 7.1.9 that the quantization noise Z is nonuniform if X is nonuniform. In this problem, we examine whether it is a reasonable approximation to model the quantization noise as uniform. Consider the special case of a Gaussian $(0, 1)$ random variable X passed through a uniform b -bit quantizer over the interval $(-r/2, r/2)$ with $r = 6$. Does a uniform approximation get better or worse as b increases? Write a MATLAB program to generate histograms for Z to answer this question.

is the temperature on the i th day of the month. Suppose that $E[T_i] = 80$ for all i , and that T_i and T_j have covariance

$$\text{Cov}[T_i, T_j] = \frac{36}{1 + |i - j|} \quad (8.67)$$

Define the daily average temperature as

$$Y = \frac{T_1 + T_2 + \cdots + T_{31}}{31}. \quad (8.68)$$

Based on this model, write a MATLAB program `p=julytemps(T)` that calculates $P[Y \geq T]$, the probability that the daily average temperature is at least T degrees.

Further Reading: [WS01] and [PP02] make extensive use of vectors and matrices. To go deeply into vector random variables, students can use [Str98] to gain a firm grasp of principles of linear algebra.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

8.1.1 ● For random variables $X_1, \dots, X_n$ in Problem 5.10.3, let $\mathbf{X} = [X_1 \ \cdots \ X_n]'$. What is $f_{\mathbf{X}}(\mathbf{x})$?

8.1.2 ● Random vector $\mathbf{X}$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} c\mathbf{a}'\mathbf{x} & \mathbf{0} \leq \mathbf{x} \leq \mathbf{1}, \\ 0 & \text{otherwise,} \end{cases}$$

where $\mathbf{a} = [a_1 \ \cdots \ a_n]'$ is a vector with each component $a_i > 0$. What is c ?

8.1.3 ■ Given $f_{\mathbf{X}}(\mathbf{x})$ with $c = 2/3$ and $a_1 = a_2 = a_3 = 1$ in Problem 8.1.2, find the marginal PDF $f_{X_3}(x_3)$.

8.1.4 ■ $\mathbf{X} = [X_1 \ X_2 \ X_3]'$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 6 & 0 \leq x_1 \leq x_2 \leq x_3 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $\mathbf{U} = [X_1 \ X_2]'$, $\mathbf{V} = [X_1 \ X_3]'$ and $\mathbf{W} = [X_2 \ X_3]'$. Find the marginal PDFs $f_{\mathbf{U}}(\mathbf{u})$, $f_{\mathbf{V}}(\mathbf{v})$ and $f_{\mathbf{W}}(\mathbf{w})$.

8.1.5 ■ A wireless data terminal has three messages waiting for transmission. After sending a message, it expects an acknowl-

edgement from the receiver. When it receives the acknowledgement, it transmits the next message. If the acknowledgement does not arrive, it sends the message again. The probability of successful transmission of a message is p independent of other transmissions. Let $\mathbf{K} = [K_1 \ K_2 \ K_3]'$ be the three-dimensional random vector in which K_i is the total number of transmissions when message i is received successfully. (K_3 is the total number of transmissions used to send all three messages.) Show that

$$P_{\mathbf{K}}(\mathbf{k}) = \begin{cases} p^3(1-p)^{k_3-3} & k_1 < k_2 < k_3; \\ & k_i \in \{1, 2, \dots\}, \\ 0 & \text{otherwise.} \end{cases}$$

8.1.6 ■ From the joint PMF $P_{\mathbf{K}}(\mathbf{k})$ in Problem 8.1.5, find the marginal PMFs

- (a) $P_{K_1, K_2}(k_1, k_2)$,
- (b) $P_{K_1, K_3}(k_1, k_3)$,
- (c) $P_{K_2, K_3}(k_2, k_3)$,
- (d) $P_{K_1}(k_1)$, $P_{K_2}(k_2)$, and $P_{K_3}(k_3)$.

8.1.7 ■ Let $\mathbf{N}$ be the r -dimensional random vector with the multinomial PMF given in

Example 5.21 with $n > r > 2$:

$$P_{\mathbf{N}}(\mathbf{n}) = \binom{n}{n_1, \dots, n_r} p_1^{n_1} \cdots p_r^{n_r}.$$

- What is the joint PMF of N_1 and N_2 ? Hint: Consider a new classification scheme with categories: s_1 , s_2 , and “other.”
- Let $T_i = N_1 + \cdots + N_i$. What is the PMF of T_i ?
- What is the joint PMF of T_1 and T_2 ?

8.1.8 ■ The random variables $Y_1, \dots, Y_4$ have the joint PDF

$$f_{\mathbf{Y}}(\mathbf{y}) = \begin{cases} 24 & 0 \leq y_1 \leq y_2 \leq y_3 \leq y_4 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PDFs $f_{Y_1, Y_4}(y_1, y_4)$, $f_{Y_1, Y_2}(y_1, y_2)$, and $f_{Y_1}(y_1)$.

8.1.9 ♦ As a generalization of the message transmission system in Problem 8.1.5, consider a terminal that has n messages to transmit. The components k_i of the n -dimensional random vector $\mathbf{K}$ are the total number of messages transmitted when message i is received successfully.

- Find the PMF of $\mathbf{K}$.
- For each $j \in \{1, 2, \dots, n-1\}$, find the marginal PMF $P_{K_1, \dots, K_j}(k_1, \dots, k_j)$.
- For each $i \in \{1, 2, \dots, n\}$, find the marginal PMF $P_{K_i}(k_i)$.

Hint: These PMFs are members of a family of discrete random variables in Appendix A.

8.2.1 ● The n components X_i of random vector $\mathbf{X}$ have $E[X_i] = 0$, $\text{Var}[X_i] = \sigma^2$. What is the covariance matrix $\mathbf{C}_{\mathbf{X}}$?

8.2.2 ● The 4-dimensional random vector $\mathbf{X}$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 1 & 0 \leq x_i \leq 1, i = 1, 2, 3, 4, \\ 0 & \text{otherwise.} \end{cases}$$

Are the four components of $\mathbf{X}$ independent random variables?

8.2.3 ● As in Example 8.1, the random vector $\mathbf{X}$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 6e^{-\mathbf{a}'\mathbf{x}} & \mathbf{x} \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

where $\mathbf{a} = [1 \ 2 \ 3]'$. Are the components of $\mathbf{X}$ independent random variables?

8.2.4 ● The PDF of the 3-dimensional random vector $\mathbf{X}$ is

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} e^{-x_3} & 0 \leq x_1 \leq x_2 \leq x_3, \\ 0 & \text{otherwise.} \end{cases}$$

Are the components of $\mathbf{X}$ independent random variables?

8.2.5 ■ The random vector $\mathbf{X}$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} e^{-x_3} & 0 \leq x_1 \leq x_2 \leq x_3, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PDFs $f_{X_1}(x_1)$, $f_{X_2}(x_2)$, and $f_{X_3}(x_3)$.

8.3.1 ● Discrete random vector $\mathbf{X}$ has PMF $P_{\mathbf{X}}(\mathbf{x})$. Prove that for an invertible matrix $\mathbf{A}$, $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ has PMF

$$P_{\mathbf{Y}}(\mathbf{y}) = P_{\mathbf{X}}(\mathbf{A}^{-1}(\mathbf{y} - \mathbf{b})).$$

8.3.2 ■ In the message transmission problem, Problem 8.1.5, the PMF for the number of transmissions when message i is received successfully is

$$P_{\mathbf{K}}(\mathbf{k}) = \begin{cases} p^3(1-p)^{k_3-3} & k_1 < k_2 < k_3; \\ & k_i \in \{1, 2, \dots\}, \\ 0 & \text{otherwise.} \end{cases}$$

Let $J_3 = K_3 - K_2$, the number of transmissions of message 3; $J_2 = K_2 - K_1$, the number of transmissions of message 2; and $J_1 = K_1$, the number of transmissions of message 1. Derive a formula for $P_{\mathbf{J}}(\mathbf{j})$, the PMF of the number of transmissions of individual messages.

8.3.3 ■ In an automatic geolocation system, a dispatcher sends a message to six trucks in a fleet asking their locations. The waiting times for responses from the six trucks

are iid exponential random variables, each with expected value 2 seconds.

- What is the probability that all six responses will arrive within 5 seconds?
- If the system has to locate all six vehicles within 3 seconds, it has to reduce the expected response time of each vehicle. What is the maximum expected response time that will produce a location time for all six vehicles of 3 seconds or less with probability of at least 0.9?

8.3.4♦♦ Let $X_1, \dots, X_n$ denote n iid random variables with PDF $f_X(x)$ and CDF $F_X(x)$. What is the probability $P[X_n = \max\{X_1, \dots, X_n\}]$?

8.4.1● Random variables X_1 and X_2 have zero expected value and variances $\text{Var}[X_1] = 4$ and $\text{Var}[X_2] = 9$. Their covariance is $\text{Cov}[X_1, X_2] = 3$.

- Find the covariance matrix of $\mathbf{X} = [X_1 \ X_2]'$.
- Find the covariance matrix of $\mathbf{Y} = [Y_1 \ Y_2]'$ given by

$$\begin{aligned} Y_1 &= X_1 - 2X_2, \\ Y_2 &= 3X_1 + 4X_2. \end{aligned}$$

8.4.2● Let $X_1, \dots, X_n$ be iid random variables with expected value 0, variance 1, and covariance $\text{Cov}[X_i, X_j] = \rho$, for $i \neq j$. Use Theorem 8.8 to find the expected value and variance of the sum $Y = X_1 + \dots + X_n$.

8.4.3● The two-dimensional random vector $\mathbf{X}$ and the three-dimensional random vector $\mathbf{Y}$ are independent and $E[\mathbf{Y}] = 0$. What is the vector cross-correlation $\mathbf{R}_{\mathbf{X}\mathbf{Y}}$?

8.4.4● The four-dimensional random vector $\mathbf{X}$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 1 & 0 \leq x_i \leq 1, i = 1, 2, 3, 4 \\ 0 & \text{otherwise.} \end{cases}$$

Find the expected value vector $E[\mathbf{X}]$, the correlation matrix $\mathbf{R}_{\mathbf{X}}$, and the covariance matrix $\mathbf{C}_{\mathbf{X}}$.

8.4.5● The random vector $\mathbf{Y} = [Y_1 \ Y_2]'$ has covariance matrix $\mathbf{C}_{\mathbf{Y}} = \begin{bmatrix} 25 & \gamma \\ \gamma & 4 \end{bmatrix}$ where γ is a constant. In terms of γ , what is the correlation coefficient ρ_{Y_1, Y_2} of Y_1 and Y_2 ? For what values of γ is $\mathbf{C}_{\mathbf{Y}}$ a valid covariance matrix?

8.4.6● In the message transmission system in Problem 8.1.5, the solution to Problem 8.3.2 is a formula for the PMF of $\mathbf{J}$, the number of transmissions of individual messages. For $p = 0.8$, find the expected value vector $E[\mathbf{J}]$, the correlation matrix $\mathbf{R}_{\mathbf{J}}$, and the covariance matrix $\mathbf{C}_{\mathbf{J}}$.

8.4.7■ In the message transmission system in Problem 8.1.5,

$$P_{\mathbf{K}}(\mathbf{k}) = \begin{cases} p^3(1-p)^{k_3-3}; & k_1 < k_2 < k_3; \\ & k_i \in \{1, 2, \dots\}, \\ 0 & \text{otherwise.} \end{cases}$$

For $p = 0.8$, find the expected value vector $E[\mathbf{K}]$, the covariance matrix $\mathbf{C}_{\mathbf{K}}$, and the correlation matrix $\mathbf{R}_{\mathbf{K}}$.

8.4.8■ Random vector $\mathbf{X} = [X_1 \ X_2]'$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 10e^{-5x_1-2x_2} & x_1 \geq 0, x_2 \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- Find $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$.
- Derive the expected value vector $\boldsymbol{\mu}_{\mathbf{X}}$ and covariance matrix $\mathbf{C}_{\mathbf{X}}$.
- Let $\mathbf{Z} = \mathbf{A}\mathbf{X}$, where $\mathbf{A} = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$. Find the covariance matrix of $\mathbf{Z}$.

8.4.9■ As in Quiz 5.10 and Example 5.23, the 4-dimensional random vector $\mathbf{Y}$ has PDF

$$f_{\mathbf{Y}}(\mathbf{y}) = \begin{cases} 4 & 0 \leq y_1 \leq y_2 \leq 1; \\ & 0 \leq y_3 \leq y_4 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the expected value vector $E[\mathbf{Y}]$, the correlation matrix $\mathbf{R}_{\mathbf{Y}}$, and the covariance matrix $\mathbf{C}_{\mathbf{Y}}$.

8.4.10 ■ $\mathbf{X} = [X_1 \ X_2]'$ is a random vector with $E[\mathbf{X}] = [0 \ 0]'$ and covariance matrix

$$\mathbf{C}_X = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

For some ω satisfying $0 \leq \omega \leq 1$, let $Y = \sqrt{\omega}X_1 + \sqrt{1-\omega}X_2$. What value (or values) of ω will maximize $E[Y^2]$?

8.4.11 ■ The two-dimensional random vector $\mathbf{Y}$ has PDF

$$f_Y(\mathbf{y}) = \begin{cases} 2 & \mathbf{y} \geq \mathbf{0}, [1 \ 1] \mathbf{y} \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the expected value vector $E[\mathbf{Y}]$, the correlation matrix $\mathbf{R}_Y$, and the covariance matrix $\mathbf{C}_Y$.

8.4.12 ♦ Let $\mathbf{X}$ be a random vector with correlation matrix $\mathbf{R}_X$ and covariance matrix $\mathbf{C}_X$. Show that $\mathbf{R}_X$ and $\mathbf{C}_X$ are both positive semidefinite by showing that for any nonzero vector $\mathbf{a}$,

$$\begin{aligned} \mathbf{a}'\mathbf{R}_X\mathbf{a} &\geq 0, \\ \mathbf{a}'\mathbf{C}_X\mathbf{a} &\geq 0. \end{aligned}$$

8.5.1 ● $\mathbf{X}$ is the 3-dimensional Gaussian random vector with expected value $\boldsymbol{\mu}_X = [4 \ 8 \ 6]'$ and covariance

$$\mathbf{C}_X = \begin{bmatrix} 4 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 4 \end{bmatrix}.$$

Calculate

- the correlation matrix, $\mathbf{R}_X$,
- the PDF of the first two components of $\mathbf{X}$, $f_{X_1, X_2}(x_1, x_2)$,
- the probability that $X_1 > 8$.

8.5.2 ● $\mathbf{X} = [X_1 \ X_2]'$ is the Gaussian random vector with $E[\mathbf{X}] = [0 \ 0]'$ and covariance matrix

$$\mathbf{C}_X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

What is the PDF of $Y = [2 \ 1] \mathbf{X}$?

8.5.3 ● Given the Gaussian random vector $\mathbf{X}$ in Problem 8.5.1, $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 1/2 & 2/3 \\ 1 & -1/2 & 2/3 \end{bmatrix}$$

and $\mathbf{b} = [-4 \ -4]'$. Calculate

- the expected value $\boldsymbol{\mu}_Y$,
- the covariance $\mathbf{C}_Y$,
- the correlation $\mathbf{R}_Y$,
- the probability that $-1 \leq Y_2 \leq 1$.

8.5.4 ● Let $\mathbf{X}$ be a Gaussian ($\boldsymbol{\mu}_X, \mathbf{C}_X$) random vector. Given a vector $\mathbf{a}$, find the expected value and variance of $Y = \mathbf{a}'\mathbf{X}$. Is Y a Gaussian random variable?

8.5.5 ■ Random variables X_1 and X_2 have zero expected value. The random vector $\mathbf{X} = [X_1 \ X_2]'$ has a covariance matrix of the form

$$\mathbf{C} = \begin{bmatrix} 1 & \alpha \\ \beta & 4 \end{bmatrix}.$$

- For what values of α and β is $\mathbf{C}$ a valid covariance matrix?
- For what values of α and β can $\mathbf{X}$ be a Gaussian random vector?
- Suppose now that α and β satisfy the conditions in part (b) and $\mathbf{X}$ is a Gaussian random vector. What is the PDF of X_2 ? What is the PDF of $W = 2X_1 - X_2$?

8.5.6 ■ The Gaussian random vector $\mathbf{X} = [X_1 \ X_2]'$ has expected value $E[\mathbf{X}] = \mathbf{0}$ and covariance matrix $\mathbf{C}_X = \begin{bmatrix} \sigma_1^2 & 1 \\ 1 & \sigma_2^2 \end{bmatrix}$.

- Under what conditions on σ_1^2 and σ_2^2 is $\mathbf{C}_X$ a valid covariance matrix?
- Suppose $\mathbf{Y} = [Y_1 \ Y_2]' = \mathbf{A}\mathbf{X}$ where $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. For what values (if any) of σ_1^2 and σ_2^2 are the components Y_1 and Y_2 independent?

8.5.7 ■ The Gaussian random vector $\mathbf{X} = [X_1 \ X_2]'$ has expected value $E[\mathbf{X}] = \mathbf{0}$ and covariance matrix $\mathbf{C}_X = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$.

- Find the PDF of $W = X_1 + 2X_2$.

- (b) Find the PDF $f_{\mathbf{Y}}(\mathbf{y})$ of $\mathbf{Y} = \mathbf{A}\mathbf{X}$ where $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.

8.5.8 ■ Let $\mathbf{X}$ be a Gaussian random vector with expected value $[\mu_1 \ \mu_2]'$ and covariance matrix

$$\mathbf{C}_{\mathbf{X}} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}.$$

Show that $\mathbf{X}$ has bivariate Gaussian PDF $f_{\mathbf{X}}(\mathbf{x}) = f_{X_1, X_2}(x_1, x_2)$ given by Definition 5.10.

8.5.9 ■ $X = [X_1 \ X_2]'$ is a Gaussian random vector with $E[\mathbf{X}] = [0 \ 0]'$ and covariance matrix $\mathbf{C}_{\mathbf{X}} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

- What conditions must a , b , c , and d satisfy?
- Under what conditions (in addition to those in part (a)) are X_1 and X_2 independent?
- Under what conditions (in addition to those in part (a)) are X_1 and X_2 identical?

8.5.10 ■ Let $\mathbf{X}$ be a Gaussian $(\boldsymbol{\mu}_{\mathbf{X}}, \mathbf{C}_{\mathbf{X}})$ random vector. Let $\mathbf{Y} = \mathbf{A}\mathbf{X}$ where $\mathbf{A}$ is an $m \times n$ matrix of rank m . By Theorem 8.11, $\mathbf{Y}$ is a Gaussian random vector. Is

$$\mathbf{W} = \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix}$$

a Gaussian random vector?

8.5.11 ■ The 2×2 matrix

$$\mathbf{Q} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

is called a rotation matrix because $\mathbf{y} = \mathbf{Q}\mathbf{x}$ is the rotation of $\mathbf{x}$ by the angle θ . Suppose $\mathbf{X} = [X_1 \ X_2]'$ is a Gaussian $(\mathbf{0}, \mathbf{C}_{\mathbf{X}})$ vector where $\mathbf{C}_{\mathbf{X}} = \text{diag}[\sigma_1^2, \sigma_2^2]$ and $\sigma_2^2 \geq \sigma_1^2$. Let $\mathbf{Y} = \mathbf{Q}\mathbf{X}$.

- Find the covariance of Y_1 and Y_2 . Show that Y_1 and Y_2 are independent for all θ if $\sigma_1^2 = \sigma_2^2$.
- Suppose $\sigma_2^2 > \sigma_1^2$. For what values θ are Y_1 and Y_2 independent?

8.5.12 ♦ $\mathbf{X} = [X_1 \ X_2]'$ is a Gaussian $(\mathbf{0}, \mathbf{C}_{\mathbf{X}})$ vector where

$$\mathbf{C}_{\mathbf{X}} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

Thus, depending on the value of the correlation coefficient ρ , the joint PDF of X_1 and X_2 may resemble one of the graphs of Figure 5.6 with $X_1 = X$ and $X_2 = Y$. Show that $\mathbf{X} = \mathbf{Q}\mathbf{Y}$, where $\mathbf{Q}$ is the $\theta = 45^\circ$ rotation matrix (see Problem 8.5.11) and $\mathbf{Y}$ is a Gaussian $(\mathbf{0}, \mathbf{C}_{\mathbf{Y}})$ vector such that

$$\mathbf{C}_{\mathbf{Y}} = \begin{bmatrix} 1 + \rho & 0 \\ 0 & 1 - \rho \end{bmatrix}.$$

This result verifies, for $\rho \neq 0$, that the PDF of X_1 and X_2 shown in Figure 5.6 is the joint PDF of two independent Gaussian random variables (with variances $1 + \rho$ and $1 - \rho$) rotated by 45° .

8.5.13 ♦ An n -dimensional Gaussian vector $\mathbf{W}$ has a block diagonal covariance matrix

$$\mathbf{C}_{\mathbf{W}} = \begin{bmatrix} \mathbf{C}_{\mathbf{X}} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_{\mathbf{Y}} \end{bmatrix},$$

where $\mathbf{C}_{\mathbf{X}}$ is $m \times m$, $\mathbf{C}_{\mathbf{Y}}$ is $(n-m) \times (n-m)$. Show that $\mathbf{W}$ can be written in terms of component vectors $\mathbf{X}$ and $\mathbf{Y}$ in the form

$$\mathbf{W} = \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix},$$

such that $\mathbf{X}$ and $\mathbf{Y}$ are independent Gaussian random vectors.

8.5.14 ♦♦ In this problem, we extend the proof of Theorem 8.11 to the case when $\mathbf{A}$ is $m \times n$ with $m < n$. For this proof, we assume $\mathbf{X}$ is an n -dimensional Gaussian vector and that we have proved Theorem 8.11 for the case $m = n$. Since the case $m = n$ is sufficient to prove that $\mathbf{Y} = \mathbf{X} + \mathbf{b}$ is Gaussian, it is sufficient to show for $m < n$ that $\mathbf{Y} = \mathbf{A}\mathbf{X}$ is Gaussian in the case when $\boldsymbol{\mu}_{\mathbf{X}} = \mathbf{0}$.

- Prove there exists an $(n-m) \times n$ matrix $\tilde{\mathbf{A}}$ of rank $n-m$ with the property that $\tilde{\mathbf{A}}\mathbf{A}' = \mathbf{0}$. Hint: Review the Gram-Schmidt procedure.

- (b) Let $\hat{\mathbf{A}} = \tilde{\mathbf{A}}\mathbf{C}_{\mathbf{X}}^{-1}$ and define the random vector

$$\bar{\mathbf{Y}} = \begin{bmatrix} \mathbf{Y} \\ \hat{\mathbf{Y}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} \\ \hat{\mathbf{A}} \end{bmatrix} \mathbf{X}.$$

Use Theorem 8.11 for the case $m = n$ to argue that $\bar{\mathbf{Y}}$ is a Gaussian random vector.

- (c) Find the covariance matrix $\bar{\mathbf{C}}$ of $\bar{\mathbf{Y}}$. Use the result of Problem 8.5.13 to show that $\mathbf{Y}$ and $\hat{\mathbf{Y}}$ are independent Gaussian random vectors.

8.6.1 ● Consider the vector $\mathbf{X}$ in Problem 8.5.1 and define $Y = (X_1 + X_2 + X_3)/3$. What is the probability that $Y > 4$?

8.6.2 ■ A better model for the sailboat race of Problem 5.10.8 accounts for the fact that all boats are subject to the same randomness of wind and tide. Suppose in the race of ten sailboats, the finishing times X_i are identical Gaussian random variables with expected value 35 minutes and standard deviation 5 minutes. However, for every pair of boats i and j , the finish times X_i and X_j have correlation coefficient $\rho = 0.8$.

- (a) What is the covariance matrix of $\mathbf{X} = [X_1 \ \cdots \ X_{10}]'$?
- (b) Let

$$Y = \frac{X_1 + X_2 + \cdots + X_{10}}{10}.$$

What are the expected value and variance of Y ? What is $P[Y \leq 25]$?

8.6.3 ■ For the vector of daily temperatures $[T_1 \ \cdots \ T_{31}]'$ and average temperature Y modeled in Quiz 8.6, we wish to estimate the probability of the event

$$A = \left\{ Y \leq 82, \min_i T_i \geq 72 \right\}.$$

To form an estimate of A , generate 10,000 independent samples of the vector $\mathbf{T}$ and calculate the relative frequency of A in those trials.

8.6.4 ♦ We continue Problem 8.6.2 where the vector $\mathbf{X}$ of finish times has correlated components. Let W denote the finish time of the winning boat. We wish to estimate $P[W \leq 25]$, the probability that the winning boat finishes in under 25 minutes. To do this, simulate $m = 10,000$ races by generating m samples of the vector $\mathbf{X}$ of finish times. Let $Y_j = 1$ if the winning time in race j is under 25 minutes; otherwise, $Y_j = 0$. Calculate the estimate

$$P[W \leq 25] \approx \frac{1}{m} \sum_{j=1}^m Y_j.$$

8.6.5 ♦♦ Write a MATLAB program that simulates m runs of the weekly lottery of Problem 7.5.9. For $m = 1000$ sample runs, form a histogram for the jackpot J .

Quiz 9.6

X is the binomial $(100, 0.5)$ random variable and Y is the discrete uniform $(0, 100)$ random variable. Calculate and graph the PMF of $W = X + Y$.

Further Reading: [Dur94] contains a concise, rigorous presentation and proof of the central limit theorem.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

9.1.1 ● X_1 and X_2 are iid random variables with variance $\text{Var}[X]$.

- (a) What is $E[X_1 - X_2]$?
- (b) What is $\text{Var}[X_1 - X_2]$?

9.1.2 ● Flip a biased coin 100 times. On each flip, $P[H] = p$. Let X_i denote the number of heads that occur on flip i . What is $P_{X_{33}}(x)$? Are X_1 and X_2 independent? Define $Y = X_1 + X_2 + \cdots + X_{100}$. Describe Y in words. What is $P_Y(y)$? Find $E[Y]$ and $\text{Var}[Y]$.

9.1.3 ■ A radio program gives concert tickets to the fourth caller with the right answer to a question. Of the people who call, 25% know the answer. Phone calls are independent of one another. The random variable N_r indicates the number of phone calls taken when the r th correct answer arrives. (If the fourth correct answer arrives on the eighth call, then $N_4 = 8$.)

- (a) What is the PMF of N_1 , the number of phone calls needed to obtain the first correct answer?
- (b) What is $E[N_1]$, the expected number of phone calls needed to obtain the first correct answer?
- (c) What is the PMF of N_4 , the number of phone calls needed to obtain the fourth correct answer? Hint: See Example 3.13.
- (d) What is $E[N_4]$? Hint: N_4 can be written as the independent sum $N_4 = K_1 + K_2 + K_3 + K_4$, where each K_i is distributed identically to N_1 .

9.1.4 ■ X_1 , X_2 and X_3 are iid continuous uniform random variables. Random variable $Y = X_1 + X_2 + X_3$ has expected value $E[Y] = 0$ and variance $\sigma_Y^2 = 4$. What is the PDF $f_{X_1}(x)$ of X_1 ?

9.1.5 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & x \geq 0, y \geq 0, x+y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

What is the variance of $W = X + Y$?

9.2.1 ● For a constant $a > 0$, a Laplace random variable X has PDF

$$f_X(x) = \frac{a}{2} e^{-a|x|}, \quad -\infty < x < \infty.$$

Calculate the MGF $\phi_X(s)$.

9.2.2 ■ Random variables J and K have the joint probability mass function

$P_{J,K}(j,k)$	$k = -1$	$k = 0$	$k = 1$
$j = -2$	0.42	0.12	0.06
$j = -1$	0.28	0.08	0.04

- (a) What is the MGF of J ?
- (b) What is the MGF of K ?
- (c) Find the PMF of $M = J + K$?
- (d) What is $E[M^4]$?

9.2.3 ■ X is the continuous uniform (a,b) random variable. Find the MGF $\phi_X(s)$. Use the MGF to calculate the first and second moments of X .

9.2.4 ♦ Let X be a Gaussian $(0, \sigma)$ random variable. Use the moment generating function to show that

$$\begin{aligned} E[X] &= 0, & E[X^2] &= \sigma^2, \\ E[X^3] &= 0, & E[X^4] &= 3\sigma^4. \end{aligned}$$

Let Y be a Gaussian (μ, σ) random variable. Use the moments of X to show that

$$\begin{aligned} E[Y^2] &= \sigma^2 + \mu^2, \\ E[Y^3] &= 3\mu\sigma^2 + \mu^3, \\ E[Y^4] &= 3\sigma^4 + 6\mu\sigma^2 + \mu^4. \end{aligned}$$

9.2.5 ♦♦ Random variable K has a discrete uniform $(1, n)$ PMF. Use the MGF $\phi_K(s)$ to find $E[K]$ and $E[K^2]$. Use the first and second moments of K to derive well-known expressions for $\sum_{k=1}^n k$ and $\sum_{k=1}^n k^2$.

9.3.1 ● N is the binomial $(100, 0.4)$ random variable. M is the binomial $(50, 0.4)$ random variable. M and N are independent. What is the PMF of $L = M + N$?

9.3.2 ● Random variable Y has the moment generating function $\phi_Y(s) = 1/(1-s)$. Random variable V has the moment generating function $\phi_V(s) = 1/(1-s)^4$. Y and V are independent. $W = Y + V$.

- What are $E[Y]$, $E[Y^2]$, and $E[Y^3]$?
- What is $E[W^2]$?

9.3.3 ● Let $K_1, K_2, \dots$ denote a sequence of iid Bernoulli (p) random variables. Let $M = K_1 + \dots + K_n$.

- Find the MGF $\phi_K(s)$.
- Find the MGF $\phi_M(s)$.
- Use the MGF $\phi_M(s)$ to find $E[M]$ and $\text{Var}[M]$.

9.3.4 ● Suppose you participate in a chess tournament in which you play n games. Since you are an average player, each game is equally likely to be a win, a loss, or a tie. You collect 2 points for each win, 1 point for each tie, and 0 points for each loss. The outcome of each game is independent of the outcome of every other game. Let X_i be the

number of points you earn for game i and let Y equal the total number of points earned over the n games.

- Find the moment generating functions $\phi_{X_i}(s)$ and $\phi_Y(s)$.
- Find $E[Y]$ and $\text{Var}[Y]$.

9.3.5 ● At time $t = 0$, you begin counting the arrivals of buses at a depot. The number of buses K_i that arrive between time $i - 1$ minutes and time i minutes has the Poisson PMF

$$P_{K_i}(k) = \begin{cases} 2^k e^{-2}/k! & k = 0, 1, 2, \dots, \\ 0 & \text{otherwise.} \end{cases}$$

$K_1, K_2, \dots$ are an iid random sequence. Let $R_i = K_1 + K_2 + \dots + K_i$ denote the number of buses arriving in the first i minutes.

- What is the moment generating function $\phi_{K_i}(s)$?
- Find the MGF $\phi_{R_i}(s)$.
- Find the PMF $P_{R_i}(r)$. Hint: Compare $\phi_{R_i}(s)$ and $\phi_{K_i}(s)$.
- Find $E[R_i]$ and $\text{Var}[R_i]$.

9.3.6 ● Suppose that during the i th day of December, the energy X_i stored by a solar collector is a Gaussian random variable with expected value $32 - i/4$ kW-hr and standard deviation of 10 kW-hr. Assuming the energy stored each day is independent of any other day, what is the PDF of Y , the total energy stored in the 31 days of December?

9.3.7 ■ $K, K_1, K_2, \dots$ are iid random variables. Use the MGF of $M = K_1 + \dots + K_n$ to prove that

- $E[M] = n E[K]$.
- $E[M^2] = n(n-1)(E[K])^2 + n E[K^2]$.

9.4.1 ■ $X_1, X_2, \dots$ is a sequence of iid random variables each with exponential PDF

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- Find $\phi_X(s)$.

- (b) Let K be a geometric random variable with PMF

$$P_K(k) = \begin{cases} (1-q)q^{k-1} & k = 1, 2, \dots, \\ 0 & \text{otherwise.} \end{cases}$$

Find the MGF and PDF of $V = X_1 + \dots + X_K$.

9.4.2 ■ In any game, the number of passes N that Donovan McNabb will throw is the Poisson (30) random variable. Each pass is completed with probability $q = 2/3$, independent of any other pass or the number of passes thrown. Let K equal the number of completed passes McNabb throws in a game. What are $\phi_K(s)$, $E[K]$, and $\text{Var}[K]$? What is the PMF $P_K(k)$?

9.4.3 ■ Suppose we flip a fair coin repeatedly. Let X_i equal 1 if flip i was heads (H) and 0 otherwise. Let N denote the number of flips needed until H has occurred 100 times. Is N independent of the random sequence $X_1, X_2, \dots$? Define $Y = X_1 + \dots + X_N$. Is Y an ordinary random sum of random variables? What is the PMF of Y ?

9.4.4 ♦ K , the number of passes that Donovan McNabb completes in any game, is the Poisson (20) random variable. If NFL yardage were measured with greater care (as opposed to always being rounded to the nearest yard), officials might discover that each completion results in a yardage gain Y that is the exponential random variable with expected value 15 yards. Let V equal McNabb's total passing yardage in a game. Find $\phi_V(s)$, $E[V]$, $\text{Var}[V]$, and (if possible) the PDF $f_V(v)$.

9.4.5 ♦ This problem continues the lottery of Problem 3.7.10 in which each ticket has 6 randomly marked numbers out of $1, \dots, 46$. A ticket is a winner if the six marked numbers match 6 numbers drawn at random at the end of a week. Suppose that following a week in which the pot carried over was r dollars, the number of tickets sold in that week, K , is the Poisson (r) random variable. What is the PMF of the number of winning

tickets? Hint: What is the probability q that an arbitrary ticket is a winner?

9.4.6 ♦ X is the Gaussian $(1, 1)$ random variable and K is a discrete random variable, independent of X , with PMF

$$P_K(k) = \begin{cases} q(1-q)^k & k = 0, 1, \dots, \\ 0 & \text{otherwise.} \end{cases}$$

Let $X_1, X_2, \dots$ denote a sequence of iid random variables each with the same distribution as X .

- What is the MGF of K ?
- What is the MGF of $R = X_1 + \dots + X_K$? Note that $R = 0$ if $K = 0$.
- Find $E[R]$ and $\text{Var}[R]$.

9.4.7 ♦♦ Let $X_1, \dots, X_n$ denote a sequence of iid Bernoulli (p) random variables and let $K = X_1 + \dots + X_n$. In addition, let M denote a binomial (n, p) random variable, independent of $X_1, \dots, X_n$. Do the random variables $U = X_1 + \dots + X_K$ and $V = X_1 + \dots + X_M$ have the same expected value? Be careful: U is *not* an ordinary random sum of random variables.

9.4.8 ♦♦ Suppose you participate in a chess tournament in which you play until you lose a game. Since you are an average player, each game is equally likely to be a win, a loss, or a tie. You collect 2 points for each win, 1 point for each tie, and 0 points for each loss. The outcome of each game is independent of the outcome of every other game. Let X_i be the number of points you earn for game i and let Y equal the total number of points earned in the tournament.

- Find the moment generating function $\phi_Y(s)$. Hint: What is $E[e^{sX_i} | N = n]$? This is not the usual random sum of random variables problem.
- Find $E[Y]$ and $\text{Var}[Y]$.

9.5.1 ● The waiting time in milliseconds, W , for accessing one record from a computer database is the continuous uniform $(0, 10)$ random variable. The read time R

(for moving the information from the disk to main memory) is 3 milliseconds. The random variable X milliseconds is the total access time (waiting time + read time) to get one block of information from the disk. Before performing a certain task, the computer must access 12 different blocks of information from the disk. (Access times for different blocks are independent of one another.) The total access time for all the information is a random variable A milliseconds.

- What is $E[X]$?
- What is $\text{Var}[X]$?
- What is $E[A]$?
- What is σ_A ? time?
- Use the central limit theorem to estimate $P[A > 116\text{ms}]$.
- Use the central limit theorem to estimate $P[A < 86\text{ms}]$.

9.5.2 ● Internet packets can be classified as video (V) or as generic data (D). Based on a lot of observations taken by the Internet service provider, we have the following probability model: $P[V] = 3/4$, $P[D] = 1/4$. Data packets and video packets occur independently of one another. The random variable K_n is the number of video packets in a collection of n packets.

- What is $E[K_{100}]$, the expected number of video packets in a set of 100 packets?
- What is $\sigma_{K_{100}}$?
- Use the central limit theorem to estimate $P[K_{100} \geq 18]$.
- Use the central limit theorem to estimate $P[16 \leq K_{100} \leq 24]$.

9.5.3 ● The duration of a cellular telephone call is an exponential random variable with expected value 150 seconds. A subscriber has a calling plan that includes 300 minutes per month at a cost of \$30.00 plus \$0.40 for each minute that the total calling time exceeds 300 minutes. In a certain month, the subscriber has 120 cellular calls.

- Use the central limit theorem to estimate the probability that the subscriber's bill is greater than \$36. (Assume that the durations of all phone calls are mutually independent and that the telephone company measures call duration exactly and charges accordingly, without rounding up fractional minutes.)
- Suppose the telephone company does charge a full minute for each fractional minute used. Re-calculate your estimate of the probability that the bill is greater than \$36.

9.5.4 ● Let $K_1, K_2, \dots$ be an iid sequence of Poisson (1) random variables. Let $W_n = K_1 + \dots + K_n$. Use the improved central limit theorem approximation to estimate $P[W_n = n]$. For $n = 4, 25, 64$, compare the approximation to the exact value of $P[W_n = n]$.

9.5.5 ● In any one-minute interval, the number of requests for a popular Web page is a Poisson random variable with expected value 300 requests.

- A Web server has a capacity of C requests per minute. If the number of requests in a one-minute interval is greater than C , the server is overloaded. Use the central limit theorem to estimate the smallest value of C for which the probability of overload is less than 0.05.
- Use MATLAB to calculate the actual probability of overload for the value of C derived from the central limit theorem.
- For the value of C derived from the central limit theorem, what is the probability of overload in a *one-second* interval?
- What is the smallest value of C for which the probability of overload in a one-second interval is less than 0.05?
- Comment on the application of the central limit theorem to estimate the overload probability in a one-second interval and in a one-minute interval.

9.5.6 ● Integrated circuits from a certain factory pass a certain quality test with probability 0.8. The outcomes of all tests are mutually independent.

- What is the expected number of tests necessary to find 500 acceptable circuits?
- Use the central limit theorem to estimate the probability of finding 500 acceptable circuits in a batch of 600 circuits.
- Use MATLAB to calculate the actual probability of finding 500 acceptable circuits in a batch of 600 circuits.
- Use the central limit theorem to calculate the minimum batch size for finding 500 acceptable circuits with probability 0.9 or greater.

9.5.7 ■ Internet packets can be classified as video (V) or as generic data (D). Based on a lot of observations taken by the Internet service provider, we have the following probability model: $P[V] = 0.8$, $P[D] = 0.2$. Data packets and video packets occur independently of one another. The random variable K_n is the number of video packets in a collection of n packets.

- What is $E[K_{48}]$, the expected number of video packets in a set of 48 packets?
- What is $\sigma_{K_{48}}$, the standard deviation of the number of video packets in a set of 48 packets?
- Use the central limit theorem to estimate $P[30 \leq K_{48} \leq 42]$, the probability of between 30 and 42 voice calls in a set of 48 calls.
- Use the De Moivre–Laplace formula to estimate $P[30 \leq K_{48} \leq 42]$.

9.5.8 ■ In the presence of a headwind of normalized intensity W , your speed on your bike is $V = 20 - 10W^3$ mi/hr. The wind intensity W is a continuous uniform $(-1, 1)$ random variable. Moreover, the wind changes every ten minutes. Let W_i denote the headwind intensity in the i th ten-minute interval. In a five-hour bike ride,

with 30 ten-minute intervals, the wind intensities $W_1, \dots, W_{30}$ are independent and identical to W . The distance you travel is

$$X = \frac{V_1 + V_2 + \cdots + V_{30}}{6}.$$

Use the CLT to estimate $P[X \geq 95]$.

9.5.9 ■ An amplifier circuit has power consumption Y that grows nonlinearly with the input signal voltage X . When the input signal is X volts, the instantaneous power consumed by the amplifier is $Y = 20 + 15X^2$ Watts. The input signal X is the continuous uniform $(-1, 1)$ random variable. Sampling the input signal every millisecond over a 100-millisecond interval yields the iid signal samples $X_1, X_2, \dots, X_{100}$. Over the 100 ms interval, you estimate the average power of the amplifier as

$$W = \frac{1}{100} \sum_{i=1}^{100} Y_i$$

where $Y_i = 20 + 15X_i^2$. Use the central limit theorem to estimate $P[W \leq 25.4]$.

9.5.10 ■ In the face of perpetually varying headwinds, cyclists Lance and Ashwin are in a 3000 mile race across America. To maintain a speed of v miles/hour in the presence of a w mi/hr headwind, a cyclist must generate a power output $y = 50 + (v + w - 15)^3$ Watts. During each mile of road, the wind speed W is the continuous uniform $(0, 10)$ random variable independent of the wind speed in any other mile.

- Lance rides at constant velocity $v = 15$ mi/hr mile after mile. Let Y denote Lance's power output over a randomly chosen mile. What is $E[Y]$?
- Ashwin is less powerful but he is able to ride at constant power $\hat{y}$ Watts in the presence of the same variable headwinds. Use the central limit theorem to find $\hat{y}$ such that Ashwin wins the race with probability $1/2$.

9.5.11 ♦ Suppose your grade in a probability course depends on 10 weekly quizzes. Each quiz has ten yes/no questions, each

worth 1 point. The scoring has no partial credit. Your performance is a model of consistency: On each one-point question, you get the right answer with probability p , independent of the outcome on any other question. Thus your score X_i on quiz i is between 0 and 10. Your average score, $X = \sum_{i=1}^{10} X_i/100$ is used to determine your grade. The course grading has simple letter grades without any curve: $A: X \geq 0.9$, $B: 0.8 \leq X < 0.9$, $C: 0.7 \leq X < 0.8$, $D: 0.6 \leq X < 0.7$ and $F: X < 0.6$. As it happens, you are a borderline B/C student with $p = 0.8$.

- What is the PMF of X_i ?
- Use the central limit theorem to estimate the probability $P[A]$ that your grade is an A .
- Suppose now that the course has “attendance quizzes.” If you attend a lecture with an attendance quiz, you get credit for a bonus quiz with a score of 10. If you are present for n bonus quizzes, your modified average

$$X' = \frac{10n + \sum_{i=1}^{10} X_i}{10n + 100}$$

is used to calculate your grade: $A: X' \geq 0.9$, $B: 0.8 \leq X' < 0.9$, and so on. Given you attend n attendance quizzes, use the central limit theorem to estimate $P[A]$.

- Now suppose there are no attendance quizzes and your week 1 quiz is scored an 8. A few hours after the week 1 quiz, you notice that a question was marked incorrectly; your quiz score should have been 9. You appeal to the annoying prof who says “Sorry, all regrade requests must be submitted immediately after receiving your score. But don’t worry, the probability it makes a difference is virtually nil.” Let U denote the event that your letter grade is unchanged because of the scoring error. Find an exact expression for $P[U]$.

9.6.1 ● W_n is the number of ones in 10^n independent transmitted bits, each equiprobably 0 or 1. For $n = 3, 4, \dots$, use the `binomialpmf` function to calculate

$$P[0.499 \leq W_n/10^n \leq 0.501].$$

What is the largest n for which your MATLAB installation can perform the calculation? Can you perform the exact calculation of Example 9.14?

9.6.2 ● Use the MATLAB `plot` function to compare the Erlang (n, λ) PDF to a Gaussian PDF with the same expected value and variance for $\lambda = 1$ and $n = 4, 20, 100$. Why are your results not surprising?

9.6.3 ● Recreate the plots of Figure 9.3. On the same plots, superimpose the PDF of Y_n , a Gaussian random variable with the same expected value and variance. If X_n denotes the binomial (n, p) random variable, explain why for most integers k , $P_{X_n}(k) \approx f_Y(k)$.

9.6.4 ● Find the PMF of $W = X_1 + X_2$ in Example 9.17 using the `conv` function.

9.6.5 ■ Use `uniform12.m` to estimate the probability of a storm surge greater than 7 feet in Example 10.4 based on:

- 1000 samples,
- 10000 samples.

9.6.6 ■ X_1 , X_2 , and X_3 are independent random variables such that X_k has PMF

$$P_{X_k}(x) = \begin{cases} 1/(10k) & x = 1, 2, \dots, 10k, \\ 0 & \text{otherwise.} \end{cases}$$

Find the PMF of $W = X_1 + X_2 + X_3$.

9.6.7 ■ Let X and Y denote independent finite random variables described by the probability and range vectors `px`, `sx` and `py`, `sy`. Write a MATLAB function

`[pw, sw]=sumfinitepmf(px, sx, py, sy)`

such that finite random variable $W = X + Y$ is described by `pw` and `sw`.

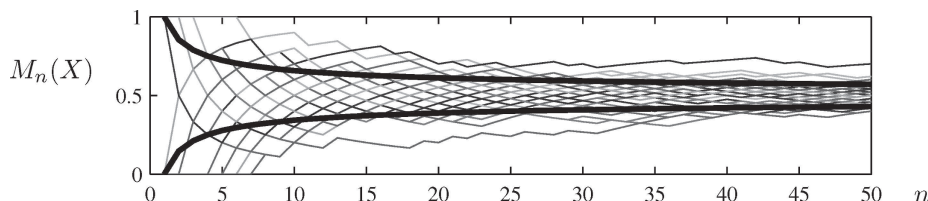


Figure 10.2 Sample output of `bernoullitraces.m`, including the deterministic standard error graphs. The graph shows how at any given step, about two thirds of the sample means are within one standard error of the true mean.

traces, such that M_k is within one standard error of the expected value p . Graph $T_k = M_k/m$ as a function of k . Explain your results.

Further Reading: [Dur94] contains concise, rigorous presentations and proofs of the laws of large numbers. [WS01] covers parameter estimation for both scalar and vector random variables and stochastic processes.

Problems

Difficulty: ● Easy ■ Moderate ◆ Difficult ◆◆ Experts Only

10.1.1 ● $X_1, \dots, X_n$ is an iid sequence of exponential random variables, each with expected value 5.

- What is $\text{Var}[M_9(X)]$, the variance of the sample mean based on nine trials?
- What is $P[X_1 > 7]$, the probability that one outcome exceeds 7?
- Use the central limit theorem to estimate $P[M_9(X) > 7]$, the probability that the sample mean of nine trials exceeds 7.

10.1.2 ● $X_1, \dots, X_n$ are independent uniform random variables with expected value $\mu_X = 7$ and variance $\text{Var}[X] = 3$.

- What is the PDF of X_1 ?
- What is $\text{Var}[M_{16}(X)]$, the variance of the sample mean based on 16 trials?
- What is $P[X_1 > 9]$, the probability that one outcome exceeds 9?
- Would you expect $P[M_{16}(X) > 9]$ to be bigger or smaller than $P[X_1 > 9]$?

To check your intuition, use the central limit theorem to estimate $P[M_{16}(X) > 9]$.

10.1.3 ■ X is a uniform $(0, 1)$ random variable. $Y = X^2$. What is the standard error of the estimate of μ_Y based on 50 independent samples of X ?

10.1.4 ■ Let $X_1, X_2, \dots$ denote a sequence of independent samples of a random variable X with variance $\text{Var}[X]$. We define a new random sequence $Y_1, Y_2, \dots$ as $Y_1 = X_1 - X_2$ and $Y_n = X_{2n-1} - X_{2n}$.

- Find $E[Y_n]$ and $\text{Var}[Y_n]$.
- Find the expected value and variance of $M_n(Y)$.

10.2.1 ● The weight of a randomly chosen Maine black bear has expected value $E[W] = 500$ pounds and standard deviation $\sigma_W = 100$ pounds. Use the Chebyshev inequality to upper bound the probability that the weight of a randomly chosen bear

is more than 200 pounds from the expected value of the weight.

10.2.2 ● For an arbitrary random variable X , use the Chebyshev inequality to show that the probability that X is more than k standard deviations from its expected value $E[X]$ satisfies

$$P[|X - E[X]| \geq k\sigma] \leq \frac{1}{k^2}.$$

For a Gaussian random variable Y , use the $\Phi(\cdot)$ function to calculate the probability that Y is more than k standard deviations from its expected value $E[Y]$. Compare the result to the upper bound based on the Chebyshev inequality.

10.2.3 ■ Elevators arrive randomly at the ground floor of an office building. Because of a large crowd, a person will wait for time W in order to board the third arriving elevator. Let X_1 denote the time (in seconds) until the first elevator arrives and let X_i denote the time between the arrival of elevator $i - 1$ and i . Suppose X_1, X_2, X_3 are independent uniform $(0, 30)$ random variables. Find upper bounds to the probability W exceeds 75 seconds using

- (a) the Markov inequality,
- (b) the Chebyshev inequality,
- (c) the Chernoff bound.

10.2.4 ■ Let X equal the arrival time of the third elevator in Problem 10.2.3. Find the exact value of $P[W \geq 75]$. Compare your answer to the upper bounds derived in Problem 10.2.3.

10.2.5 ■ In a game with two dice, the event *snake eyes* refers to both six-sided dice showing one spot. Let R denote the number of dice rolls needed to observe the third occurrence of *snake eyes*. Find

- (a) the upper bound to $P[R \geq 250]$ based on the Markov inequality,
- (b) the upper bound to $P[R \geq 250]$ based on the Chebyshev inequality,
- (c) the exact value of $P[R \geq 250]$.

10.2.6 ■ Use the Chernoff bound to show that the Gaussian $(0, 1)$ random variable Z satisfies

$$P[Z \geq c] \leq e^{-c^2/2}.$$

For $c = 1, 2, 3, 4, 5$, use Table 4.2 and Table 4.3 to compare the Chernoff bound to the true value: $P[Z \geq c] = Q(c)$.

10.2.7 ■ Use the Chernoff bound to show for a Gaussian (μ, σ) random variable X that

$$P[X \geq c] \leq e^{-(c-\mu)^2/2\sigma^2}.$$

Hint: Apply the result of Problem 10.2.6.

10.2.8 ■ Let K be a Poisson random variable with expected value α . Use the Chernoff bound to find an upper bound to $P[K \geq c]$. For what values of c do we obtain the trivial upper bound $P[K \geq c] \leq 1$?

10.2.9 ■ In a subway station, there are exactly enough customers on the platform to fill three trains. The arrival time of the n th train is $X_1 + \cdots + X_n$ where $X_1, X_2, \dots$ are iid exponential random variables with $E[X_i] = 2$ minutes. Let W equal the time required to serve the waiting customers. Find $P[W > 20]$.

10.2.10 ■ Let $X_1, \dots, X_n$ be independent samples of a random variable X . Use the Chernoff bound to show that $M_n(X) = (X_1 + \cdots + X_n)/n$ satisfies

$$P[M_n(X) \geq c] \leq \left(\min_{s \geq 0} e^{-sc} \phi_X(s) \right)^n.$$

10.3.1 ● Let $X_1, X_2, \dots$ denote an iid sequence of random variables, each with expected value 75 and standard deviation 15.

- (a) How many samples n do we need to guarantee that the sample mean $M_n(X)$ is between 74 and 76 with probability 0.99?
- (b) If each X_i has a Gaussian distribution, how many samples n' would we need to guarantee $M_{n'}(X)$ is between 74 and 76 with probability 0.99?

10.3.2 Let X_A be the indicator random variable for event A with probability $P[A] = 0.8$. Let $\hat{P}_n(A)$ denote the relative frequency of event A in n independent trials.

- (a) Find $E[X_A]$ and $\text{Var}[X_A]$.
- (b) What is $\text{Var}[\hat{P}_n(A)]$?
- (c) Use the Chebyshev inequality to find the confidence coefficient $1 - \alpha$ such that $\hat{P}_{100}(A)$ is within 0.1 of $P[A]$. In other words, find α such that

$$P \left[\left| \hat{P}_{100}(A) - P[A] \right| \leq 0.1 \right] \geq 1 - \alpha.$$

- (d) Use the Chebyshev inequality to find out how many samples n are necessary to have $\hat{P}_n(A)$ within 0.1 of $P[A]$ with confidence coefficient 0.95. In other words, find n such that

$$P \left[\left| \hat{P}_n(A) - P[A] \right| \leq 0.1 \right] \geq 0.95.$$

10.3.3 $X_1, X_2, \dots$ is a sequence of iid Bernoulli $(1/2)$ random variables. Consider the random sequence $Y_n = X_1 + \dots + X_n$.

- (a) What is $\lim_{n \rightarrow \infty} P[|Y_{2n} - n| \leq \sqrt{n/2}]$?
- (b) What does the weak law of large numbers say about Y_{2n} ?

10.3.4 In communication systems, the error probability $P[E]$ may be difficult to calculate; however it may be easy to derive an upper bound of the form $P[E] \leq \epsilon$. In this case, we may still want to estimate $P[E]$ using the relative frequency $\hat{P}_n(E)$ of E in n trials. In this case, show that

$$P \left[\left| \hat{P}_n(E) - P[E] \right| \geq c \right] \leq \frac{\epsilon}{nc^2}.$$

10.3.5 A factory manufactures chocolate chip cookies on an assembly line. Each cookie is sprinkled with K chips from a very large vat of chips, where K is Poisson with $E[K] = 10$, independent of the number on any other cookie. Imagine you are a chip in the vat and you are sprinkled onto a cookie. Let J denote the number of chips (including you) in your cookie. What is the PMF of J ?

Hint: Suppose n cookies have been made such that N_k cookies have k chips. You are just one of the $\sum_{k=0}^{\infty} kN_k$ chips used in the n cookies.

10.3.6 In this problem, we develop a weak law of large numbers for a correlated sequence $X_1, X_2, \dots$ of identical random variables. In particular, each X_i has expected value $E[X_i] = \mu$, and the random sequence has covariance function

$$C_X[m, k] = \text{Cov}[X_m, X_{m+k}] = \sigma^2 a^{|k|}$$

where a is a constant such that $|a| < 1$. For this correlated random sequence, we can define the sample mean of n samples as

$$M_n = \frac{X_1 + \dots + X_n}{n}.$$

- (a) Use Theorem 9.2 to show that

$$\text{Var}[X_1 + \dots + X_n] \leq n\sigma^2 \left(\frac{1+a}{1-a} \right).$$

- (b) Use the Chebyshev inequality to show that for any $c > 0$,

$$P[|M_n - \mu| \geq c] \leq \frac{\sigma^2(1+a)}{n(1-a)c^2}.$$

- (c) Use part (b) to show that for any $c > 0$,

$$\lim_{n \rightarrow \infty} P[|M_n - \mu| \geq c] = 0.$$

10.3.7 In the Gaussian Movie DataBase (GMDB), reviewers like you rate movies with Gaussian scores. In particular, the first person to rate a movie assigns a Gaussian $(q, 1)$ review score X_1 , where r_0 is the true “quality” of the movie. After n reviews, a movie’s rating is $R_n = \sum_{i=1}^n X_i/n$. Strangely enough, in the GMDB, reviewers are influenced by prior reviews; if after $n-1$ reviews a movie is rated $R_{n-1} = r$, the n th review n will rate the movie X_n , a Gaussian $(r, 1)$ random variable, *conditionally* independent of $X_1, \dots, X_{n-1}$ given $R_{n-1} = r$.

- (a) Find $E[R_n]$.

- (b) Find the PDF $f_{R_n}(r)$. Hint: You may have unresolved parameters in this answer.
- (c) Find $\text{Var}[R_n]$. Hint: Find $E[R_n^2 | R_{n-1}]$.
- (d) Interpret your results as $n \rightarrow \infty$? Does the law of large numbers apply here?

10.4.1 ● When X is Gaussian, verify Equation (10.24), which states that the sample mean is within one standard error of the expected value with probability 0.68.

10.4.2 ■ Suppose the sequence of estimates $\hat{R}_n$ is biased but asymptotically unbiased. If $\lim_{n \rightarrow \infty} \text{Var}[\hat{R}_n] = 0$, is the sequence $\hat{R}_n$ consistent?

10.4.3 ■ An experimental trial produces random variables X_1 and X_2 with correlation $r = E[X_1 X_2]$. To estimate r , we perform n independent trials and form the estimate

$$\hat{R}_n = \frac{1}{n} \sum_{i=1}^n X_1(i) X_2(i),$$

where $X_1(i)$ and $X_2(i)$ are samples of X_1 and X_2 on trial i . Show that if $\text{Var}[X_1 X_2]$ is finite, then $\hat{R}_1, \hat{R}_2, \dots$ is an unbiased, consistent sequence of estimates of r .

10.4.4 ■ An experiment produces random vector $\mathbf{X} = [X_1 \ \cdots \ X_k]'$ with expected value $\boldsymbol{\mu}_{\mathbf{X}} = [\mu_1 \ \cdots \ \mu_k]'$. The i th component of $\mathbf{X}$ has variance $\text{Var}[X_i] = \sigma_i^2$. To estimate $\boldsymbol{\mu}_{\mathbf{X}}$, we perform n independent trials such that $\mathbf{X}(i)$ is the sample of $\mathbf{X}$ on trial i , and we form the vector mean

$$\mathbf{M}(n) = \frac{1}{n} \sum_{i=1}^n \mathbf{X}(i).$$

- (a) Show $\mathbf{M}(n)$ is unbiased by showing $E[\mathbf{M}(n)] = \boldsymbol{\mu}_{\mathbf{X}}$.
- (b) Show that the sequence of estimates $\mathbf{M}_n$ is consistent by showing that for any constant $c > 0$,

$$\lim_{n \rightarrow \infty} P \left[\max_{j=1, \dots, k} |M_j(n) - \mu_j| \geq c \right] = 0.$$

Hint: Let $A_i = \{|M_i(n) - \mu_i| \geq c\}$ and apply the union bound (see Problem 1.3.11) to upper bound $P[A_1 \cup A_2 \cup \cdots \cup A_k]$. Then apply the Chebyshev inequality.

10.4.5 ♦ Given the iid samples $X_1, X_2, \dots$ of X , define the sequence $Y_1, Y_2, \dots$ by

$$Y_k = \left(X_{2k-1} - \frac{X_{2k-1} + X_{2k}}{2} \right)^2 + \left(X_{2k} - \frac{X_{2k-1} + X_{2k}}{2} \right)^2.$$

Note that each Y_k is an example of V_2' , an estimate of the variance of X using two samples, given in Theorem 10.13. Show that if $E[X^k] < \infty$ for $k = 1, 2, 3, 4$, then the sample mean $M_n(Y)$ is a consistent, unbiased estimate of $\text{Var}[X]$.

10.4.6 ♦♦ An experiment produces a Gaussian random vector $\mathbf{X} = [X_1 \ \cdots \ X_k]'$ with $E[\mathbf{X}] = \mathbf{0}$ and correlation matrix $\mathbf{R} = E[\mathbf{X}\mathbf{X}']$. To estimate $\mathbf{R}$, we perform n independent trials, yielding the iid sample vectors $\mathbf{X}(1), \mathbf{X}(2), \dots, \mathbf{X}(n)$, and form the sample correlation matrix

$$\hat{\mathbf{R}}(n) = \frac{1}{n} \sum_{m=1}^n \mathbf{X}(m) \mathbf{X}'(m).$$

- (a) Show $\hat{\mathbf{R}}(n)$ is unbiased by showing $E[\hat{\mathbf{R}}(n)] = \mathbf{R}$.
- (b) Show that the sequence of estimates $\hat{\mathbf{R}}(n)$ is consistent by showing that every element $\hat{R}_{ij}(n)$ of the matrix $\hat{\mathbf{R}}$ converges to R_{ij} . That is, show that for any $c > 0$,

$$\lim_{n \rightarrow \infty} P \left[\max_{i,j} |\hat{R}_{ij} - R_{ij}| \geq c \right] = 0.$$

Hint: Extend the technique used in Problem 10.4.4. You will need to use the result of Problem 7.6.4 to show that $\text{Var}[X_i X_j]$ is finite.

10.5.1 ● $X_1, \dots, X_n$ are n independent identically distributed samples of random

variable X with PMF

$$P_X(x) = \begin{cases} 0.1 & x = 0, \\ 0.9 & x = 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) How is $E[X]$ related to $P_X(1)$?
 (b) Use Chebyshev's inequality to find the confidence level α such that $M_{90}(X)$, the estimate based on 90 observations, is within 0.05 of $P_X(1)$. In other words, find α such that

$$P[|M_{90}(X) - P_X(1)| \geq 0.05] \leq \alpha.$$

- (c) Use Chebyshev's inequality to find out how many samples n are necessary to have $M_n(X)$ within 0.03 of $P_X(1)$ with confidence level 0.1. In other words, find n such that

$$P[|M_n(X) - P_X(1)| \geq 0.03] \leq 0.1.$$

10.5.2 ● X is a Bernoulli random variable with unknown success probability p . Using 100 independent samples of X , find a confidence interval estimate of p with confidence coefficient 0.99. If $M_{100}(X) = 0.06$, what is our interval estimate?

10.5.3 ● In n independent experimental trials, the relative frequency of event A is $\hat{P}_n(A)$. How large should n be to ensure that the confidence interval estimate

$$\hat{P}_n(A) - 0.05 \leq P[A] \leq \hat{P}_n(A) + 0.05$$

has confidence coefficient 0.9?

10.5.4 ■ When we perform an experiment, event A occurs with probability $P[A] = 0.01$. In this problem, we estimate $P[A]$ using $\hat{P}_n(A)$, the relative frequency of A over n independent trials.

- (a) How many trials n are needed so that the interval estimate

$$\hat{P}_n(A) - 0.001 < P[A] < \hat{P}_n(A) + 0.001$$

has confidence coefficient $1 - \alpha = 0.99$?

- (b) How many trials n are needed so that the probability $\hat{P}_n(A)$ differs from $P[A]$ by more than 0.1% is less than 0.01?

10.6.1 ● Graph one trace of the sample mean of the Poisson (1) random variable. Calculate (using a central limit theorem approximation) and graph the corresponding 0.9 confidence interval estimate.

10.6.2 ● X is the Bernoulli (1/2) random variable. The sample mean $M_n(X)$ has standard error

$$e_n = \sqrt{\frac{\text{Var}[X]}{n}} = \frac{1}{2\sqrt{n}}.$$

The probability that $M_n(X)$ is within one standard error of p is

$$p_n = P\left[\frac{1}{2} - \frac{1}{2\sqrt{n}} \leq M_n(X) \leq \frac{1}{2} + \frac{1}{2\sqrt{n}}\right].$$

Use the `binomialcdf` function to calculate the exact probability p_n as a function of n . What is the source of the unusual sawtooth pattern? Compare your results to the solution of Quiz 10.6.

10.6.3 ■ Recall that an exponential (λ) random variable X has

$$E[X] = 1/\lambda, \\ \text{Var}[X] = 1/\lambda^2.$$

Thus, to estimate λ from n independent samples $X_1, \dots, X_n$, either of the following techniques should work.

- (a) Calculate the sample mean $M_n(X)$ and form the estimate $\hat{\lambda} = 1/M_n(X)$.
 (b) Calculate the unbiased variance estimate $V'_n(X)$ of Theorem 10.13 and form the estimate $\tilde{\lambda} = 1/\sqrt{V'_n(X)}$.

Use MATLAB to simulate the calculation $\hat{\lambda}$ and $\tilde{\lambda}$ for $m = 1000$ experimental trials to determine which estimate is better.

10.6.4 ♦ $\mathbf{X}$ is 10-dimensional Gaussian ($\mathbf{0}, \mathbf{I}$) random vector. Since $E[\mathbf{X}] = \mathbf{0}$, $\mathbf{R}_{\mathbf{X}} = \mathbf{C}_{\mathbf{X}} = \mathbf{I}$. We will use the method of Problem 10.4.6 and estimate $\mathbf{R}_{\mathbf{X}}$ using the

sample correlation matrix

$$\hat{\mathbf{R}}(n) = \frac{1}{n} \sum_{m=1}^n \mathbf{X}(m) \mathbf{X}'(m).$$

For $n \in \{10, 100, 1000, 10,000\}$, construct a MATLAB simulation to estimate

$$\mathrm{P} \left[\max_{i,j} \left| \hat{R}_{ij} - I_{ij} \right| \geq 0.05 \right].$$

10.6.5 ♦ In terms of parameter a , random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < a - 1, \\ 1 - \frac{1}{[x - (a - 2)]^2} & x \geq a - 1. \end{cases}$$

- (a) Show that $\mathrm{E}[X] = a$ by showing that $\mathrm{E}[X - (a - 2)] = 2$.
 - (b) Generate $m = 100$ traces of the sample mean $M_n(X)$ of length $n = 1000$. Do you observe convergence of the sample mean to $\mathrm{E}[X] = a$?
-

Further Reading: [Kay98] provides detailed, readable coverage of hypothesis testing. [Hay01] presents detection of digital communications signals as a hypothesis test. A collection of challenging homework problems for sections 11.3 and 11.4 are based on bit detection for code division multiple access (CDMA) communications systems. The authoritative treatment of this subject can be found in [Ver98].

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

11.1.1 ● Let L equal the number of flips of a coin up to and including the first flip of heads. Devise a significance test for L at level $\alpha = 0.05$ to test the hypothesis H that the coin is fair. What are the limitations of the test?

11.1.2 ● A course has two recitation sections that meet at different times. On the midterm, the average for section 1 is 5 points higher than the average for section 2. A logical conclusion is that the TA for section 1 is better than the TA for section 2. Using words rather than math, give reasons why this might be the wrong conclusion.

11.1.3 ● Under the null hypothesis H_0 that traffic is typical, the number of call attempts in a 1-second interval (during rush hour) at a mobile telephone switch is a Poisson random variable N with $E[N] = 2.5$. Over a T -second period, the measured call rate is $M = (N_1 + \cdots + N_T)/T$, where $N_1, \dots, N_T$ are iid Poisson random variables identical to N . However, whenever there is unusually heavy traffic (resulting from an accident or bad weather or some other event), the measured call rate M is higher than usual. Based on the observation M , design a significance test to reject the null hypothesis H_0 that traffic is typical at a significance level $\alpha = 0.05$. Justify your choice of the rejection region R . Hint: You may use a Gaussian (central limit theorem) approximation for calculating probabilities with respect to M . How does your test depend on the observation period T ? Explain your answer.

11.1.4 ● A cellular telephone company is upgrading its network to a new (N) transmission system one area at a time, but they

do not announce where the upgrades take place. You have the task of determining whether certain areas have been upgraded. You have decided to use an application in your smartphone to measure the ping time (how long it takes to receive a response to a certain message) in each area. The new system is faster than the old (O) one. It has on average shorter ping times. The probability model for the ping time in milliseconds of the new system is the exponential (60) random variable. Perform a ping test and reject the null hypothesis that the area has the new system if the ping time is greater than t_0 ms.

- Write a formula for α , the significance of the test as a function of t_0 .
- What is the value of t_0 that produces a significance level $\alpha = 0.05$?

11.1.5 ● When a pacemaker factory is operating normally (the null hypothesis H_0), a randomly selected pacemaker fails a “drop test” with probability $q_0 = 10^{-4}$. Each day, an inspector randomly tests pacemakers. Design a significance test for the null hypothesis with significance level $\alpha = 0.01$. Note that drop testing of pacemakers is expensive because the pacemakers that are tested must be discarded. Thus the significance test should try to minimize the number of pacemakers tested.

11.1.6 ■ Let K be the number of heads in $n = 100$ flips of a coin. Devise significance tests for the hypothesis H that the coin is fair such that

- The significance level $\alpha = 0.05$ and the rejection set R has the form $\{|K - E[K]| > c\}$.

- (b) The significance level $\alpha = 0.01$ and the rejection set R has the form $\{K > c'\}$.

11.1.7 ■ When a chip fabrication facility is operating normally, the lifetime of a microchip operated at temperature T , measured in degrees Celsius, is given by an exponential (λ) random variable X with expected value $E[X] = 1/\lambda = (200/T)^2$ years. Occasionally, the chip fabrication plant has contamination problems and the chips tend to fail much more rapidly. To test for contamination problems, each day m chips are subjected to a one-day test at $T = 100^\circ\text{C}$. Based on the number N of chips that fail in one day, design a significance test for the null hypothesis test H_0 that the plant is operating normally.

- Suppose the rejection set of the test is $R = \{N > 0\}$. Find the significance level of the test as a function of m , the number of chips tested.
- How many chips must be tested so that the significance level is $\alpha = 0.01$.
- If we raise the temperature of the test, does the number of chips we need to test increase or decrease?

11.1.8 ■ A group of n people form a football pool. The rules of this pool are simple: 16 football games are played each week. Each contestant must pick the winner of each game against a point *spread*. The contestant who picks the most games correctly over a 16-week season wins the pool. The spread is a point difference d such that picking the favored team is a winning pick only if that team wins by more than d points; otherwise, the pick of the opposing team is a winner. Each pool contestant can study the teams' past histories, performance trends, official injury reports, coaches' weekly press conferences, chat room gossip and any other wisdom that might help in placing a winning bet.

After m weeks, contestant i will have picked W_i games correctly out of $16m$ games. For example, suppose that after $m = 14$ weeks, $16(14) = 224$ games have

been played and that the leader (call him Narayan) has picked 119 games correctly. Does the pool leader Narayan have skills or is he just lucky?

- To address this question, design a significance test to determine whether the pool leader actually has any skill at picking games. Let H_0 denote the null hypothesis that *all* players, including the leader, pick winners in each game with probability $p = 1/2$, independent of the outcome of any other game. Based on the observation of W , the number of winning picks by the pool leader after m weeks of the season, design a one-sided significance test for hypothesis H_0 at significance level $\alpha = 0.05$. You may use a central limit theorem approximation for binomial PMFs as needed.
- Given that Narayan is the leader with 119 winning picks in $m = 14$ weeks in a pool with $n = 38$ contestants, do you reject or accept hypothesis H_0 ?
- How does the significance test depend on picks being made against the point spread?

11.1.9 ♦ A class has $2n$ (a large number) students. The students are separated into two groups A and B , each with n students. Group A students take exam A and earn iid scores $X_1, \dots, X_n$. Group B students take exam B , earning iid scores $Y_1, \dots, Y_n$. The two exams are similar but different; however, the exams were designed so that a student's score X on exam A or Y on exam B have the same expected value and variance $\sigma^2 = 100$. For each exam, we form the sample mean statistic

$$M_A = \frac{X_1 + \dots + X_n}{n},$$

$$M_B = \frac{Y_1 + \dots + Y_n}{n}.$$

Based on the statistic $D = M_A - M_B$, use the central limit theorem to design a significance test at significance level $\alpha = 0.05$ for the hypothesis H_0 that a student's score

on the two exams has the same expected value μ and variance $\sigma^2 = 100$. What is the rejection region if $n = 100$? Make sure to specify any additional assumptions that you need to make; however, try to make as few additional assumptions as possible.

11.2.1 ● In a random hour, the number of call attempts N at a telephone switch has a Poisson distribution with an expected value of either α_0 (hypothesis H_0) or α_1 (hypothesis H_1). For a priori probabilities $P[H_i]$, find the MAP and ML hypothesis testing rules given the observation of N .

11.2.2 ■ The ping time, in milliseconds of a new transmission system, described in Problem 11.1.4 is the exponential (60) random variable N . The ping time of an old system is an exponential random variable O with expected value $\mu_O > 60$ ms. The null hypothesis of a binary hypothesis test is H_0 : The transmission system is the new system. The alternative hypothesis is H_1 : The transmission system is the old system. The probability of a new system is $P[N] = 0.8$. The probability of an old system is $P[O] = 0.2$. A binary hypothesis test measures T milliseconds, the result of one ping test. The decision is H_0 if $T \leq t_0$ ms. Otherwise, the decision is H_1 .

- Write a formula for the false alarm probability as a function of t_0 and μ_O .
- Write a formula for the miss probability as a function of t_0 and μ_O .
- Calculate the maximum likelihood decision time $t_0 = t_{ML}$ for $\mu_O = 120$ ms and $\mu_O = 200$ ms.
- Do you think that t_{MAP} , the maximum a posteriori decision time, is greater than or less than t_{ML} ? Explain your answer.
- Calculate the maximum a posteriori probability decision time $t_0 = t_{MAP}$ for $\mu_O = 120$ ms and $\mu_O = 200$ ms.
- Draw the receiver operating curves for $\mu_O = 120$ ms and $\mu_O = 200$ ms.

11.2.3 ■ An automatic doorbell system rings a bell whenever it detects someone at

the door. The system uses a photodetector such that if a person is present, hypothesis H_1 , the photodetector output N is a Poisson random variable with an expected value of 1300 photons. Otherwise, if no one is there, hypothesis H_0 , the photodetector output is a Poisson random variable with an expected value of 1000. Devise a Neyman–Pearson test for the presence of someone outside the door such that the false alarm probability is $\alpha \leq 10^{-6}$. What is minimum value of P_{MISS} ?

11.2.4 ■ In the radar system of Example 11.4, $P[H_1] = 0.01$. In the case of a false alarm, the system issues an unnecessary alert at the cost of $C_{10} = 1$ unit. The cost of a miss is $C_{01} = 10^4$ units because the target could cause a lot of damage. When the target is present, the voltage is $X = 4 + N$, a Gaussian (4, 1) random variable. When there is no target present, the voltage is $X = N$, the Gaussian (0, 1) random variable. In a binary hypothesis test, the acceptance sets are $A_0 = \{X \leq x_0\}$ and $A_1 = \{X > x_0\}$.

- For the MAP hypothesis test, find the decision threshold $x_0 = x_{MAP}$, the error probabilities P_{FA} and P_{MISS} , and the average cost $E[C]$.
- Compare the MAP test performance against the minimum cost hypothesis test.

11.2.5 ■ In the radar system of Example 11.4, show that the ROC in Figure 11.2 is the result of a Neyman–Pearson test. That is, show that the Neyman–Pearson test is a threshold test with acceptance set $A_0 = \{X \leq x_0\}$. How is x_0 related to the false alarm probability α ?

11.2.6 ♦ A system administrator (and part-time spy) at a classified research facility wishes to use a gateway router for covert communication of research secrets to an outside accomplice. The sysadmin covertly communicates a bit W for every n transmitted packets. To signal $W = 0$, the router does nothing while n regular packets are sent out through the gateway as a

Poisson process of rate λ_0 packets/sec. To signal $W = 1$ the sysadmin injects additional fake outbound packets so that n outbound packets are sent as a Poisson process of rate $2\lambda_0$. The secret communication bits are equiprobable in that $P[W = 1] = P[W = 0] = 1/2$. The sysadmin's accomplice (outside the gateway) monitors the outbound packet transmission process by observing the vector $\mathbf{X} = [X_1, X_2, \dots, X_n]$ of packet interarrival times and guessing the bit W every n packets.

- Find the conditional PDFs $f_{\mathbf{X}|W=0}(\mathbf{x})$ and $f_{\mathbf{X}|W=1}(\mathbf{x})$.
- What are the MAP and ML hypothesis tests for the accomplice to guess either hypothesis H_0 that $W = 0$ or hypothesis H_1 that $W = 1$?
- Let $\hat{W}$ denote the decision of the ML hypothesis test. Use the Chernoff bound to upper bound the error probability $P[\hat{W} = 0|W = 1]$.

11.2.7 ♦ The ping time, in milliseconds, of a new transmission system, described in Problem 11.1.4 is the exponential (60) random variable N . The ping time of an old system is the exponential (120) random variable O . The null hypothesis of a binary hypothesis test is H_0 : The transmission system is the new system. The alternative hypothesis is H_1 : The transmission system is the old system. The probability of a new system is $P[N] = 0.8$. The probability of an old system $P[O] = 0.2$. A binary hypothesis test performs k ping tests and calculates $M_n(T)$, the sample mean of the ping time. The decision is H_0 if $M_n(T) \leq t_0$ ms. Otherwise, the decision is H_1 .

- Use the central limit theorem to write a formula for the false alarm probability as a function of t_0 and k .
- Use the central limit theorem to write a formula for the miss probability as a function of t_0 and k .
- Calculate the maximum likelihood decision time, $t_0 = t_{ML}$, for $k = 9$ ping tests.

- Calculate the maximum a posteriori probability decision time, $t_0 = t_{MAP}$ for $k = 9$ ping tests.
- Draw the receiver operating curves for $k = 9$ ping tests and $k = 16$ ping tests.

11.2.8 ♦ In this problem, we perform the old/new detection test of Problem 11.2.7, except now we monitor k ping tests and observe whether each ping lasts longer than t_0 ms. The random variable M is the number of pings that last longer than t_0 ms. The decision is H_0 if $M \leq m_0$. Otherwise, the decision is H_1 .

- Write a formula for the false alarm probability as a function of t_0 , m_0 , and n .
- Find the maximum likelihood decision number $m_0 = m_{ML}$ for $t_0 = 4.5$ ms and $k = 16$ ping tests.
- Find the maximum a posteriori probability decision number $m_0 = m_{MAP}$ for $t_0 = 4.5$ ms and $k = 16$ ping tests.
- Draw the receiver operating curves for $t_0 = 90$ ms and $t_0 = 60$ ms. In both cases let $k = 16$ ping tests.

11.2.9 ♦ A binary communication system has transmitted signal X , the Bernoulli (1/2) random variable. At the receiver, we observe $Y = VX + W$, where V is a “fading factor” and W is additive noise. Note that V and W are exponential (1) random variables and that X , V , and W are mutually independent. Given the observation Y , we must guess whether $X = 0$ or $X = 1$ was transmitted. Use a binary hypothesis test to determine the rule that minimizes the probability P_e of a decoding error. For the optimum decision rule, calculate P_e .

11.2.10 ♦ In a BPSK amplify-and-forward relay system, a source transmits a random bit $V \in \{-1, 1\}$ every T seconds to a destination receiver via a set of n relay transmitters. $V = 1$ and $V = -1$ are equally likely. In this communication system, the source transmits during the time period $(0, T/2)$

such that relay i receives

$$X_i = \alpha_i V + W_i, \quad i = 1, 2, \dots, n,$$

where the W_i are iid Gaussian $(0, 1)$ random variables representing relay i receiver noise. In the time interval $(T/2, T)$, each relay node amplifies and forwards the received source signal. The destination receiver obtains the vector $\mathbf{Y} = [Y_1 \ \dots \ Y_n]'$ such that

$$Y_i = \beta_i X_i + Z_i, \quad i = 1, 2, \dots, n,$$

where the Z_i are also iid Gaussian $(0, 1)$ random variables. In the following, assume that the parameters α_i and β_i are all non-negative. Also, let H_0 denote the hypothesis that $V = -1$ and H_1 the hypothesis $V = 1$.

- (a) Suppose you build a suboptimal detector based on the sum $Y = \sum_{i=1}^n Y_i$. If $Y > 0$, the receiver guesses H_1 ; otherwise the receiver guesses H_0 . What is the probability of error P_e for this receiver?
- (b) Based on the observation $\mathbf{Y}$, now suppose the destination receiver detector performs a MAP test for hypotheses H_0 or H_1 . What is the MAP detector rule? Simplify your answer as much as possible. Hint: First find the likelihood functions $f_{\mathbf{Y}|H_i}(y)$.
- (c) What is the probability of bit error P_e^* of the MAP detector?
- (d) Compare the two detectors when $n = 4$ and

$$\begin{aligned} (\alpha_1, \beta_1) &= (1, 1), & (\alpha_2, \beta_2) &= (10, 1), \\ (\alpha_3, \beta_3) &= (1, 10), & (\alpha_4, \beta_4) &= (10, 10). \end{aligned}$$

In general, what's bad about the sub-optimal detector?

11.2.11♦ In a BPSK communication system, a source wishes to communicate a random bit $X \in \{-1, 1\}$ to a receiver. Inputs $X = 1$ and $X = -1$ are equally likely. In this system, the source transmits X multiple times. In the i th transmission, the receiver observes $Y_i = X + W_i$, where the W_i

are iid Gaussian $(0, 1)$ noises, independent of X .

- (a) After n transmissions of X , you observe $\mathbf{Y} = \mathbf{y} = [y_1 \ \dots \ y_n]'$. Find $P[X = 1 | \mathbf{Y} = \mathbf{y}]$. Express your answer in terms of the likelihood ratio

$$L(\mathbf{y}) = \frac{f_{\mathbf{Y}|X}(\mathbf{y}|1)}{f_{\mathbf{Y}|X}(\mathbf{y}|1)}.$$

- (b) Suppose after n transmissions, the receiver observes $\mathbf{Y} = \mathbf{y}$ and decides

$$X^* = \begin{cases} 1 & P[X = 1 | \mathbf{Y} = \mathbf{y}] > 1/2, \\ -1 & \text{otherwise.} \end{cases}$$

Find the probability of error $P_e = P[X^* \neq X]$ in terms of the $\Phi(\cdot)$ function. Hint: For P_e calculations, symmetry implies $P_e = P[X^* \neq X | X = 1]$.

- (c) Now suppose the system uses ARQ as follows. If $|\hat{X}_n(\mathbf{y})| < 1 - \epsilon$, the receiver requests that X be retransmitted; otherwise the receiver guesses what is transmitted. In particular, if $\hat{X}_n(\mathbf{y}) > 1 - \epsilon$, the receiver guesses $X^* = 1$. If $\hat{X}_n(\mathbf{y}) < -1 + \epsilon$, the receiver guesses $X^* = -1$. Following the receiver's guess, the transmitter starts sending a new bit. Find upper and lower bounds to $P_e = P[X^* \neq X]$. That is, find ϵ_1 and ϵ_2 such that

$$\epsilon_1 \leq P_e \leq \epsilon_2.$$

11.2.12♦ Suppose in the disk drive factory of Example 11.8, we can observe K , the number of failed devices out of n devices tested. As in the example, let H_i denote the hypothesis that the failure rate is q_i .

- (a) Assuming $q_0 < q_1$, what is the ML hypothesis test based on an observation of K ?
- (b) What are the conditional probabilities of error $P_{\text{FA}} = P[A_1 | H_0]$ and $P_{\text{MISS}} = P[A_0 | H_1]$? Calculate these probabilities for $n = 500$, $q_0 = 10^{-4}$, $q_1 = 10^{-2}$.
- (c) Compare this test to that considered in Example 11.8. Which test is more

reliable? Which test is easier to implement?

11.2.13♦ Consider a binary hypothesis test in which there is a cost associated with each type of decision. In addition to the cost C'_{10} for a false alarm and C'_{01} for a miss, we also have the costs C'_{00} for correctly deciding hypothesis H_0 and the C'_{11} for correctly deciding hypothesis H_1 . Based on the observation of a continuous random vector $\mathbf{X}$, design the hypothesis test that minimizes the total expected cost

$$\begin{aligned} E[C'] = & P[A_1|H_0] P[H_0] C'_{10} \\ & + P[A_0|H_0] P[H_0] C'_{00} \\ & + P[A_0|H_1] P[H_1] C'_{01} \\ & + P[A_1|H_1] P[H_1] C'_{11}. \end{aligned}$$

Show that the decision rule that minimizes total cost is the same as decision rule of the minimum cost test in Theorem 11.3, with the costs C_{01} and C_{10} replaced by the differential costs $C'_{01} - C'_{11}$ and $C'_{10} - C'_{00}$.

11.3.1● In a ternary amplitude shift keying (ASK) communications system, there are three equally likely transmitted signals $\{s_0, s_1, s_2\}$. These signals are distinguished by their amplitudes such that if signal s_i is transmitted, the receiver output will be

$$X = a(i - 1) + N,$$

where a is a positive constant and N is a Gaussian $(0, \sigma_N)$ random variable. Based on the output X , the receiver must decode which symbol s_i was transmitted.

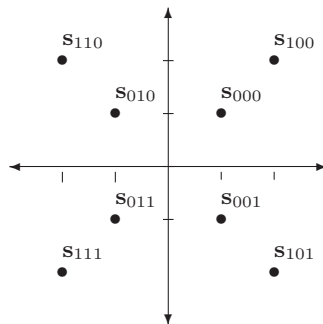
- What are the acceptance sets A_i for the hypotheses H_i that s_i was transmitted?
- What is $P[D_E]$, the probability that the receiver decodes the wrong symbol?

11.3.2● A multilevel QPSK communications system transmits three bits every unit of time. For each possible sequence ijk of three bits, one of eight symbols, $\{s_{000}, s_{001}, \dots, s_{111}\}$, is transmitted. When signal s_{ijk} is transmitted, the receiver out-

put is

$$\mathbf{X} = \mathbf{s}_{ijk} + \mathbf{N},$$

where $\mathbf{N}$ is a Gaussian $(\mathbf{0}, \sigma^2 \mathbf{I})$ random vector. The two-dimensional signal vectors $\mathbf{s}_{000}, \dots, \mathbf{s}_{111}$ are



Let H_{ijk} denote the hypothesis that s_{ijk} was transmitted. The receiver output $\mathbf{X} = [X_1 \ X_2]'$ is used to decide the acceptance sets $\{A_{000}, \dots, A_{111}\}$. If all eight symbols are equally likely, sketch the acceptance sets.

11.3.3■ An M -ary quadrature amplitude modulation (QAM) communications system can be viewed as a generalization of the QPSK system described in Example 11.13. In the QAM system, one of M equally likely symbols $s_0, \dots, s_{M-1}$ is transmitted every unit of time. When symbol s_i is transmitted, the receiver produces the two-dimensional vector output

$$\mathbf{X} = \mathbf{s}_i + \mathbf{N},$$

where $\mathbf{N}$ has iid Gaussian $(0, \sigma^2)$ components. Based on the output $\mathbf{X}$, the receiver must decide which symbol was transmitted. Design a hypothesis test that maximizes the probability of correctly deciding what symbol was sent. Hint: Following Example 11.13, describe the acceptance set in terms of the vectors

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{s}_i = \begin{bmatrix} s_{i1} \\ s_{i2} \end{bmatrix}.$$

11.3.4 ■ Suppose a user of the multilevel QPSK system needs to decode only the third bit k of the message ijk . For $k = 0, 1$, let H_k denote the hypothesis that the third bit was k . What are the acceptance sets A_0 and A_1 ? What is $P[B_3]$, the probability that the third bit is in error?

11.3.5 ♦ The QPSK system of Example 11.13 can be generalized to an M -ary phase shift keying (MPSK) system with $M > 4$ equally likely signals. The signal vectors are $\{\mathbf{s}_0, \dots, \mathbf{s}_{M-1}\}$, where

$$\mathbf{s}_i = \begin{bmatrix} s_{i1} \\ s_{i2} \end{bmatrix} = \begin{bmatrix} \sqrt{E} \cos \theta_i \\ \sqrt{E} \sin \theta_i \end{bmatrix}$$

and $\theta_i = 2\pi i/M$. When the i th message is sent, the received signal is $\mathbf{X} = \mathbf{s}_i + \mathbf{N}$ where $\mathbf{N}$ is a Gaussian $(\mathbf{0}, \sigma^2 \mathbf{I})$ noise vector.

- Sketch the acceptance set A_i for the hypothesis H_i that $\mathbf{s}_i$ was transmitted.
- Find the largest value of d such that

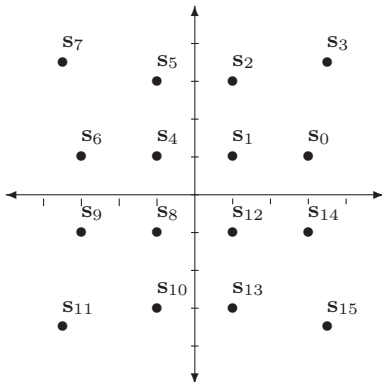
$$\{\mathbf{x} \mid \|\mathbf{x} - \mathbf{s}_i\| \leq d\} \subset A_i.$$

- Use d to find an upper bound for the probability of error.

11.3.6 ♦ A modem uses QAM (see Problem 11.3.3) to transmit one of 16 symbols, $s_0, \dots, s_{15}$, every 1/600 seconds. When signal s_i is transmitted, the receiver output is

$$\mathbf{X} = \mathbf{s}_i + \mathbf{N}.$$

The signal vectors $\mathbf{s}_0, \dots, \mathbf{s}_{15}$ are



- Sketch the acceptance sets based on the receiver outputs X_1, X_2 . Hint: Apply the solution to Problem 11.3.3.
- Let H_i be the event that symbol s_i was transmitted and let C be the event that the correct symbol is decoded. What is $P[C|H_1]$?
- Argue that $P[C] \geq P[C|H_1]$.

11.3.7 ♦ For the QPSK communications system of Example 11.13, identify the acceptance sets for the MAP hypothesis test when the symbols are not equally likely. Sketch the acceptance sets when $\sigma = 0.8$, $E = 1$, $P[H_0] = 1/2$, and $P[H_1] = P[H_2] = P[H_3] = 1/6$.

11.3.8 ♦ In a code division multiple access (CDMA) communications system, k users share a radio channel using a set of n -dimensional code vectors $\{\mathbf{S}_1, \dots, \mathbf{S}_k\}$ to distinguish their signals. The dimensionality factor n is known as the processing gain. Each user i transmits independent data bits X_i such that the vector $\mathbf{X} = [X_1 \ \dots \ X_k]'$ has iid components with $P_{X_i}(1) = P_{X_i}(-1) = 1/2$. The received signal is

$$\mathbf{Y} = \sum_{i=1}^k X_i \sqrt{p_i} \mathbf{S}_i + \mathbf{N},$$

where $\mathbf{N}$ is a Gaussian $(\mathbf{0}, \sigma^2 \mathbf{I})$ noise vector. From the observation $\mathbf{Y}$, the receiver performs a multiple hypothesis test to decode the data bit vector $\mathbf{X}$.

- Show that in terms of vectors,

$$\mathbf{Y} = \mathbf{S} \mathbf{P}^{1/2} \mathbf{X} + \mathbf{N},$$

where $\mathbf{S}$ is an $n \times k$ matrix with i th column $\mathbf{S}_i$ and $\mathbf{P}^{1/2} = \text{diag}[\sqrt{p_1}, \dots, \sqrt{p_k}]$ is a $k \times k$ diagonal matrix.

- Given $\mathbf{Y} = \mathbf{y}$, show that the MAP and ML detectors for $\mathbf{X}$ are the same and are given by

$$\mathbf{x}^*(\mathbf{y}) = \arg \min_{\mathbf{x} \in B_k} \|\mathbf{y} - \mathbf{S} \mathbf{P}^{1/2} \mathbf{x}\|.$$

where B_k is the set of all k dimensional vectors with ± 1 elements.

- (c) How many hypotheses does the ML detector need to evaluate?

11.3.9 ♦ For the CDMA communications system of Problem 11.3.8, a detection strategy known as *decorrelation* applies a transformation to $\mathbf{Y}$ to generate

$$\tilde{\mathbf{Y}} = (\mathbf{S}'\mathbf{S})^{-1}\mathbf{S}'\mathbf{Y} = \mathbf{P}^{1/2}\mathbf{X} + \tilde{\mathbf{N}}$$

where $\tilde{\mathbf{N}} = (\mathbf{S}'\mathbf{S})^{-1}\mathbf{S}'\mathbf{N}$ is still a Gaussian noise vector with expected value $E[\tilde{\mathbf{N}}] = \mathbf{0}$. Decorrelation separates the signals in that the i th component of $\tilde{\mathbf{Y}}$ is

$$\tilde{Y}_i = \sqrt{p_i}X_i + \tilde{N}_i,$$

which is the same as a single-user receiver output of the binary communication system of Example 11.6. For equally likely inputs $X_i = 1$ and $X_i = -1$, Example 11.6 showed that the optimal (minimum probability of bit error) decision rule based on the receiver output $\tilde{Y}_i$ is

$$\hat{X}_i = \text{sgn}(\tilde{Y}_i).$$

Although this technique requires the code vectors $\mathbf{S}_1, \dots, \mathbf{S}_k$ to be linearly independent, the number of hypotheses that must be tested is greatly reduced in comparison to the optimal ML detector introduced in Problem 11.3.8. In the case of linearly independent code vectors, is the decorrelator optimal? That is, does it achieve the same bit error rate (BER) as the optimal ML detector?

11.4.1 ● A wireless pressure sensor (buried in the ground) reports a discrete random variable X with range $S_X = \{1, 2, \dots, 20\}$ to signal the presence of an object. Given an observation X and a threshold x_0 , we decide that an object is present (hypothesis H_1) if $X > x_0$; otherwise we decide that no object is present (hypothesis H_0). Under

hypothesis H_i , X has conditional PMF

$$P_{X|H_i}(x) = \begin{cases} \frac{(1-p_i)p_i^{x-1}}{1-p_i^{20}} & x = 1, 2, \dots, 20, \\ 0 & \text{otherwise,} \end{cases}$$

where $p_0 = 0.99$ and $p_1 = 0.9$. Calculate and plot the false alarm and miss probabilities as a function of the detection threshold x_0 . Calculate the discrete receiver operating curve (ROC) specified by x_0 .

11.4.2 ● For the binary communications system of Example 11.7, graph the error probability P_{ERR} as a function of p , the probability that the transmitted signal is 0. For the signal-to-noise voltage ratio, consider $v/\sigma \in \{0.1, 1, 10\}$. What values of p minimize P_{ERR} ? Why are those values not practical?

11.4.3 ● For the squared distortion communications system of Example 11.14 with $v = 1.5$ and $d = 0.5$, find the value of T that minimizes P_e .

11.4.4 ■ A poisonous gas sensor reports continuous random variable X . In the presence of toxic gases, hypothesis H_1 ,

$$f_{X|H_1}(x) = \begin{cases} (x/8)e^{-x^2/16} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

In the absence of dangerous gases, X has conditional PDF

$$f_{X|H_0}(x) = \begin{cases} (1/2)e^{-x/2} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Devise a hypothesis test that determines the presence of poisonous gases. Plot the false alarm and miss probabilities for the test as a function of the decision threshold. Lastly, plot the corresponding receiver operating curve.

11.4.5 ♦ Simulate the M -ary PSK system in Problem 11.3.5 for $M = 8$ and $M = 16$. Let $\hat{P}_{\text{ERR}}$ denote the relative frequency of symbol errors in the simulated transmission in 10^5 symbols. For each value of M , graph $\hat{P}_{\text{ERR}}$, as a function of the signal-to-noise power ratio (SNR) $\gamma = E/\sigma^2$. Consider

$10 \log_{10} \gamma$, the SNR in dB, ranging from 0 to 30 dB.

11.4.6♦♦ In this problem, we evaluate the bit error rate (BER) performance of the CDMA communications system introduced in Problem 11.3.8. In our experiments, we will make the following additional assumptions.

- In practical systems, code vectors are generated pseudorandomly. We will assume the code vectors are random. For each transmitted data vector $\mathbf{X}$, the code vector of user i will be $\mathbf{S}_i = \frac{1}{\sqrt{n}} [S_{i1} \ S_{i2} \ \cdots \ S_{in}]'$, where the components S_{ij} are iid random variables such that $P_{S_{ij}}(1) = P_{S_{ij}}(-1) = 1/2$. Note that the factor $1/\sqrt{n}$ is used so that each code vector $\mathbf{S}_i$ has length 1: $\|\mathbf{S}_i\|^2 = \mathbf{S}_i' \mathbf{S}_i = 1$.
 - Each user transmits at 6dB SNR. For convenience, assume $P_i = p = 4$ and $\sigma^2 = 1$.
- (a) Use MATLAB to simulate a CDMA system with processing gain $n = 16$. For each experimental trial, generate a random set of code vectors $\{\mathbf{S}_i\}$, data vector $\mathbf{X}$, and noise vector $\mathbf{N}$. Find the ML estimate $\mathbf{x}^*$ and count the number of bit errors; i.e., the number of positions in which $x_i^* \neq X_i$. Use the relative frequency of bit errors as an estimate of the probability of bit error. Consider $k = 2, 4, 8, 16$ users. For each value of k , perform enough trials so that bit errors are generated on 100 independent trials. Explain why your simulations take so long.
- (b) For a simpler detector known as the matched filter, when $\mathbf{Y} = \mathbf{y}$, the detector decision for user i is

$$\hat{x}_i = \text{sgn}(\mathbf{S}_i' \mathbf{y})$$

where $\text{sgn}(x) = 1$ if $x > 0$, $\text{sgn}(x) = -1$ if $x < 0$, and otherwise $\text{sgn}(x) = 0$. Compare the bit error rate of the matched filter and the maximum likelihood detectors. Note that the matched

filter is also called a single user detector since it can detect the bits of user i without the knowledge of the code vectors of the other users.

11.4.7♦♦ For the CDMA system in Problem 11.3.8, we wish to use MATLAB to evaluate the bit error rate (BER) performance of the decorrelator introduced Problem 11.3.9. In particular, we want to estimate P_e , the probability that for a set of randomly chosen code vectors, that a randomly chosen user's bit is decoded incorrectly at the receiver.

- (a) For a k user system with a fixed set of code vectors $\mathbf{S}_1, \dots, \mathbf{S}_k$, let $\mathbf{S}$ denote the matrix with $\mathbf{S}_i$ as its i th column. Assuming that the matrix inverse $(\mathbf{S}'\mathbf{S})^{-1}$ exists, write an expression for $P_{e,i}(\mathbf{S})$, the probability of error for the transmitted bit of user i , in terms of $\mathbf{S}$ and the $Q(\cdot)$ function. For the same fixed set of code vectors $\mathbf{S}$, write an expression for P_e , the probability of error for the bit of a randomly chosen user.
- (b) In the event that $(\mathbf{S}'\mathbf{S})^{-1}$ does not exist, we assume the decorrelator flips a coin to guess the transmitted bit of each user. What are $P_{e,i}$ and P_e in this case?
- (c) For a CDMA system with processing gain $n = 32$ and k users, each with SNR 6 dB, write a MATLAB program that averages over randomly chosen matrices $\mathbf{S}$ to estimate P_e for the decorrelator. Note that unlike the case for Problem 11.4.6, simulating the transmission of bits is not necessary. Graph your estimate $\hat{P}_e$ as a function of k .

11.4.8♦♦ Simulate the multi-level QAM system of Problem 11.3.4. Estimate the probability of symbol error and the probability of bit error as a function of the noise variance σ^2 .

11.4.9♦♦ In Problem 11.4.5, we used simulation to estimate the probability of *symbol* error. For transmitting a binary bit stream

over an MPSK system, we set each $M = 2^N$ and each transmitted symbol corresponds to N bits. For example, for $M = 16$, we map each four-bit input $b_3b_2b_1b_0$ to one of 16 symbols. A simple way to do this is binary index mapping: transmit $\mathbf{s}_i$ when $b_3b_2b_1b_0$ is the binary representation of i . For example, the bit input 1100 is mapped to the transmitted signal $\mathbf{s}_{12}$. Symbol errors in the communication system cause bit errors. For example if $\mathbf{s}_1$ is sent but noise causes $\mathbf{s}_2$ to be decoded, the input bit sequence $b_3b_2b_1b_0 = 0001$ is decoded as $\hat{b}_3\hat{b}_2\hat{b}_1\hat{b}_0 = 0010$, resulting in 2 correct bits and 2 bit errors. In this problem, we use MATLAB to investigate how the mapping of bits to symbols affects the probability of bit error. For our preliminary investigation, it will be sufficient to map the three bits $b_2b_1b_0$ to the $M = 8$ PSK system of Problem 11.3.5.

- (a) Find the acceptance sets $\{A_0, \dots, A_7\}$.
- (b) Simulate m trials of the transmission of symbol $\mathbf{s}_0$. Estimate the probabilities $\{P_{0j} | j = 0, 1, \dots, 7\}$, that the receiver output is $\mathbf{s}_j$ when $\mathbf{s}_0$ was sent. By symmetry, use the set $\{P_{0j}\}$ to determine P_{ij} for all i and j .

- (c) Let $\mathbf{b}(i) = [b_2(i) \ b_1(i) \ b_0(i)]$ denote the input bit sequence that is mapped to $\mathbf{s}_i$. Let d_{ij} denote the number of bit positions in which $\mathbf{b}(i)$ and $\mathbf{b}(j)$ differ. For a given mapping, the bit error rate (BER) is

$$\text{BER} = \frac{1}{M} \sum_i \sum_j P_{ij} d_{ij}.$$

- (d) Estimate the BER for the binary index mapping.
- (e) The Gray code is perhaps the most commonly used mapping:

$\mathbf{b}$	000	001	010	011	100	101	110	111
$\mathbf{s}_i$	$\mathbf{s}_0$	$\mathbf{s}_1$	$\mathbf{s}_3$	$\mathbf{s}_2$	$\mathbf{s}_7$	$\mathbf{s}_6$	$\mathbf{s}_4$	$\mathbf{s}_5$

Does the Gray code reduce the BER compared to the binary index mapping?

11.4.10 ♦♦ Continuing Problem 11.4.9, in the mapping of the bit sequence $b_2b_1b_0$ to the symbols $\mathbf{s}_i$, we wish to determine the probability of error for each input bit b_i . Let q_i denote the probability that bit b_i is decoded in error. Determine q_0 , q_1 , and q_2 for both the binary index mapping as well as the Gray code mapping.

Quiz 12.5

Estimate the Gaussian (0,1) random variable X using the observation vector $\mathbf{Y} = \mathbf{1}X + \mathbf{W}$, where $\mathbf{1}$ is the vector of 20 1's. The noise vector $\mathbf{W} = [W_0 \ \cdots \ W_{19}]'$ is independent of X , has zero expected value, and has a correlation matrix with i, j th entry $R_{\mathbf{W}}(i, j) = c^{|i-j|-1}$. Find $\hat{X}_L(\mathbf{Y})$, the linear MMSE estimate of X given $\mathbf{Y}$. For c in the range $0 < c < 1$, what value of c minimizes the mean square error of the estimate?

Further Reading: The final chapter of [WS01] presents the basic theory of estimation of random variables as well as extensions to stochastic process estimation in the time domain and frequency domain.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

12.1.1 ● Generalizing the solution of Example 12.2, let the call duration T be an exponential (λ) random variable. For $t_0 > 0$, show that the minimum mean square error estimate of T , given that $T > t_0$ is

$$\hat{T} = t_0 + E[T].$$

12.1.2 ● X and Y have the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6(y-x) & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_X(x)$?
- (b) What is the blind estimate $\hat{x}_B$?
- (c) What is the minimum mean square error estimate of X given $X < 0.5$?
- (d) What is $f_Y(y)$?
- (e) What is the blind estimate $\hat{y}_B$?
- (f) What is the minimum mean square error estimate of Y given $Y > 0.5$?

12.1.3 ■ X and Y have the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_X(x)$?
- (b) What is the blind estimate $\hat{x}_B$?

- (c) What is the minimum mean square error estimate of X given $X > 1/2$?
- (d) What is $f_Y(y)$?
- (e) What is the blind estimate $\hat{y}_B$?
- (f) What is the minimum mean square error estimate of Y given $X > 1/2$?

12.1.4 ● X and Y have the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6(y-x) & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_{X|Y}(x|y)$?
- (b) What is $\hat{x}_M(y)$, the MMSE estimate of X given $Y = y$?
- (c) What is $f_{Y|X}(y|x)$?
- (d) What is $\hat{y}_M(x)$, the MMSE estimate of Y given $X = x$?

12.1.5 ■ X and Y have the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_{X|Y}(x|y)$?
- (b) What is $\hat{x}_M(y)$, the MMSE estimate of X given $Y = y$?
- (c) What is

$$e^*(0.5) = E[(X - \hat{x}_M(0.5))^2 | Y = 0.5],$$

the minimum mean square error of the estimate of X given $Y = 0.5$?

12.1.6 ■ A signal X and noise Z are independent Gaussian $(0, 1)$ random variables, and $Y = X + Z$ is a noisy observation of the signal X . Usually, we want to use Y to estimate of X ; however, in this problem we will use Y to estimate the noise Z .

- Find $\hat{Z}(Y)$, the MMSE estimator of Z given Y .
- Find the mean squared error $e = E[(Z - \hat{Z}(Y))^2]$.

12.1.7 ♦ Random variable $Y = X - Z$ is a noisy observation of the continuous random variable X . The noise Z has zero expected value and unit variance and is independent of X . Find the conditional expectation $E[X|Y]$.

12.1.8 ♦ In a BPSK communication system, a source wishes to communicate a random bit X to a receiver. The possible inputs $X = 1$ and $X = -1$ are equally likely. In this system, the source transmits X multiple times. In the i th transmission, the receiver observes $Y_i = X + W_i$. After n transmissions of X , the receiver has observed $\mathbf{Y} = \mathbf{y} = [y_1 \ \cdots \ y_n]'$.

- Find $\hat{X}_n(\mathbf{y})$, the MMSE estimate of X given the observation $\mathbf{Y} = \mathbf{y}$. Express your answer in terms of the likelihood ratio

$$L(\mathbf{y}) = \frac{f_{\mathbf{Y}|X}(\mathbf{y}|1)}{f_{\mathbf{Y}|X}(\mathbf{y}|-1)}.$$

- Simplify your answer when the W_i are iid Gaussian $(0, 1)$ random variables, independent of X .

12.2.1 ■ Random variables X and Y have joint PMF

$P_{X,Y}(x, y)$	$y = -3$	$y = -1$	$y = 1$	$y = 3$
$x = -1$	1/6	1/8	1/24	0
$x = 0$	1/12	1/12	1/12	1/12
$x = 1$	0	1/24	1/8	1/6

- Find the marginal probability mass functions $P_X(x)$ and $P_Y(y)$.
- Are X and Y independent?
- Find $E[X]$, $\text{Var}[X]$, $E[Y]$, $\text{Var}[Y]$, and $\text{Cov}[X, Y]$.
- Let $\hat{X}(Y) = aY + b$ be a linear estimator of X . Find a^* and b^* , the values of a and b that minimize the mean square error e_L .
- What is e_L^* , the minimum mean square error of the optimum linear estimate?
- Find $P_{X|Y}(x|-3)$, the conditional PMF of X given $Y = -3$.
- Find $\hat{x}_M(-3)$, the optimum (nonlinear) mean square estimator of X given $Y = -3$.
- Find the mean square error

$$e^*(-3) = E[(X - \hat{x}_M(-3))^2 | Y = -3]$$

of this estimate.

12.2.2 ● A telemetry voltage V , transmitted from a position sensor on a ship's rudder, is a random variable with PDF

$$f_V(v) = \begin{cases} 1/12 & -6 \leq v \leq 6, \\ 0 & \text{otherwise.} \end{cases}$$

A receiver in the ship's control room receives $R = V + X$. The random variable X is a Gaussian $(0, \sqrt{3})$ noise voltage that is independent of V . The receiver uses R to calculate a linear estimate of the telemetry voltage: $\hat{V} = aR + b$. Find

- the expected received voltage $E[R]$,
- the variance $\text{Var}[R]$ of the received voltage,
- the covariance $\text{Cov}[V, R]$ of the transmitted and received voltages,
- a^* and b^* , the optimum coefficients in the linear estimate,
- e_L^* , the minimum mean square error of the estimate.

12.2.3 Random variables X and Y have joint PMF given by the following table:

$P_{X,Y}(x,y)$	$y = -1$	$y = 0$	$y = 1$
$x = -1$	3/16	1/16	0
$x = 0$	1/6	1/6	1/6
$x = 1$	0	1/8	1/8

We estimate Y by $\hat{Y}_L(X) = aX + b$.

- Find a and b to minimize the mean square estimation error.
- What is the minimum mean square error e_L^* ?

12.2.4 The random variables X and Y have the joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} 2(y+x) & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

What is $\hat{X}_L(Y)$, the linear minimum mean square error estimate of X given Y ?

12.2.5 For random variables X and Y from Problem 12.1.4, find $\hat{X}_L(Y)$, the linear minimum mean square error estimator of X given Y .

12.2.6 Random variable X has a second-order Erlang PDF

$$f_X(x) = \begin{cases} \lambda x e^{-\lambda x} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Given $X = x$, Y is a uniform $(0, x)$ random variable. Find

- the MMSE estimate of Y given $X = x$, $\hat{y}_M(x)$,
- the MMSE estimate of X given $Y = y$, $\hat{x}_M(y)$,
- the LMSE estimate of Y given X , $\hat{Y}_L(X)$,
- the LMSE estimate of X given Y , $\hat{X}_L(Y)$.

12.2.7 Random variable R has an exponential PDF with expected value 1. Given $R = r$, X has an exponential PDF with expected value $1/r$. Find

- the MMSE estimate of R given $X = x$, $\hat{r}_M(x)$,
- the MMSE estimate of X given $R = r$, $\hat{x}_M(r)$,
- the LMSE estimate of R given X , $\hat{R}_L(X)$,
- the LMSE estimate of X given R , $\hat{X}_L(R)$.

12.2.8 For random variables X and Y , we wish to use Y to estimate X . However, our estimate must be of the form $\hat{X} = aY$.

- Find a^* , the value of a that minimizes the mean square error $e = E[(X - aY)^2]$.
- For $a = a^*$, what is the minimum mean square error e^* ?
- Under what conditions is $\hat{X}$ the LMSE estimate of X ?

12.2.9 Here are four different joint PMFs:

$P_{X,Y}(x,y)$	$x = -1$	$x = 0$	$x = 1$
$y = -1$	1/9	1/9	1/9
$y = 0$	1/9	1/9	1/9
$y = 1$	1/9	1/9	1/9

$P_{U,V}(u,v)$	$u = -1$	$u = 0$	$u = 1$
$v = -1$	0	0	1/3
$v = 0$	0	1/3	0
$v = 1$	1/3	0	0

$P_{S,T}(s,t)$	$s = -1$	$s = 0$	$s = 1$
$t = -1$	1/6	0	1/6
$t = 0$	0	1/3	0
$t = 1$	1/6	0	1/6

$P_{Q,R}(q,r)$	$q = -1$	$q = 0$	$q = 1$
$r = -1$	1/12	1/12	1/6
$r = 0$	1/12	1/6	1/12
$r = 1$	1/6	1/12	1/12

- For each pair of random variables, indicate whether the two random variables are independent, and compute the correlation coefficient ρ .

- (b) Compute the least mean square linear estimator $\hat{U}_L(V)$ of U given V . What is the mean square error? Do the same for the pairs X, Y , Q, R , and S, T .

12.2.10 ♦ Random variable $Y = X - Z$ is a noisy observation of the continuous random variable X . The noise Z has zero expected value and unit variance and is independent of X . Consider the following argument:

Since $X = Y + Z$, we see that if $Y = y$, then $X = y + Z$. Thus, by Theorem 6.4, the conditional PDF of X given $Y = y$ is $f_{X|Y}(x|y) = f_Z(x - y)$. It follows that

$$\begin{aligned} E[X|Y = y] &= \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx \\ &= \int_{-\infty}^{\infty} x f_Z(x - y) dx. \end{aligned}$$

With the variable substitution, $z = x - y$,

$$\begin{aligned} E[X|Y = y] &= \int_{-\infty}^{\infty} (z + y) f_Z(z) dz \\ &= E[Z] + y = y. \end{aligned}$$

We conclude that $E[X|Y] = Y$. Since $E[X|Y]$ is optimal in the mean square sense, we conclude that the optimal linear estimator $\hat{X}(Y) = \hat{a}Y$ must satisfy $\hat{a} = 1$.

Prove that this conclusion is wrong. What is the error in the above argument? Hint: Find the LMSE estimator $\hat{X}_L(Y) = aY$.

12.3.1 ■ Suppose that in Quiz 12.3, R , measured in meters, has a uniform PDF over $[0, 1000]$. Find the MAP estimate of R given $X = x$. In this case, are the MAP and ML estimators the same?

12.3.2 ♦ Let R be an exponential random variable with expected value $1/\mu$. If $R = r$, then over an interval of length T , the number of phone calls N that arrive at a telephone switch has a Poisson PMF with expected value rT .

- (a) Find the MMSE estimate of N given R .
 (b) Find the MAP estimate of N given R .
 (c) Find the ML estimate of N given R .

12.3.3 ♦ Let R be an exponential random variable with expected value $1/\mu$. If $R = r$, then over an interval of length T , the number of phone calls N that arrive at a telephone switch has a Poisson PMF with expected value rT .

- (a) Find the MMSE estimate of R given N .
 (b) Find the MAP estimate of R given N .
 (c) Find the ML estimate of R given N .

12.3.4 ♦ Flip a coin n times. For each flip, the probability of heads is $Q = q$ independent of all other flips. Q is a uniform $(0, 1)$ random variable. K is the number of heads in n flips.

- (a) What is the ML estimator of Q given K ?
 (b) What is the PMF of K ? What is $E[K]$?
 (c) What is the conditional PDF $f_{Q|K}(q|k)$?
 (d) Find the MMSE estimator of Q given $K = k$.

12.4.1 ● You would like to know a sample value of X , a Gaussian $(0, 4)$ random variable. However, you only can observe noisy observations of the form $Y_i = X + N_i$. In terms of a vector of noisy observations, you observe

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} X + \begin{bmatrix} N_1 \\ N_2 \end{bmatrix},$$

where N_1 is a Gaussian $(0, 1)$ random variable and N_2 is a Gaussian $(0, 2)$ random variable. Under the assumption that X , N_1 , and N_2 are mutually independent, answer the following questions:

- (a) Suppose you use Y_1 as an estimate of X . The error in the estimate is $D_1 = Y_1 - X$. What are the expected error $E[D_1]$ and the expected squared error $E[D_1^2]$?
 (b) Suppose we use $Y_3 = (Y_1 + Y_2)/2$ as an estimate of X . The error for this estimate is $D_3 = Y_3 - X$. Find the expected squared error $E[D_3^2]$. Is Y_3 or Y_1 a better estimate of X ?

- (c) Let $Y_4 = \mathbf{A}\mathbf{Y}$ where $\mathbf{A} = \begin{bmatrix} a & 1-a \end{bmatrix}$ is a 1×2 matrix. Let $D_4 = Y_4 - X$ denote the error in using Y_4 as an estimate for X . In terms of a , what is the expected squared error $E[D_4^2]$? What value of a minimizes $E[D_4^2]$?

12.4.2 $\bullet$ $\mathbf{X}$ is a three-dimensional random vector with $E[\mathbf{X}] = \mathbf{0}$ and autocorrelation matrix $\mathbf{R}_{\mathbf{X}}$ with elements $r_{ij} = (-0.80)^{|i-j|}$. Use X_1 and X_2 to form a linear estimate of X_3 : $\hat{X}_3 = a_1X_2 + a_2X_1$.

- (a) What are the optimum coefficients $\hat{a}_1$ and $\hat{a}_2$ and corresponding minimum mean square error e_L^* ?
- (b) Use X_2 to form a linear estimate of X_3 : $\hat{X}_3 = aX_2 + b$. What are the optimum coefficients a^* and b^* and corresponding minimum mean square error e_L^* ?

12.4.3 $\blacksquare$ $\mathbf{X}$ is a 3-dimensional random vector with $E[\mathbf{X}] = \mathbf{0}$ and autocorrelation matrix $\mathbf{R}_{\mathbf{X}}$ with elements

$$R_{\mathbf{X}}(i, j) = 1 - 0.25|i - j|.$$

$\mathbf{Y}$ is a two-dimensional random vector with

$$Y_1 = X_1 + X_2, \quad Y_2 = X_2 + X_3.$$

Use $\mathbf{Y}$ to form $\hat{X}_1 = \begin{bmatrix} a_1 & a_2 \end{bmatrix} \mathbf{Y}$, a linear estimate of X_1 .

- (a) Find the optimum coefficients $\hat{a}_1$ and $\hat{a}_2$ and the minimum mean square error e_L^* .
- (b) Use Y_1 to form a linear estimate of X_1 : $\hat{X}_1 = aY_1 + b$. What are the optimum coefficients a^* and b^* ? What is the minimum mean square error e_L^* ?

12.4.4 $\blacksquare$ $\mathbf{X}$ is a three-dimensional random vector with $E[\mathbf{X}] = \mathbf{0}$ and correlation matrix $\mathbf{R}_{\mathbf{X}}$ with elements

$$R_{\mathbf{X}}(i, j) = 1 - 0.25|i - j|.$$

$\mathbf{W}$ is a two-dimensional random vector, independent of $\mathbf{X}$, with $E[\mathbf{W}] = \mathbf{0}$, $E[W_1W_2] = 0$, and $E[W_1^2] = E[W_2^2] = 0.1$.

$\mathbf{Y}$ is a two-dimensional random vector with

$$\begin{aligned} Y_1 &= X_1 + X_2 + W_1, \\ Y_2 &= X_2 + X_3 + W_2. \end{aligned}$$

Use $\mathbf{Y}$ to form $\hat{X}_1 = \begin{bmatrix} a_1 & a_2 \end{bmatrix} \mathbf{Y}$, a linear estimate of X_1 .

- (a) Find the optimum coefficients $\hat{a}_1$ and $\hat{a}_2$ and minimum mean square error e_L^* .
- (b) Use Y_1 to form a linear estimate of X_1 : $\hat{X}_1 = aY_1 + b$. What are the optimum coefficients a^* and b^* ? What is the minimum mean square error e_L^* ?

12.4.5 $\blacksquare$ Suppose

$$Y_k = q_0 + q_1k + q_2k^2 + Z_k,$$

where $q_0 + q_1k + q_2k^2$ is an unknown quadratic function of k and Z_k is a sequence of iid Gaussian $(0, 1)$ noise random variables. We wish to estimate the unknown parameters q_0 , q_1 , and q_2 of the quadratic function. Suppose we assume q_0 , q_1 , and q_2 are samples of iid Gaussian $(0, 1)$ random variables. Find the optimum linear estimator $\hat{\mathbf{Q}}(\mathbf{Y})$ of $\mathbf{Q} = \begin{bmatrix} q_0 & q_1 & q_2 \end{bmatrix}'$ given the observation $\mathbf{Y} = \begin{bmatrix} Y_1 & \cdots & Y_n \end{bmatrix}'$.

12.4.6 $\blacksquare$ $\mathbf{X}$ is a three-dimensional random vector with $E[\mathbf{X}] = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}'$ and correlation matrix $\mathbf{R}_{\mathbf{X}}$ with elements

$$R_{\mathbf{X}}(i, j) = 1 - 0.25|i - j|.$$

$\mathbf{W}$ is a two-dimensional random vector, independent of $\mathbf{X}$, with $E[\mathbf{W}] = \mathbf{0}$, $E[W_1W_2] = 0$, and

$$E[W_1^2] = E[W_2^2] = 0.1.$$

$\mathbf{Y}$ is a two-dimensional random vector with

$$\begin{aligned} Y_1 &= X_1 + X_2 + W_1, \\ Y_2 &= X_2 + X_3 + W_2. \end{aligned}$$

Use $\mathbf{Y}$ to form a linear estimate of X_1 :

$$\hat{X}_1 = \begin{bmatrix} \hat{a}_1 & \hat{a}_2 \end{bmatrix} \mathbf{Y} + \hat{b}.$$

- (a) What are the optimum coefficients $\hat{a}_1$, $\hat{a}_2$, and $\hat{b}$?

- (b) Find the MMSE e_L^* .
- (c) Use Y_1 to form a linear estimate of X_1 : $\hat{X}_1 = aY_1 + b$. What are the optimum coefficients a^* and b^* ? What is the minimum mean square error e_L^* ?

12.4.7 ■ When X and Y have expected values $\mu_X = \mu_Y = 0$, Theorem 12.3 says that $\hat{X}_L(Y) = \rho_{X,Y} \frac{\sigma_X}{\sigma_Y} Y$. Show that this result is a special case of Theorem 12.8 when random vector $\mathbf{Y}$ is the one-dimensional random variable Y .

12.4.8 ■ Prove the following theorem: $\mathbf{X}$ is an n -dimensional random vector with $E[\mathbf{X}] = \mathbf{0}$ and autocorrelation matrix $\mathbf{R}_X$ with elements $r_{ij} = c^{|i-j|}$, where $|c| < 1$. The optimum linear estimator of X_n ,

$$\hat{X}_n = a_1 X_{n-1} + a_2 X_{n-2} + \cdots + a_{n-1} X_1,$$

is $\hat{X}_n = cX_{n-1}$. The minimum mean square estimation error is $e_L^* = 1 - c^2$. Hint: Consider the $n-1$ equations $\partial e_L / \partial a_i = 0$.

12.4.9 ♦ In the CDMA multiuser communications system introduced in Problem 11.3.8, each user i transmits an independent data bit X_i such that the vector $\mathbf{X} = [X_1 \cdots X_n]'$ has iid components with $P_{X_i}(1) = P_{X_i}(-1) = 1/2$. The received signal is

$$\mathbf{Y} = \sum_{i=1}^k X_i \sqrt{p_i} \mathbf{S}_i + \mathbf{N},$$

where $\mathbf{N}$ is a Gaussian $(\mathbf{0}, \sigma^2 \mathbf{I})$ noise.

- (a) Based on the observation $\mathbf{Y}$, find the LMSE estimate $\hat{X}_i(\mathbf{Y}) = \hat{\mathbf{a}}_i' \mathbf{Y}$ of X_i .
- (b) Let $\hat{\mathbf{X}} = [\hat{X}_1 \cdots \hat{X}_k]'$ denote the vector of LMSE estimates of bits transmitted by users $1, \dots, k$. Show that

$$\hat{\mathbf{X}} = \mathbf{P}^{1/2} \mathbf{S}' (\mathbf{S} \mathbf{P} \mathbf{S}' + \sigma^2 \mathbf{I})^{-1} \mathbf{Y}.$$

12.5.1 ● Continuing Example 12.10, the 21-dimensional vector $\mathbf{X}$ has correlation ma-

trix $\mathbf{R}_X$ with i, j th element

$$R_X(i, j) = \frac{\sin(\phi_0 \pi |i - j|)}{\phi_0 \pi |i - j|}.$$

We use the observation vector $\mathbf{Y} = \mathbf{Y}_{(n)} = [Y_1 \cdots Y_n]'$ to estimate $X = X_{21}$. Find the LMSE estimate $\hat{X}_L(\mathbf{Y}_{(n)}) = \hat{\mathbf{a}}_{(n)}' \mathbf{Y}_{(n)}$. Graph the mean square error $e_L^*(n)$ as a function of the number of observations n for $\phi_0 \in \{0.1, 0.5, 0.9\}$. Interpret your results. Does smaller ϕ_0 or larger ϕ_0 yield better estimates?

12.5.2 ● Repeat Problem 12.5.1 when

$$R_X(i, j) = \cos(\phi_0 \pi |i - j|).$$

12.5.3 ■ In a variation on Example 12.10, we use the observation vector $\mathbf{Y} = \mathbf{Y}_{(n)} = [Y_1 \cdots Y_n]'$ to estimate $X = X_1$. The 21-dimensional vector $\mathbf{X}$ has correlation matrix $\mathbf{R}_X$ with i, j th element

$$R_X(i, j) = r_{|i-j|}.$$

Find the LMSE estimate $\hat{X}_L(\mathbf{Y}_{(n)}) = \hat{\mathbf{a}}_{(n)}' \mathbf{Y}_{(n)}$. Graph the mean square error $e_L^*(n)$ as a function of the number of observations n , and interpret your results for the cases

- (a) $r_{|i-j|} = \frac{\sin(0.1\pi|i-j|)}{0.1\pi|i-j|}$,
- (b) $r_{|i-j|} = \cos(0.5\pi|i-j|)$.

12.5.4 ♦♦ In the k user CDMA system employing LMSE receivers in Problem 12.4.9, the receiver employs the LMSE bit estimate $\hat{X}_i$ to implement the bit decision rule $\hat{X}_i = \text{sgn}(\hat{X}_i)$ for user i . Using the approach in Problem 11.4.6, construct a simulation to estimate the BER for a system with processing gain $n = 32$, with each user operating at 6 dB SNR. Graph your results as a function of k for $k = 1, 2, 4, 8, 16, 32$. Make sure to average your results over the choice of code vectors $\mathbf{S}_i$.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

13.1.1 ● For Example 13.4, define a set of random variables that could produce the sample function $m(t, s)$. Do not duplicate the set listed in Example 13.7.

13.1.2 ● For the random processes of Examples 13.3, 13.4, 13.5, and 13.6, identify whether the process is discrete-time or continuous-time, discrete-value or continuous-value.

13.1.3 ■ Let $Y(t)$ denote the random process corresponding to the transmission of one symbol over the QPSK communications system of Example 13.6. What is the sample space of the underlying experiment? Sketch the ensemble of sample functions.

13.1.4 ■ In a binary phase shift keying (BPSK) communications system, one of two equally probable bits, 0 or 1, is transmitted every T seconds. If the k th bit is $j \in \{0, 1\}$, the waveform $x_j(t) = \cos(2\pi f_0 t + j\pi)$ is transmitted over the interval $[(k-1)T, kT]$. Let $X(t)$ denote the random process in which three symbols are transmitted in the interval $[0, 3T]$. Assuming f_0 is an integer multiple of $1/T$, sketch the sample space and corresponding sample functions of the process $X(t)$.

13.1.5 ■ True or false: For a continuous-value random process $X(t)$, the random variable $X(t_0)$ is always a continuous random variable.

13.2.1 ■ Let W be an exponential random variable with PDF

$$f_W(w) = \begin{cases} e^{-w} & w \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Find the CDF $F_{X(t)}(x)$ of the time-delayed ramp process $X(t) = t - W$.

13.2.2 ■ In a production line for 10 kHz oscillators, the output frequency of each oscillator is a random variable W uniformly distributed between 9980 Hz and 1020 Hz. The frequencies of different oscillators are independent. The oscillator com-

pany has an order for one part in 10^4 oscillators with frequency between 9999 Hz and 10,001 Hz. A technician takes one oscillator per minute from the production line and measures its exact frequency. (This test takes one minute.) The random variable T_r minutes is the elapsed time at which the technician finds r acceptable oscillators.

- Find p , the probability that an single oscillator has one-part-in- 10^4 accuracy.
- What is $E[T_1]$ minutes, the expected time for the technician to find the first one-part-in- 10^4 oscillator?
- What is the probability that the technician will find the first one-part-in- 10^4 oscillator in exactly 20 minutes?
- What is $E[T_5]$, the expected time of finding the fifth one-part-in- 10^4 oscillator?

13.2.3 ■ For the random process of Problem 13.2.2, what is the conditional PMF of T_2 given T_1 ? If the technician finds the first oscillator in 3 minutes, what is $E[T_2|T_1 = 3]$, the conditional expected value of the time of finding the second one-part-in- 10^4 oscillator?

13.2.4 ♦ Let $X(t) = e^{-(t-T)}u(t-T)$ be an exponential pulse with a random delay T . The delay T has a PDF $f_T(t)$. Find the PDF of $X(t)$.

13.3.1 ● Suppose that at the equator, we can model the noontime temperature in degrees Celsius, X_n , on day n by a sequence of iid Gaussian random variables with expected value 30 degrees and standard deviation of 5 degrees. A new random process $Y_k = [X_{2k-1} + X_{2k}]/2$ is obtained by averaging the temperature over two days. Is Y_k an iid random sequence?

13.3.2 ■ For the equatorial noontime temperature sequence X_n of Problem 13.3.1, a second sequence of averaged temperatures is $W_n = [X_n + X_{n-1}]/2$. Is W_n an iid random sequence?

13.3.3■ Let Y_k denote the number of failures between successes $k - 1$ and k of a Bernoulli (p) random process. Also, let Y_1 denote the number of failures before the first success. What is the PMF $P_{Y_k}(y)$? Is Y_k an iid random sequence?

13.4.1● The arrivals of new telephone calls at a telephone switching office is a Poisson process $N(t)$ with an arrival rate of $\lambda = 4$ calls per second. An experiment consists of monitoring the switching office and recording $N(t)$ over a 10-second interval.

- What is $P_{N(1)}(0)$, the probability of no phone calls in the first second of observation?
- What is $P_{N(1)}(4)$, the probability of exactly four calls arriving in the first second of observation?
- What is $P_{N(2)}(2)$, the probability of exactly two calls arriving in the first two seconds?

13.4.2● Queries presented to a computer database are a Poisson process of rate $\lambda = 6$ queries per minute. An experiment consists of monitoring the database for m minutes and recording $N(m)$, the number of queries presented. The answer to each of the following questions can be expressed in terms of the PMF $P_{N(m)}(k) = P[N(m) = k]$.

- What is the probability of no queries in a one-minute interval?
- What is the probability of exactly six queries arriving in a one-minute interval?
- What is the probability of exactly three queries arriving in a one-half-minute interval?

13.4.3● At a successful garage, there is always a backlog of cars waiting to be serviced. The service times of cars are iid exponential random variables with a mean service time of 30 minutes. Find the PMF of $N(t)$, the number of cars serviced in the first t hours of the day.

13.4.4● The count of students dropping the course “Probability and Stochastic Processes” is known to be a Poisson process of

rate 0.1 drops per day. Starting with day 0, the first day of the semester, let $D(t)$ denote the number of students that have dropped after t days. What is $P_{D(t)}(d)$?

13.4.5● Customers arrive at the Veryfast Bank as a Poisson process of rate λ customers per minute. Each arriving customer is immediately served by a teller. After being served, each customer immediately leaves the bank. The time a customer spends with a teller is called the service time. If the service time of a customer is exactly two minutes, what is the PMF of the number of customers $N(t)$ in service at the bank at time t ?

13.4.6● Given a Poisson process $N(t)$, identify which of the following are Poisson processes..

- $N(2t)$,
- $N(t/2)$,
- $2N(t)$,
- $N(t)/2$,
- $N(t+2)$,
- $N(t) - N(t-1)$.

13.4.7● Starting at any time t , the number N_τ of hamburgers sold at a White Castle in the τ minute interval $(t, t + \tau)$ has the Poisson PMF

$$P_{N_\tau}(n) = \begin{cases} (10\tau)^n e^{-10\tau} / n! & n = 0, 1, \dots \\ 0 & \text{otherwise} \end{cases}$$

- Find the expected number $E[N_{60}]$ of hamburgers sold in one hour (60 minutes).
- What is the probability that no hamburgers are sold in the 10-minute interval starting at 12 noon?
- You arrive at the White Castle at 12 noon. You wait a random time W (minutes) until you see a hamburger sold. What is the PDF of W ? Hint: Find $P[W > w]$.

13.4.8■ A sequence of queries are made to a database system. The response time of the system, T seconds, is the exponential $(1/8)$ random variable. As soon as the system responds to a query, the next query is

made. Assuming the first query is made at time zero, let $N(t)$ denote the number of queries made by time t .

- (a) What is $P[T \geq 4]$, the probability that a single query will last at least four seconds?
- (b) If the database user has been waiting five seconds for a response, what is $P[T \geq 13 | T \geq 5]$, the probability that the user will wait at least eight more seconds?
- (c) What is the PMF of $N(t)$?

13.4.9 ■ The proof of Theorem 13.3 neglected the first interarrival time X_1 . Show that X_1 has an exponential (λ) PDF.

13.4.10 ♦ $U_1, U_2, \dots$ are independent identically distributed uniform random variables with parameters 0 and 1.

- (a) Let $X_i = -\ln U_i$. What is $P[X_i > x]$?
- (b) What kind of random variable is X_i ?
- (c) Given a constant $t > 0$, let N denote the value of n , such that

$$\prod_{i=1}^n U_i \geq e^{-t} > \prod_{i=1}^{n+1} U_i.$$

Note that we define $\prod_{i=1}^0 U_i = 1$. What is the PMF of N ?

13.5.1 ● Customers arrive at a casino as a Poisson process of rate 100 customers per hour. Upon arriving, each customer must flip a coin, and only those customers who flip heads actually enter the casino. Let $N(t)$ denote the process of customers entering the casino. Find the PMF of N , the number of customers who arrive between 5 PM and 7 PM.

13.5.2 ● A subway station carries both blue (B) line and red (R) line trains. Red line trains and blue line trains arrive as independent Poisson processes with rates $\lambda_R = 0.05$ and $\lambda_B = 0.15$ trains/min, respectively. You arrive at a random time t and wait until a red train arrives. Let N denote the number of blue line trains that pass through the

station while you are waiting. What is the PMF $P_N(n)$?

13.5.3 ● A subway station carries both blue (B) line and red (R) line trains. Red line trains and blue line trains arrive as independent Poisson processes with rates $\lambda_R = 0.15$ and $\lambda_B = 0.30$ trains/min respectively. You arrive at the station at random time t and watch the trains for one hour.

- (a) What is the PMF of N , the number of trains that you count passing through the station?
- (b) Given that you see $N = 30$ trains, what is the conditional PMF of R , the number of red trains that you see?

13.5.4 ■ Buses arrive at a bus stop as a Poisson process of rate $\lambda = 1$ bus/minute. After a very long time t , you show up at the bus stop.

- (a) Let X denote the interarrival time between two bus arrivals. What is the PDF $f_X(x)$?
- (b) Let W equal the time you wait after time t until the next bus arrival. What is the PDF $f_W(w)$?
- (c) Let V equal the time (in minutes) that has passed since the most recent bus arrival. What is the PDF $f_V(v)$?
- (d) Let U equal the time gap between the most recent bus arrival and the next bus arrival. What is the PDF of U ?

13.5.5 ■ For a Poisson process of rate λ , the Bernoulli arrival approximation assumes that in any very small interval of length Δ , there is either 0 arrivals with probability $1 - \lambda\Delta$ or 1 arrival with probability $\lambda\Delta$. Use this approximation to prove Theorem 13.7.

13.5.6 ♦ Continuing Problem 13.4.5, suppose each service time is either one minute or two minutes equiprobably, independent of the arrival process or the other service times. What is the PMF of the number of customers $N(t)$ in service at the bank at time t ?

13.5.7♦ Ten runners compete in a race starting at time $t = 0$. The runners' finishing times $R_1, \dots, R_{10}$ are iid exponential random variables with expected value $1/\mu = 10$ minutes.

- What is the probability that the last runner will finish in less than 20 minutes?
- What is the PDF of X_1 , the finishing time of the winning runner?
- Find the PDF of $Y = R_1 + \dots + R_{10}$.
- Let $X_1, \dots, X_{10}$ denote the runners' interarrival times at the finish line. Find the joint PDF $f_{X_1, \dots, X_{10}}(x_1, \dots, x_{10})$.

13.5.8♦♦ Let N denote the number of arrivals of a Poisson process of rate λ over the interval $(0, T)$. Given $N = n$, let $S_1, \dots, S_n$ denote the corresponding arrival times. Prove that

$$f_{S_1, \dots, S_n | N}(S_1, \dots, S_n | n) = \begin{cases} n!/T^n & 0 \leq s_1 < \dots < s_n \leq T, \\ 0 & \text{otherwise.} \end{cases}$$

Conclude that, given $N(T) = n$, $S_1, \dots, S_n$ are the order statistics of a collection of n uniform $(0, T)$ random variables. (See Problem 5.10.11.)

13.6.1● Over the course of a day, the stock price of a widely traded company can be modeled as a Brownian motion process where $X(0)$ is the opening price at the morning bell. Suppose the unit of time t is an hour, the exchange is open for eight hours, and the standard deviation of the daily price change (the difference between the opening bell and closing bell prices) is $1/2$ point. What is the Brownian motion parameter α ?

13.6.2■ $X_0, X_1, \dots$ is an iid Gaussian $(0, 1)$ random sequence. The random sequence Y_n is defined by $Y_0 = 0$ and $Y_{n+1} = X_{n+1} + Y_n$. Find the autocorrelation function $R_Y[n, k]$.

13.6.3■ Let $X(t)$ be a Brownian motion process with variance $\text{Var}[X(t)] = \alpha t$. For a

constant $c > 0$, determine whether $Y(t) = X(ct)$ is a Brownian motion process.

13.6.4♦ For a Brownian motion process $X(t)$, let $X_0 = X(0), X_1 = X(1), \dots$ represent samples of a Brownian motion process with variance αt . The discrete-time continuous-value process $Y_1, Y_2, \dots$ defined by $Y_n = X_n - X_{n-1}$ is called an *increments process*. Show that Y_n is an iid random sequence.

13.6.5♦ This problem works out the missing steps in the proof of Theorem 13.8. For $\mathbf{W}$ and $\mathbf{X}$ as defined in the proof of the theorem, show that $\mathbf{W} = \mathbf{A}\mathbf{X}$. What is the matrix $\mathbf{A}$? Use Theorem 8.11 to find $f_{\mathbf{W}}(\mathbf{w})$.

13.7.1● X_n is an iid random sequence with expected value $E[X_n] = \mu_X$ and variance $\text{Var}[X_n] = \sigma_X^2$. What is the autocovariance $C_X[m, k]$?

13.7.2■ For the time-delayed ramp process $X(t)$ from Problem 13.2.1, find for any $t \geq 0$:

- The expected value function $\mu_X(t)$,
- The autocovariance function $C_X(t, \tau)$.
Hint: $E[W] = 1$ and $E[W^2] = 2$.

13.7.3■ A simple model (in degrees Celsius) for the daily temperature process $C(t)$ of Example 13.3 is

$$C_n = 16 \left[1 - \cos \frac{2\pi n}{365} \right] + 4X_n,$$

where $X_1, X_2, \dots$ is an iid random sequence of Gaussian $(0, 1)$ random variables.

- What is $E[C_n]$?
- Find the autocovariance function $C_C[m, k]$.
- Why is this model overly simple?

13.7.4♦ A different model for the daily temperature process $C(n)$ of Example 13.3 is

$$C_n = \frac{1}{2}C_{n-1} + 4X_n,$$

where $C_0, X_1, X_2, \dots$ is an iid random sequence of Gaussian $(0, 1)$ random variables.

- (a) Find the mean and variance of C_n .
- (b) Find the autocovariance $C_C[m, k]$.
- (c) Is this a plausible model for the daily temperature over the course of a year?
- (d) Would $C_1, \dots, C_{31}$ constitute a plausible model for the daily temperature for the month of January?

13.7.5 ■ For a Poisson process $N(t)$ of rate λ , show that for $s < t$, the autocovariance is $C_N(s, t) = \lambda s$. If $s > t$, what is $C_N(s, t)$? Is there a general expression for $C_N(s, t)$?

13.7.6 ♦ $N(t)$ is a Poisson process of rate $\lambda = 1$ and $X_0, X_1, X_2, \dots$ is an iid sequence of Gaussian $(0, \sigma)$ random variables that are independent of $N(t)$. Consider the process $\{Y(t) | t \geq 0\}$ defined by $Y(t) = X_{N(t)}$. Find the expected value $\mu_Y(t) = E[Y(t)]$ and the covariance function $C_Y(t, \tau)$. (Assume $|\tau| < t$.)

13.7.7 ♦ X_n is an iid random sequence with $E[X_n] = 0$ and $\text{Var}[X_n] = 3$. Find the autocorrelation function $C_Y[n, k]$ of the process $Y_n = X_{n-1}X_n$.

13.8.1 ● For an arbitrary constant a , let $Y(t) = X(t + a)$. If $X(t)$ is a stationary random process, is $Y(t)$ stationary?

13.8.2 ● $\mathbf{X} = [X_1 \ X_2]'$ has expected value $E[\mathbf{X}] = \mathbf{0}$ and covariance matrix

$$\mathbf{C}_X = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

Does there exist a stationary process $X(t)$ and time instances t_1 and t_2 such that $\mathbf{X}$ is actually a pair of observations $[X(t_1) \ X(t_2)]'$ of the process $X(t)$?

13.8.3 ● For an arbitrary constant a , let $Y(t) = X(at)$. If $X(t)$ is a stationary random process, is $Y(t)$ stationary?

13.8.4 ● Let $X(t)$ be a stationary continuous-time random process. By

sampling $X(t)$ every Δ seconds, we obtain the discrete-time random sequence $Y_n = X(n\Delta)$. Is Y_n a stationary sequence?

13.8.5 ● Given a stationary random sequence X_n , we can *subsample* X_n by extracting every k th sample: $Y_n = X_{kn}$. Is Y_n a stationary random sequence?

13.8.6 ■ Let A be a nonnegative random variable that is independent of any collection of samples $X(t_1), \dots, X(t_k)$ of a stationary random process $X(t)$. Is $Y(t) = AX(t)$ a stationary random process?

13.8.7 ♦ Let $g(x)$ be deterministic function. If $X(t)$ is a stationary random process, is $Y(t) = g(X(t))$ a stationary process?

13.9.1 ● Which of the following are valid autocorrelation functions?

$$\begin{aligned} R_1(\tau) &= \delta(\tau) & R_2(\tau) &= \delta(\tau) + 10 \\ R_3(\tau) &= \delta(\tau - 10) & R_4(\tau) &= \delta(\tau) - 10 \end{aligned}$$

13.9.2 ● Let A be a nonnegative random variable that is independent of any collection of samples $X(t_1), \dots, X(t_k)$ of a wide sense stationary random process $X(t)$. Is $Y(t) = A + X(t)$ a wide sense stationary process?

13.9.3 ● True or False: If X_n is a wide sense stationary random sequence with $E[X_n] = 0$, then $Y_n = X_n - X_{n-1}$ is a wide sense stationary random sequence.

13.9.4 ● Let X_n denote an iid sequence of Bernoulli ($p = 1/2$) random variables. Find the autocorrelation function $R_X[n, k]$ and the autocovariance function $C_X[n, k]$.

13.9.5 ● X_n is an iid sequence with $E[X_n] = \mu$ and $\text{Var}[X_n] = \sigma^2$. Find the autocorrelation function $R_X[n, k]$.

13.9.6 ■ $X(t)$ and $Y(t)$ are independent wide sense stationary processes. Determine if these processes are wide-sense stationary:

- (a) $V(t) = X(t) + Y(t)$,
- (b) $W(t) = X(t)Y(t)$.

13.9.7 ■ True or False: If X_n is a wide sense stationary random sequence with $E[X_n] = 0$, then $Y_n = X_n + (-1)^{n-1}X_{n-1}$ is a wide sense stationary random sequence.

13.9.8 ■ Consider the random process

$$W(t) = X \cos(2\pi f_0 t) + Y \sin(2\pi f_0 t),$$

where X and Y are uncorrelated random variables, each with expected value 0 and variance σ^2 . Find the autocorrelation $R_W(t, \tau)$. Is $W(t)$ wide sense stationary?

13.9.9 ■ $X(t)$ is a wide sense stationary random process with average power equal to 1. Let Θ denote a random variable with uniform distribution over $[0, 2\pi]$ such that $X(t)$ and Θ are independent.

- What is $E[X^2(t)]$?
- What is $E[\cos(2\pi f_c t + \Theta)]$?
- Let $Y(t) = X(t) \cos(2\pi f_c t + \Theta)$. What is $E[Y(t)]$?
- What is the average power of $Y(t)$?

13.9.10 ■ Prove the properties of $R_X[n]$ given in Theorem 13.12.

13.9.11 ♦ Let X_n be a wide sense stationary random sequence with expected value μ_X and autocovariance $C_X[k]$. For $m = 0, 1, \dots$, we define

$$\bar{X}_m = \frac{1}{2m+1} \sum_{n=-m}^m X_n$$

as the sample mean process. Prove that if $\sum_{k=-\infty}^{\infty} C_X[k] < \infty$, then $\bar{X}_0, \bar{X}_1, \dots$ is an unbiased consistent sequence of estimates of μ_X .

13.9.12 ♦ Determine whether each of these statements is true or false:

- If X_n and Y_n are independent stationary processes, then $V_n = X_n/Y_n$ is wide-sense stationary.
- If X_n and Y_n are independent wide sense stationary processes, then $W_n = X_n/Y_n$ is wide-sense stationary.

13.10.1 ● $X(t)$ and $Y(t)$ are independent wide sense stationary processes with expected values μ_X and μ_Y and autocorrelation functions $R_X(\tau)$ and $R_Y(\tau)$, respectively. Let $W(t) = X(t)Y(t)$.

- Find μ_W and $R_W(t, \tau)$ and show that $W(t)$ is wide sense stationary.
- Are $W(t)$ and $X(t)$ jointly wide sense stationary?

13.10.2 ■ $X(t)$ is a wide sense stationary random process. For each process $X_i(t)$ defined below, determine whether $X_i(t)$ and $X(t)$ are jointly wide sense stationary.

- $X_1(t) = X(t + a)$
- $X_2(t) = X(at)$

13.10.3 ■ $X(t)$ is a wide sense stationary stochastic process with autocorrelation function

$$R_X(\tau) = 10 \sin(2\pi 1000\tau) / (2\pi 1000\tau).$$

The process $Y(t)$ is a version of $X(t)$ delayed by 50 microseconds: $Y(t) = X(t - t_0)$ where $t_0 = 5 \times 10^{-5}$ s.

- Derive the autocorrelation function of $Y(t)$.
- Derive the cross-correlation function of $X(t)$ and $Y(t)$.
- Is $Y(t)$ wide-sense stationary?
- Are $X(t)$ and $Y(t)$ jointly wide sense stationary?

13.11.1 ● A stationary Gaussian process $X(t)$ is observed at times t_1 and t_2 to form the random vector $\mathbf{X} = [X(t_1) \ X(t_2)]'$ with expected value $E[\mathbf{X}] = \mathbf{0}$ and covariance matrix $\mathbf{C}_X = \begin{bmatrix} \sigma_1^2 & 1 \\ 1 & \sigma_2^2 \end{bmatrix}$. What is the range of valid values (if any) of σ_1^2 and σ_2^2 ?

13.11.2 ■ Given a Gaussian process $X(t)$, identify which of the following, if any, are Gaussian processes.

- $2X(t)$,
- $X(t/2)$,
- $X(t)/2$,
- $X(t) - X(t-1)$,
- $X(2t)$.

13.11.3 ■ A white Gaussian noise process $N(t)$ with autocorrelation $R_N(\tau) = \alpha\delta(\tau)$ is passed through an integrator yielding the output

$$Y(t) = \int_0^t N(u) du.$$

Find $E[Y(t)]$ and the autocorrelation function $R_Y(t, \tau)$. Show that $Y(t)$ is a nonstationary process.

13.11.4 ■ Let $X(t)$ be a Gaussian process with mean $\mu_X(t)$ and autocovariance $C_X(t, \tau)$. In this problem, we verify that the for two samples $X(t_1), X(t_2)$, the multivariate Gaussian density reduces to the bivariate Gaussian PDF. In the following steps, let σ_i^2 denote the variance of $X(t_i)$ and let $\rho = C_X(t_1, t_2 - t_1)/(\sigma_1\sigma_2)$ equal the correlation coefficient of $X(t_1)$ and $X(t_2)$.

- Find the covariance matrix $\mathbf{C}$ and show that the determinant is $|\mathbf{C}| = \sigma_1^2\sigma_2^2(1 - \rho^2)$.
- Show that the inverse of the correlation matrix is

$$\mathbf{C}^{-1} = \frac{1}{1 - \rho^2} \begin{bmatrix} \frac{1}{\sigma_1^2} & \frac{-\rho}{\sigma_1\sigma_2} \\ \frac{-\rho}{\sigma_1\sigma_2} & \frac{1}{\sigma_2^2} \end{bmatrix}.$$

- Now show that the multivariate Gaussian density for $X(t_1), X(t_2)$ is the bivariate Gaussian density.

13.11.5 ■ Show that the Brownian motion process is a Gaussian random process. Hint: For $\mathbf{W}$ and $\mathbf{X}$ as defined in the proof of the Theorem 13.8, find matrix $\mathbf{A}$ such that $\mathbf{W} = \mathbf{A}\mathbf{X}$ and then apply Theorem 8.11.

13.11.6 ♦♦ Let $X_1, X_2, \dots$ denote a sequence of iid Gaussian $(0, 1)$ random variables. Let $N(t)$ denote a Poisson process of rate λ that is independent of the sequence X_n . Consider the random process

$$Y(t) = \sum_{n=0}^{N(t)} X_n.$$

- Find the conditional CDF

$$F_{Y(t)|N(t)}(y|n) = P[Y(t) \leq y|N(t) = n].$$

Express your answer in terms of the $\Phi(\cdot)$ function.

- Is $Y(t)$ a Gaussian process?
- Is $Y(t)$ a stationary process?
- Is $Y(t)$ wide-sense stationary?

13.12.1 ● Write a MATLAB program that generates and graphs the noisy cosine sample paths $X_{cc}(t)$, $X_{dc}(t)$, $X_{cd}(t)$, and $X_{dd}(t)$ of Figure 13.3. Note that the mathematical definition of $X_{cc}(t)$ is

$$X_{cc}(t) = 2\cos(2\pi t) + N(t), \quad -1 \leq t \leq 1.$$

Note that $N(t)$ is a white noise process with autocorrelation $R_N(\tau) = 0.01\delta(\tau)$. Practically, the graph of $X_{cc}(t)$ in Figure 13.3 is a sampled version $X_{cc}[n] = X_{cc}(nT_s)$, where the sampling period is $T_s = 0.001$ s. In addition, the discrete-time functions are obtained by subsampling $X_{cc}[n]$. In subsampling, we generate $X_{dc}[n]$ by extracting every k th sample of $X_{cc}[n]$; see Problem 13.8.5. In terms of MATLAB, which starts indexing a vector $\mathbf{x}$ with first element $\mathbf{x}(1)$,

$$\mathbf{X}_{dc}(\mathbf{n}) = \mathbf{X}_{cc}(1 + (\mathbf{n}-1)\mathbf{k}).$$

The discrete-time graphs of Figure 13.3 used $k = 100$.

13.12.2 ● For the telephone switch of Example 13.28, we can estimate the expected number of calls in the system, $E[M(t)]$, after T minutes using the time average estimate

$$\overline{M}_T = \frac{1}{T} \sum_{k=1}^T M(k).$$

Perform a 600-minute switch simulation and graph the sequence $\overline{M}_1, \overline{M}_2, \dots, \overline{M}_{600}$. Does it appear that your estimates are converging? Repeat your experiment ten times and interpret your results.

13.12.3 ● A particular telephone switch handles only automated junk voicemail calls that arrive as a Poisson process of rate

$\lambda = 100$ calls per minute. Each automated voicemail call has duration of exactly one minute. Use the method of Problem 13.12.2 to estimate the expected number of calls $E[M(t)]$. Do your results differ very much from those of Problem 13.12.2?

13.12.4 ■ Recall that for a rate λ Poisson process, the expected number of arrivals in $[0, T]$ is λT . Inspection of the code for `poissonarrivals(lambda, T)` will show that initially $n = \lceil 1.1\lambda T \rceil$ arrivals are generated. If $S_n > T$, the program stops and returns $\{S_j | S_j \leq T\}$. Otherwise, if $S_n < T$, then we generate an additional n arrivals and check if $S_{2n} > T$. This process may be repeated an arbitrary number of times k until $S_{kn} > T$. Let K equal the number of times this process is repeated. What is $P[K = 1]$? What is the disadvantage of choosing larger n so as to increase $P[K = 1]$?

13.12.5 ■ In this problem, we employ the result of Problem 13.5.8 as the basis for a function `s=newarrivals(lambda, T)` that generates a Poisson arrival process. The program `newarrivals.m` should do the following:

- Generate a sample value of N , a Poisson (λT) random variable.
- Given $N = n$, generate $\{U_1, \dots, U_n\}$, a set of n uniform $(0, T)$ random variables.
- Sort $\{U_1, \dots, U_n\}$ from smallest to largest and return the vector of sorted elements.

Write the program `newarrivals.m` and experiment to find out whether this program is any faster than `poissonarrivals.m`.

13.12.6 ♦ Suppose the Brownian motion process with $\alpha = 1$ is constrained by barriers. That is, we wish to generate a process $Y(t)$ such that $-b \leq Y(t) \leq b$ for a constant $b > 0$. Build a simulation of this system. Estimate $P[Y(t) = b]$.

13.12.7 ♦ For the departure process $D(t)$ of Example 13.28, let D_n denote the time of the n th departure. The n th *interdeparture time* is then $V_n = D_n - D_{n-1}$. From a sample path containing 1000 departures, estimate the PDF of V_n . Is it reasonable to model V_n as an exponential random variable? What is the mean interdeparture time?